SEPARATION PROCESSES IN THE FOOD AND BIOTECHNOLOGY INDUSTRIES

Principles and Applications

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Preface

This book concentrates on the more recent methods and techniques for separating food components and products of the biotechnology industry. Each chapter deals with a specific type or area of application and includes information on the basic principles, industrial equipment available, commercial applications and an overview of current research and development.

The introductory chapter gives a brief overview of food composition and properties, and some of the heat and mass transfer considerations in batch and continuous processes. Separations from solids, liquids and gases are briefly discussed. A summary is provided of the more conventional separation techniques such as screening, filtration and centrifugation, and techniques for removing water, such as evaporation, freeze-concentration and dehydration. However, the main emphasis is on separation processes, which have received less attention in textbooks on food-engineering and food-processing operations. It is hoped that this book will complement and supplement many of these excellent texts. Chapter 2 deals with the use of supercritical fluids for extraction processes, with special reference to carbon dioxide. Chapter 3 deals with pressure-activated membrane techniques, and covers the general principles, reviews the applications of reverse osmosis, and serves as an introduction to Chapters 4 and 5, which deal specifically with the principles and applications of ultrafiltration and microfiltration respectively. The separation and recovery of charged particles by ion exchange and electrodialysis is covered in Chapter 6. Chapter 7 discusses innovative separation processes, and reviews some of the methods being actively investigated, some of which are now coming into industrial practice. Much of the emphasis in these chapters is on the separation and recovery of proteins and biologically active ingredients. Chapter 8 is specifically on the methods available for fractionating fat, and covers the upsurge in interest and recent developments in this area. The book concludes with a chapter on solids separation processes, with special reference to particulates. The physical properties which influence the separation are reviewed, together with sieving, screening and air classification. Wet processing methods for extraction are discussed, together with some miscellaneous applications such as dehulling, peeling and cleaning.

Much of the emphasis is on extraction of macromolecules, increasing the added value of foods and recovering valuable components from by-products and fermentation media. Many of the methods discussed are now in commercial practice, whilst others are being vigorously researched.

A. S. Grandison and M. J. Lewis

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Chapter 1

Separation processes – an overview

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1.1 FOODS - THE RAW MATERIAL

Food and drink play a vital role in all our lives, providing us with the nutrients essential for all our daily activities, including cell maintenance, growth and reproduction. Although foods are commonplace and much taken for granted, their composition and structure are by no means simple. Firstly, all foods are chemical in nature. For most foods the principal component is water and this water plays an important role in the overall behaviour of that food. One of the most important branches of separation is the removal of water, to save transportation costs and improve microbial stability.

The other components can be classified into major components, such as protein, fat or lipid, sugars, starch and fibre. The minor components include the minerals, which are known collectively as ash, vitamins and organic acids. Information on food composition and the amounts of major and minor components can be found in the *Composition of Foods Tables* (Paul and Southgate, 1978). Table 1.1. illustrates just some of the composition data that is available, for a selection of foods.

Food composition tables are useful in that they provide an average composition. However, some of their limitations are illustrated below, taking milk as an example. It should be noted that similar points could be made about most other foods.

Milk is extremely complex in terms of its chemical composition, containing protein, fat, carbohydrate, minerals and vitamins. There are many different proteins, which can be subdivided into the whey proteins, which are in true solution in the aqueous phase, and the caseins, which are in the colloidal form. The fat itself is a complex mixture of triglycerides and, being immiscible with water, is dispersed as small droplets, stabilised by a membrane, within the milk. The vitamins are classified as water or fat soluble, depending on which phase they most associate with. Some of the minerals, such as calcium and phosphorus, partition between the aqueous phase and the colloidal casein and play a major role in the stability of the colloidal dispersion. In addition, there are many other components present in trace amounts, which may affect its delicate flavour

Table 1.1. Composition of foods (weight/100 g)

	Milk	Apple	Peas	Flour	Beef	Cod
water (g)	87.6 (87.8)	85.6	78.5	13.0	74.0	82.1
protein (g)	3.3 (3.2)	0.3	5.8	9.8	20.3	17.4
fat (g)	3.8 (3.9)	tr.	0.4	1.2	4.6	0.7
sugar (g)	4.7 (4.8)	9.2	4.0	1.7	0.0	0.0
starch (g)	0.0 (0.0)	0.4	6.6	78.4	—	—
fibre (g)	0.0 (0.0)	2.4	5.2	3.4	—	
potassium (mg)	150 (140)	120	340	140	350	320
sodium (mg)	50 (55)	2	1	2	61	77
calcium (mg)	120 (115)	4	15	150	7	16
iron (mg)	0.05 (0.06)	0.3	1.9	2.2	2.1	0.3
phosphorus (mg)	95 (92)	16	100	130	180	170
vitamin C (mg)	1.50 (1.0)	15	25	—	—	—
vitamin B1 (mg)	0.04 (0.06)	0.04	0.32	0.33	0.07	0.08
vitamin B6 (mg)	0.04 (0.06)	0.03	0.16	0.15	0.32	0.33
vitamin D (ug)	0.03 (0.03)	—	—	—	tr	tr
vitamin E (mg)	0.10 (0.09)	0.2	tr	tr	0.15	0.44

* flour – household plain

tr – trace

These values are taken from Paul and Southgate (1978). Figures in parentheses are for milk, taken from McCance and Widdowson's *Composition of Foods Tables* (5th edn) (1991), Royal Society of Chemistry, MAFF. There are slight differences between the reported results.

and processing characteristics and nutritional value, such as trace minerals, organic acids and non-protein nitrogen compounds such as peptides, urea and amino acids. Walstra and Jenness (1984) have listed over 60 components present in milk, at levels that can be readily detected. Milk is also potentially a very unstable material. For example the protein can be made to coagulate by a variety of methods, including heating, addition of the enzyme rennet, acid, salts and ethanol. Also the fat globules rise to the surface under the influence of gravity.

Superimposed on this complex composition is the fact that it is subject to wide variation. Milks from different species differ markedly, and many types of milk other than cow's are consumed worldwide, e.g. sheep, goat, buffalo, camel. Within the same species there are large differences between breeds, and even between individual animals in the same herd. In addition to this, and of prime importance to the milk-processing industry, milk from the same animals is subject to wide seasonal variation, reflecting the change in the animals' diet throughout the year, and the stage of lactation. Factors relating to the handling of milk, such as the pH or the amount of dissolved oxygen, are also important to its stability.

Foods may also be contaminated with matter from their production environment, i.e. soil, water and farmyard. For example milk may be contaminated with dirt, straw, antibiotics, growth hormones, heavy metals, or radionuclides.

In chemical terms alone, there is a great deal of scope for separating the components in milk and some examples are listed:

- water removal to produce evaporated or dried products;
- fat separation to produce creams and butter;
- protein separation to produce cheese or protein concentrates;
- calcium removal to improve stability;
- lactose removal, as a specialised ingredient or for low-lactose products;
- removal of components responsible for tainting raw milk or the cooked flavour of heat-treated milk products;
- removal of radionuclides from milk.

In plant products pesticides and herbicides may additionally be present. Some foods, particularly of plant origin, also contain natural toxins, for example oxalic acid in rhubarb, and trypsin inhibitors, phytates and haemagglutinins in many legumes, cyanogenic glycosides in cassava and glucosinolates in rapeseed (Watson, 1987; Jones, 1992). However, the activity of most of these is reduced during normal processing and cooking methods.

Foods also contain active enzyme systems. For example, raw milk contains phosphatase, lipases and proteases, xanthine oxidase and many others. Fruits and vegetables contain polyphenol oxidases and peroxidases, both of which cause colour changes in foods, particularly browning, and lipoxygenases, which produce rancid off-flavours (Nagodawithana and Reed, 1993).

Therefore foods and wastes produced during food processing provide the raw material for extraction of enzymes and other important biochemicals with a range of applications, especially in the food and pharmaceuticals industries. Some examples are listed in Table 1.2. In the biotechnology industry, similar components may be produced by fermentation or enzymatic reactions and require extraction and purification. Perhaps the simplest example is alcohol, produced by a yeast fermentation, where the alcohol concentration that can be produced is limited to about 15 to 20%, as it inhibits further yeast metabolism. Alcohol can be recovered and concentrated by distillation. For low-alcohol or alcohol-free beers and wines, there is a requirement to remove alcohol. Again distillation or membrane techniques can be used.

A wide range of food additives and medical compounds are produced by fermentation; these include many enzymes, such as proteases for milk clotting or detergent cleaners, amino acids such as glutamic acid for monosodium glutamate (MSG) production, aspartic acid and phenylalanine for aspartame, and lysine for nutritional supplements, organic acids such as citric, gluconic and lactic, and hydrocolloids, such as xanthan gum for stabilising or thickening foods, and a wide range of antibiotics and other medicinal compounds.

In most cases it is necessary to purify these materials from dilute raw materials, which often requires sophisticated separation techniques. In fact a large proportion of the activities of the biotechnology industry is concerned with separations of this nature, which is known as downstream processing. In general, the products produced by bio-processing applications are more valuable than food products, and it is economically feasible to apply more complex separation techniques.

Table 1.2. Biochemicals extracted from foods and by products

Source	Product	Application
Papaya	Papain	Meat tenderisation Beer haze removal
Calf stomach	Rennet	Cheesemaking
Barley	Amylase	Glucose syrup production Baking
Pancreas	Insulin	Control of diabetes
Connective tissue	Gelatin	Gelling agent
Egg	Lysozyme	Food preservative
Soybean	Lecithin	Emulsifier
Horseradish	Peroxidase	Diagnostics
Milk	Lactoperoxidase	Antibacterial
Egg	Ovotransferrin	Antibacterial

Most foods also come contaminated with microorganisms, derived from the environment where they are produced, such as soil, water or the farmyard. These will cause food to spoil or decay, or in the case of pathogenic organisms, cause food poisoning, either directly or by producing toxins. Their activity needs to be controlled. Foods can be pasteurised, blanched, sterilised, and irradiated to control such activity. For liquid products microorganisms can also be removed by membrane sterilisation techniques.

However, it is not only the chemical nature of the food that is important; the organisation and structure of components, and hence the physical properties, are vital considerations to the application of separation techniques. For example, the composition of apples as shown in Table 1.1 appears to be relatively simple. However, to fabricate (create) an apple in the laboratory from these components would be technically impossible. Large differences occur between apples in terms of their colour, flavour and texture which are not apparent from composition tables. Similar considerations apply to many other raw materials. Unfortunately for the food processor, nature does not provide materials of uniform chemical or physical properties. Foods have important physical properties, which will influence the separation technique that is to be selected; some of these are listed in Table 1.3. In addition, the structure of both raw materials and processed foods is very varied. They may exist as emulsions or colloids. They may be non-homogeneous on a macroscopic or microscopic scale, possessing fibrous structure and cellular structure, or layered structures such as areas of fat in meat.

Foods are found as solids or liquids, but gas is frequently incorporated. This may be desirable, as in processed foods such as ice cream, bread or carbonated drinks. However, it may be desirable to remove dissolved gases from liquids such as oxygen or cellular gases from fruit and vegetables before certain processing operations.

This brief introduction has aimed to illustrate the diverse nature of foods and related biological materials, and give an insight into their composition and structure. It is this

complexity and diversity which provides the scope and potential for separating selected components from foods.

Table 1.3. Examples of physical properties of foods, and separation processes to which they relate

Physical property	Separation technique
Size, size distribution, shape	Screening, air classification
Density	Centrifugation
Viscosity	Liquid extraction processes
Rheological	Expression
Surface properties	Froth flotation
Thermal properties	Evaporation, drying
Electrical	Electrostatic sorting
Diffusional	Extraction
	Membrane separations
Solubility	Solvent extraction
	Thermal denaturation
Optical	Reflectance (colour) sorting

1.2 SEPARATION TECHNIQUES

1.2.1 Introduction

Separation of one or more components from a complex mixture is a requirement for many operations in the food and biotechnology industries. The components in question range from particulate materials down to small molecules. The separations usually aim to achieve removal of specific components, in order to increase the added value of the products, which may be the residue, the extracted components or both. All separations rely on exploiting differences in physical or chemical properties of the mixture of components. Some of the more common properties involved in separation processes are particle or molecular size and shape, density, solubility and electrostatic charge. These properties are discussed in more detail elsewhere (Mohsenin, 1980, 1984; Lewis, 1990). In some operations, more than one of these properties are involved. However, most of the processes involved are of a physical nature.

Separation from solids or liquids involves the transfer of selected components across the boundary of the food. In many processes another stream or phase is involved, for example in extraction processes. However, this is not always so, for example expression, centrifugation or filtration. In expression, fruit juice or oil is squeezed from the food by application of pressure. In centrifugation, fat can be separated from water due to their density differences, by the application of a centrifugal force. In filtration there is a physical barrier to the transfer of certain components and the liquid is forced through the barrier by pressure, whilst the solids are retained. The resistance to flow will change throughout the filtration process, due to solids build-up. It can be seen that main driving

forces in these applications are pressure and density differences. As for all processes, separation rates are very important and these are affected by the size of the driving forces involved.

In situations where a second phase or stream is involved, mass-transfer considerations become important; these involve the transfer of components within the food to the boundary, the transfer across the boundary and into the bulk of the extraction solvent. It is also important to increase the interfacial area exposed to the solvent. Therefore, size reduction, interfacial phenomena, turbulence and diffusivities all play a role in these processes. In many applications this additional stream is a liquid, either water or an organic solvent; more recently supercritical fluids, such as carbon dioxide, have been investigated (see Chapter 2). However, in hot-air drying the other phase is hot air, which supplies the energy and removes the water. Mass-transfer considerations are important also in some membrane applications and adsorption processes, where the additional stream is a solid. Other examples of driving force are concentration differences and chemical potential, which are involved in these operations (Loncin and Merson, 1979; Gekas, 1992).

In some processes, both heat and mass transfer processes are involved. This is especially so for separation reactions involving a change of phase, such as evaporation or sublimation. Heat is required to cause vaporisation for evaporation, dehydration and distillation processes. Water has a much higher latent heat of vaporisation (2257 kJ/kg) than most other organic solvents. With solid foods the rate of heat transfer through the food may limit the overall process; for example in freeze-drying the process is usually limited by rate of heat transfer through the dry layer.

Separation processes may be batch or continuous. A single separation process, for example a batch extraction, involves the contact of the solvent with the food. Initially concentration gradients are high and the rate of extraction is also high. The extraction rate falls exponentially and eventually an equilibrium state is achieved when the rate becomes zero. The extraction process may be accelerated by size reduction, inducing turbulence and increasing the extraction temperature. Equilibrium is achieved either when all the material has been extracted, in situations where the volume of solvent is well in excess of the solute or when the solvent becomes saturated with the solute, i.e. when the solubility limit has been achieved, when there is an excess of solute over the solvent. However, the attainment of equilibrium may take some considerable time. Batch reactions may operate far away from equilibrium or close to it.

Equilibrium data is important in that it provides information on the best conditions that can be achieved at the prevailing conditions. Equilibrium data is usually determined at fixed conditions of temperature and pressure. Some important types of equilibrium data are:

- solubility data for extraction processes;
- vapour/liquid equilibrium data for fractional distillation;
- partition data for selective extraction processes;
- water sorption data for drying.

Continuous processes may be single- or multiple-stage processes. The stages themselves may be discrete entities, for example a series of stirred tank reactors, or there may

be many stages built into one unit of equipment, for example a distillation column or a screw extractor. The flow of the two streams can either be co-current or counter-current, although counter-current is normally favoured as it tends to give a more uniform driving force over the length of the reactor as well as a higher average driving force over the reactor. In some instances a combination of co-current and counter-current may be used; for example in hot air drying, the initial process is co-current to take advantage of the high initial driving rates, whereas the final drying is counter-current to permit drying to a lower moisture content.

Continuous equipment usually operates under steady state conditions, i.e. the driving force changes over the length of the equipment, but at any particular location it remains constant with time. However, when the equipment is first started, it may take some time to achieve steady-state. During this transition period it is said to be operating under unsteady state conditions. In continuous processes the flow may be either streamline or turbulent. Consideration should be taken of residence times and distribution of residence times within the separation process; the two extremes of behaviour are plug flow, with no distribution of residence times, through to a well-mixed situation, with an infinite distribution of residence times. More detailed analysis of residence time distributions is provided by Levenspiel (1972).

How close the process operates to equilibrium depends upon the operating conditions, flow rates of the two phases, time and temperature. These conditions affect the efficiency of the process, such as the recovery and the size of equipment required.

Finally, all rates of reaction are temperature dependent. Physical processes are no exception, although activation energies are usually much lower than for chemical reaction rates. Using higher temperatures normally increases separation rates.

However, there are limitations with biological materials: higher temperatures increase degradation reactions, causing colour and flavour changes, enzyme inactivation, protein denaturation, loss of functionality and a reduction in nutritional value. Safety issues with respect to microbial growth may also need to be considered.

A brief overview of separation methods is now considered in this chapter, based primarily on the nature of the material or stream subjected to the separation process, i.e. whether it is solid, liquid or gaseous. Other possible classifications are based on unit operations; e.g. filtration, evaporation, dehydration etc. or types of phase contact, such as solid–liquid or gas–liquid contacting processes.

More detailed descriptions of conventional techniques can be found elsewhere – e.g. Brennan *et al.* (1990), Perry and Green (1984), King (1982).

1.2.2 Separations from solids

Most solid foods are particulate in nature, with particles having a large variety of shapes and sizes. Separations involving solids fall into two categories. The first is where it is required to separate or segregate the particles; such processes are classified as solid–solid separations. The second is where the requirement may be to selectively remove one or several components from the solid matrix. Other processes involving solids are concerned with the removal of discrete solid particles from either liquids or gases and vapours (but these will be discussed in other sections).

Solid–solid separations

Separations can be achieved on the basis of particle size from the sorting of large food units down to the molecular level. Shape, and other factors such as electrostatic charge or degree of hydration, may also affect these separations. Screening of materials through perforated beds (e.g. wire mesh or silk screens) produces materials of more uniform particle size. Screening contributes to sorting and grading of many foods, in particular fruits, vegetables and cereals. Cleaning of particulate materials or powders in the dry state can be achieved using screens in two ways. Dedusting is the removal of undersize contaminants from larger particles, e.g. beans or cereals. Scalping is the removal of oversize contaminants from powders or small particulate materials, e.g. sugar, flour. A wide range of geometric designs exists, including flat bed and rotary fixed aperture screens, and numerous variable aperture designs are available (Slade, 1967; Brennan *et al.*, 1990).

Differences in aerodynamic properties can be exploited in the cleaning, sorting and grading of particulate food raw materials (e.g. cereals, peas, nuts, flour) in the dry state. Many designs of aspirator have evolved in which the feed is applied to controlled velocity air streams where separation into two or more fractions is effected. Alternatively, differences in hydrodynamic properties can be used in the sorting of foods such as apples or peas.

A combination of particle size and density may be used to separate solids by settlement. If the solids are suspended in a fluid (liquid or gas), separation may be achieved on the basis that larger, more dense particles will settle more rapidly than smaller, less dense ones. This may be aided by the application of centrifugal force in air classification, as discussed in Chapter 9.

Differences in buoyancy between solid particles is the basis of flotation washing of some foods. For example, heavy debris, such as stones or bruised and rotten fruit, may be removed from sound fruit by fluming the dirty produce over a series of weirs.

Froth flotation depends on the differential wetting of particles. In the case of separating peas from weed seeds, the mixture is immersed in a dilute mineral oil emulsion through which air is blown. The contaminating seeds float on the foam and may be skimmed off. On a smaller scale, the method can be used to separate materials which react selectively with a surfactant, such as heavy metals, from a mixture. Surface active agents, such as proteins and other foam-inducing materials, can be separated in a similar manner. These techniques are commonly used in effluent treatment.

Operations where the outer surface of the food is removed also fall into this category. Examples include dehulling of cereals and legumes and peeling of fruit and vegetables (see Chapter 9). Cereals may be cleaned and sorted on the basis of shape to remove contaminants of similar size. Examples of this are disc and cylinder sorters which employ indentations of particular shape to pick up the corresponding food particles.

A range of equipment is also available to separate food units on the basis of photometric, magnetic and electrostatic properties. Red and green tomatoes, or blackened beans or nuts may be separated automatically on the basis of reflectance properties. Magnetic cleaning is used to remove ferrous metal particles from foods, and thus to protect both the consumer and processing equipment. Electrostatic properties may be

exploited in separating seeds which may be of similar size and shape, or in the cleaning of tea.

More detailed information on solid–solid separations is provided in Chapter 9.

Separation from the solid matrix

Many plant materials contain valuable liquid components such as oils or juices in the cellular structure. These may be separated from the pulped raw material by the use of presses, in a process known as expression. Batch type hydraulic systems or continuous roller, screw or belt systems are available for different applications such as fruit juice, wine and cane sugar production, or extraction of oil from seeds. Expression of fruit juices may be aided by the use of enzymes to improve efficiency of expression and to control the pectin level. Some of the physical properties related to expression processes are discussed by Schwartzberg (1983).

An alternative system to recover components from within a solid matrix is extraction, which relies on the use of differential solubilities for extraction of soluble solids such as sugar from sugar beet, coffee from roasted ground beans, juices from fruit and vegetables and from materials during the manufacture of instant tea. The most common extraction material is hot or superheated water. However, organic solvents are used, e.g. hexane for oil extraction and methylene chloride to extract caffeine from tea and coffee. The use of supercritical fluids such as carbon dioxide is covered in detail in Chapter 2. Extraction processes as equilibrium stage processes are covered in more detail by Brennan *et al.* (1990), Loncin and Merson (1979), Perry and Green (1984).

Many oil extraction processes employ expression, followed by solvent extraction, to obtain a high recovery of oil. The crude oil is then subjected to a series of refining processes, involving degumming, decolorisation and deodorisation to remove undesirable components.

Water, the most common component of most foods, can be removed from solids by the process of dehydration; in this case thermal energy is required to effect evaporation of the water, and this is usually supplied by hot air. Hot air drying is classified as liquid phase drying and results in shrinkage and case-hardening and loss of some volatiles of foods. Types of drier include overdraught, throughdraught, fluidised bed and pneumatic driers. These are described in more detail by Brennan *et al.* (1990), Mujumdar (1987). Freeze-drying, whereby the food is frozen and then subjected to a vacuum, provides a method which reduces shrinkage, case-hardening and flavour loss. Sublimation occurs during freeze-drying. Here conditions are controlled such that water is removed directly from its solid phase to its vapour phase, without passing through the liquid state. To achieve this, the water vapour pressure must be kept below the triple point pressure (4.6 mm Hg) (Mellor, 1978; Dalgleish, 1990).

The removal of air from fruit and vegetables, prior to heat treatment in sealed containers, is of paramount importance to prevent excessive strain on the seams during the sterilisation and subsequent cooling. This is accomplished by blanching, using steam or hot water. Nutrient losses due to leaching are minimised using steam (Selman, 1987).

1.2.3 Separations from liquids

This section will cover those situations where the separation takes place from a fluid, i.e. a substance which flows when it is subject to a shear stress. An important physical property is the viscosity of the fluid, which is defined as the ratio of the shear stress to shear rate. Viscosity and its measurement is discussed in more detail by Lewis (1990).

Solid components may be present dissolved in the liquid, in a colloidal dispersion or in suspension. For example, milk contains lactose, minerals and whey proteins in true solution, casein and calcium phosphate as a colloidal dispersion and fat globules dispersed in the aqueous phase. There may also be sediment resulting from other contaminants of the milk. The objective of the separation may be to remove any of these components.

Liquid–solid separations

Liquid–solid separation applies to operations where discrete solids are removed from the liquid. There are a number of ways of achieving this and these will be briefly reviewed.

Conventional filtration systems separate suspended particles of solids from liquids on the basis of particle size. The liquid component is passed through a porous membrane or septum which retains the solid material either as a filter cake on the upstream surface, or within its structure, or both. Filter media may be rigid, such as wire mesh or porous ceramics, or flexible, such as woven textiles, and are available in a variety of geometric shapes and pore sizes. In practice, the flow of filtrate may be brought about by gravity, the application of pressure greater than atmospheric upstream of the filter (pressure filtration), applying a vacuum downstream (vacuum filtration) or by means of centrifugal force (centrifugal filtration). The theory and equipment for industrial filtration are fully described by Brennan *et al.* (1990). Applications can be divided into those where a slurry containing large amounts of insoluble solids is separated into a solid cake and a liquid, either of which may be the desired product; alternatively clarification is the removal of small quantities (<2%) of suspended solids from a valuable liquid.

Filtration finds applications throughout the food and biotechnology industries. Sugar juices from cane or beet are filtered to remove high levels of insoluble solids, and are frequently clarified at a later stage. Filtration is employed at various stages during the refining of edible oils. In the brewing industry filtration of mash, yeast recovery after fermentation and clarification of beer are carried out. Filtration is used during the manufacture of numerous other foods, e.g. vinegar, starch and sugar syrups, fruit juices, wine, canning brines. In biotechnology, filtration is carried out to clarify and recover cells from fermentation broths.

More recently, membranes with much smaller pores have been introduced. Microfiltration involves the removal of very fine particles or the separation of microorganisms and sterilisation of fluids (see Chapter 5). Ultrafiltration membranes permit the passage of water and components of low molecular weight in a fluid but reject macromolecules such as protein or starch.

Solids may be separated from liquids on the basis of particle size and density using settlement, or using centrifugation. Settlement is a slow process because it relies on the influence of gravity, but is widely used in water and effluent treatment processes. In

centrifugal classification a suspension of insoluble solids (not more than about 1%) is subjected to cyclic motion in a bowl, which subjects the particles to a centrifugal force, many times in excess of the gravitational force. The more dense solid is retained on the inner surface of the bowl while the liquid is tapped off at the centre. An alternative is to use a filtering centrifuge in which the bowl wall is perforated so the liquid is forced out through the wall. The size of the perforations determines what portion of solids is retained in the bowl. Various designs of centrifuge are available for numerous applications such as removal of solids from dairy fluids, oils, juices, beverages, fermentation broths, or dewatering of sugar crystals and corn starches. Such separations may be carried out on a batchwise basis, although automatic and continuous centrifuges are available.

Solid–liquid separation techniques have been covered in more detail by Purchas and Wakeman (1986) and Brennan *et al.* (1990).

Immiscible liquids

Centrifugation in cylindrical bowls provides the simplest method to separate immiscible liquids of different densities. As the dense and lighter liquid streams are removed throughout, the operation can be carried out on a continuous basis. Either tubular-bowl or disc-

Cascades of separation vessels

Fractionation of a mixture can be effected on a 'time-controlled' basis, using the pilot plant described above, by collecting fractions at constant temperature and pressure during the course of the extraction. This is because the more soluble components of the mixture are preferentially extracted in the early stages of the extraction. However, because the solubilities of less soluble components may be very low, this mode of operation can be highly inefficient in terms of time. It is therefore preferable to gradually increase the 'solvent power' of the NCF by progressively changing the conditions of temperature and/or pressure in the extraction vessel. One such procedure involves progressively increasing the extraction pressure in stages such that the more soluble components are removed first. To achieve good separation between stages it would be necessary to continue the extraction at each pressure until equilibrium was approached and the rate of removal of components began to level off. The separation vessel would then have to be emptied, which could involve halting the extraction. This would clearly give rise to a very protracted process.

A simple solution to this problem is to introduce the fractionation at the separation stage of the process by using a cascade of separation vessels as shown in Fig. 2.18. Here the sample is extracted under extreme conditions (high pressure/temperature) at which all extractable components are removed in the extractor. The CO_2 stream then passes to the first separation vessel where the conditions P_1, T_1 are set to precipitate the first fraction of least soluble components. The output stream from the first separator is then passed to a second vessel at lower pressure P_2, where the second fraction precipitates. The fractionation thereby takes place down a 'cascade' of separation vessels set at progressively decreasing pressures. Because separation is much more rapid than extraction the process is not rate-limited. Care must, however, be taken to avoid carrying-over of precipitated material between separation vessels.

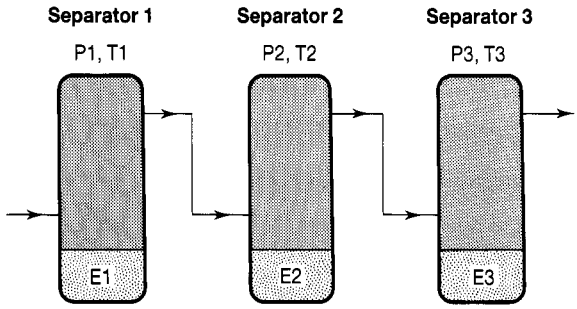

Fig. 2.18. Separation stage configuration for NCF fractionation using a 'cascade' of separation vessels (E = extract).

Zosel's 'hot finger' fractionation column

The anomalous effect of temperature at low pressures in SCFs has been introduced previously (Section 2.3.1) in which increasing the temperature results in a drop in solubility. If the temperature rise is sufficiently high, this behaviour can also be extended

to fluids at higher pressures. Zosel (1978) has exploited this effect in an elegantly simple fractionation apparatus (Fig. 2.19) in which fractionation and extraction take place in a single high-pressure vessel.

The mixture to be fractionated is placed in the bottom of the vessel and a 'hot finger' probe at the top serves to establish a temperature gradient throughout the fluid in the column. Ignoring the small pressure gradient required to maintain flow, the vessel can be considered to be at constant pressure. The density of the fluid and solubility of components are therefore uniquely determined by temperature. Solvation in the high-temperature (low-density) region in the vicinity of the probe will be at a minimum and only the most soluble components will remain in solution and be able to pass through to the separation vessel. Less soluble components will precipitate and remain under reflux in the lower portion of the column. Fractions can be removed from the column in order of decreasing solubility by progressively lowering the temperature of the probe and/or increasing the pressure.

The potential of the apparatus has been demonstrated in the fractionation of fish oils into over 50 discrete fractions.

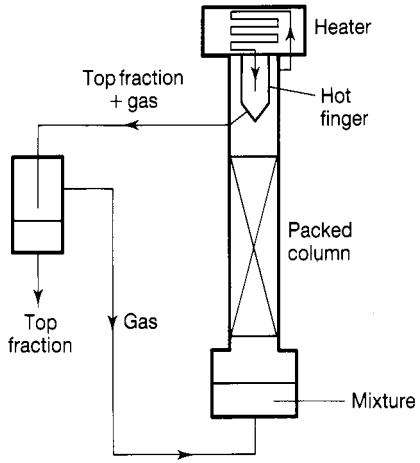

Fig. 2.19. An NCF fractionation column using a 'hot finger' to control selectivity (Zosel, 1978).

2.6 APPLICATIONS

The last decade has seen a rapid growth in research activity in the area of SCF extraction. Although many glowing reviews and hundreds of publications have appeared covering a wide variety of applications, there are surprisingly few processes and plant operating on a commercial scale. It is possible that many of the early publications overstressed the potential advantages of SCFs without addressing the limitations. It is certainly the case that many reports of applications do not differentiate between the extraction/identification of trace amounts of components and realistic quantities on which a process could be

based. Without the necessary data it is often difficult or impossible to quantify the efficiency of a reported SCF extraction process in terms of mass transfer and throughput of CO_2. Against this background it is not surprising that many popular misconceptions abound regarding SCFs.

SCF extraction is an expensive process and, although offering many advantages (Table 2.3), should not be used simply because it is a 'novel' technique. Unless use is being made of its unique features there is no rationale for its implementation if a cheaper separation process can meet the requirements of the separation as effectively. The one clear advantage that CO_2 does offer for food applications is its lack of toxicity. In the current climate of growing consumer concern regarding food safety this feature will undoubtedly promote its use in the food industry.

The following review of applications includes examples of processes currently operating on a commercial scale and applications that could soon be implemented. Examples of applications showing novel features that serve to illustrate the potential of SCF extraction are also included.

Table 2.3. Criteria determining implementation of SCF CO_2 extraction

Criteria	Advantages	Disadvantages
Economic	Can save energy Cheap Stable market	High capital cost
Safety	Non-toxic* Non-flammable	High pressure
Physical properties	Enhanced control through T and P (fractionation*) High vapour pressure enables separation at low T* Low viscosity can provide enhanced mass transfer	Low solubilities
Chemical properties	Non-oxidative Environment*	pH effects and hydrate formation when water present

* = implication for improved quality.

2.6.1 Decaffeination of coffee and tea

Over the past decade there has been a growing consumer aversion to the levels of stimulants in beverages and there is now a large market in decaffeinated products. Although there are some notable differences, the conventional solvent and NCF extraction processes for coffee and tea share many common features.

In the production of coffee 'green' beans are roasted to generate the coffee oils that later impart flavour to the infusion. To avoid co-extraction of flavour components in the decaffeination process green beans are therefore generally extracted. Moist beans are used since it is found that dry beans do not allow effective extraction. The crucial role of water in the process has not been unambiguously established but is thought to 'free' the caffeine from adsorption on the surface, reduce complexation with other molecules and reduce the tortuosity factor by swelling the cellular structure. In the conventional decaffeination process, organic solvents such as methylene chloride or ethyl acetate are employed to reduce the level of caffeine from approximately 1% w/w to 0.06%. Heightened awareness of the potentially harmful effects of residual levels of these solvents has provided some impetus to examine alternative safer solvents such as NCF CO_2. Extensive kinetic studies of the infusion of coffee beans into water (Spiro *et al.*, 1984) and organic solvents (Bischel, 1979) have been reported.

Judging by the scale of its implementation, decaffeination of coffee is one of the most successful commercial applications of CO_2 extraction with large plants operating in West Germany (Bremerhaven, 27.3 million kg per year) and the USA (Dallas). At first sight CO_2 extraction would not appear to be promising since the solubility of caffeine in NCF CO_2 is relatively low (< 0.2% w/w; Ebelling and Franck, 1984), but this is offset by the high added value of the process. Moreover, CO_2 provides a very selective solvent for decaffeination which does not remove as many of the desirable flavour–precursor components as alternative organic solvents (e.g. ethyl acetate).

Selectivity for caffeine is probably greatest in NCL CO_2, though solubility is low (approx. 0.05%). Increasing pressure and temperature serves to increase the solubility of caffeine but reduces the separation factor. In some processes for production of decaffeinated instant coffee, flavour components that are co-extracted are later separated and added back at the drying stage (Roselius *et al.*, 1974). It is often stated that the water added in decaffeination serves to increase the solubility of caffeine in NCF CO_2. This hypothesis is not borne out by experimental evidence, since increasing the levels of water to saturation does not significantly affect the solubility of caffeine in SCF CO_2 (Moulson, 1988).

A comprehensive review of the patent literature on decaffeination (and other NCF extraction processes) is given in an excellent treatise on NCF extraction by McHugh and Krukonis (1986). Most of the methods employed share common features. In a typical process (Zosel, 1974), wet green beans are contacted with SCF CO_2 in an extraction vessel and the extracted caffeine is removed by partitioning to a coexisting aqueous phase. Other methods of separation of caffeine from the CO_2 stream involve adsorption onto active carbon (Roselius *et al.*, 1979). The latter approach is less desirable since the caffeine, which can be sold as a by-product, is destroyed by burning during the reactivation of the carbon adsorbant. Some patents (Prasad *et al.*, 1981; Margolis and Chiovini, 1981) describe techniques for the extraction of aqueous roasted coffee extracts in a continuous counter-current extraction process. One of the more novel claims in the patent literature involves mixing the beans with active carbon and carrying out the extraction and separation in one vessel (Zosel, 1981). This not only saves on capital costs, but also promotes rapid mass transfer. The activated carbon (containing extracted caffeine) is later removed by sieving.

Decaffeination illustrates well the adverse effects of solute adsorption since it is observed that considerably more CO_2 is required to effect decaffeination than would be expected on solubility grounds alone. This is because caffeine adsorbs onto the beans and prevents equilibrium solubility being reached in the extraction vessel (McHugh and Krukonis, 1986).

Decaffeination of coffee represents one of the most widely patented applications of NCF extraction with innumerable variations and permutations claimed. In contrast the NCF decaffeination of tea has been less well documented (Vitzthum and Hubert, 1979). This may well be due to the more delicate flavour profile of tea, which is more susceptible to damage during the extraction process. The higher levels of caffeine in tea (3% compared to 1% for coffee) may also be a contributing factor.

2.6.2 Seed oil extraction

Extraction and processing of seed oils is a large-scale commercial operation with high throughput. The type of oilseeds processed depends almost entirely on regional agricultural policies; in the USA soya oil is by far the largest commodity, whereas in Canada rapeseed oil is more common. However, seed oils all contain the same basic triglyceride units, though the distribution of individual carboxylic acids in the triglycerides is a unique feature of each oil which imparts individual characteristics. The overall solubility of seed oils in NCF CO_2 does not appear to vary much and basic principles and conditions established for the extraction of one oil translate reasonably well to another.

In the conventional process for oilseed extraction the pretreated seeds are extracted using hexane. This also removes phospholipids (lecithin) which, although beneficial to health, present physical problems when the oil is used for cooking. In the refining process a degumming stage is therefore required to remove the phospholipids from the oil which is then bleached (to reduce colour) and deodorised.

Most of the pioneering work on the SCF CO_2 extraction of seed oils has been carried out in the USA and West Germany. The solubility of soya oil in SCF CO_2 has been measured at high pressure (Friedrich *et al.*, 1982; Stahl *et al.*, 1980, 1983a, 1984) and there has been some success in fitting solubility data using simple EOS models (Klein and Schultz, 1989). The most important feature of Friedrich's measurements was that at high pressures and temperatures (800 bar, 70°C) the oil became completely miscible with CO_2. This suggested the possibility of an efficient high-pressure SCF extraction process (Friedrich and Pryde, 1984). Under these extreme conditions the rate of extraction from soya flake was found to be rapid, and nearly complete extraction could be achieved in 20 min. Moreover, since the solubility falls markedly at lower pressures, most of the oil could be separated from the CO_2 stream without having to undergo complete decompression. Other investigators of the rate of NCF extraction of oilseeds have worked at lower pressures (e.g. King *et al.*, 1987; Brunner, 1984).

Probably the most significant feature of the SCF extraction of oilseeds is that phospholipids are not co-extracted (Friedrich and Pryde, 1984). This eliminates the need for chemical degumming of the oil. If the phospholipids are required as a separate commodity they can be extracted from the seeds in a secondary extraction with a solvent

such as hexane. Oil extracted with SCF CO_2 is also often reported as having a lighter colour.

Although showing many advantages over conventional processing techniques (List and Friedrich, 1985) NCF extraction is not currently economically viable for large-scale oilseed extraction. This is due to the low bulk value of seed oils, high plant cost and inconvenience of batch processing large quantities of solid materials under high-pressure conditions. As a result there has been much interest and speculation concerning the development of systems for continuously feeding solids into high-pressure vessels. Although a successful design was originally implemented for transferring coal slurries at low pressure (Blisset *et al.*, 1979) there appears to have been little success in producing a working system for operating under the higher pressures involved in SCF extraction, though many designs have been proposed (Stahl and Stadler, 1984).

Implementation of SCF extraction of bulk commodity seed oils will probably have to await the development of continuous processing plant or the tightening of legislation governing the use of petrochemical solvents. SCF extraction is, however, appropriate for smaller-scale processes involving high-value oils such as evening primrose (Tolboe *et al.*, 1988) and wheatgerm (Christianson *et al.*, 1984) for which there is a growing market in the health sector.

2.6.3 Purification of lecithin

Lecithin is an important emulsifying agent in the food and pharmaceutical industries. Crude lecithin as produced in the degumming of seed oils is a darkly coloured, highly viscous substance with a 'grease-like' consistency. After extraction with acetone, which removes associated oil and other components, the purified lecithin is obtained as a pale yellow powder with a high phospholipid content (95%). The lecithin residue contains significant levels of residual solvent which are removed at high temperatures at which further degradation of the product can occur.

The large differential in solubility between triglyceride oils and phospholipid suggests the feasibility of using NCF CO_2 to purify lecithin. The main problem here is the highly viscous state of lecithin, which presents enormous difficulties in its transfer into extraction vessels, and achieving good solvent contact therein.

Stahl and Stadler (1984) have proposed a process for the continuous extraction of lecithin which involves high-pressure extrusion of the crude lecithin into an extraction vessel through a small nozzle. Another solution to this problem, incorporating some ingenious features, has been reported by Peter *et al.* (1989). The possibility of using NCFs to reduce the viscosity of liquids has been previously mentioned (Section 2.2.1). However, because CO_2 is not very soluble in lecithin it is not possible to use SCF CO_2 to reduce its viscosity. Instead, NCL propane was employed by Peter to thin the lecithin which was then pumped into an extraction vessel containing SCF CO_2 at 55°C, 80 bar, where most of the lecithin was selectively precipitated. By employing temperatures at which the lecithin can be removed as a liquid the plant could be made to operate continuously and at low pressures. The paper presents a model example of an SCF process development study including phase equilibrium measurements, plant design and cost analysis.

2.6.4 Lowering cholesterol levels in foods

Although the correlation between dietary intake and levels of cholesterol in the blood is not universally accepted, initial suspicions have led to strong consumer aversion to high-cholesterol foods. This has been compounded by advertising campaigns claiming low levels of cholesterol, even in products that would not be expected to contain it.

On solubility grounds there would appear to be a good chance of selectively extracting cholesterol from oils and fats using NCF CO_2. Chrastll (1982) has measured and collated the solubilities of a variety of food components in NCF CO_2 and correlated his data using a simple 'mass action' model. Even from the limited pressure range used in this study it is clear that the solubility of cholesterol is significantly greater than that of triglyceride oils.

Krukonis (1988) has tested the feasibility of removing cholesterol from butter, egg yolk and beef tallow by measuring the partition coefficients of the individual components in SCF CO_2. Selectivity at 60°C and 150 bar was found to be greatest for egg yolk (12.2) and least for butter (3.4). This trend is in accord with the relative solubilities of the component oils in SCF CO_2. In these trials 90% removal of the cholesterol from butter was reported with an overall yield of 70% of cholesterol-reduced product. Distribution coefficients of cholesterol between milk fat and SCF CO_2 reported by Bradley (1989) support the viability of selective separation. Studies on the effect of extraction conditions upon the composition of SCF CO_2 extracted egg yolk powder (Froning *et al.*, 1990) are in broad agreement with available solubility data.

2.6.5 Fractionation of high-value oils and fats

Butterfat

Milk fat contains fatty acids representing a wide range of molecular weight and unsaturation. The acid residues are distributed in a 'random' fashion in the constituent triglyceride oils. Conventional fractionation techniques (e.g. crystallisation – see Chapter 8) are unable to effectively concentrate a short chain acid fraction (C_4–C_{10}) since these acids are combined in the triglyceride structure with higher molecular weight acids (C_{14}, C_{16}, C_{18}) which have a controlling influence on the separation. The fractionation of butterfat is of considerable interest in the dairy industry for manipulation of physical properties (e.g. spreadability, mouthfeel) and functionality in other milk products and foodstuffs. Short-chain fatty acids are also of physiological importance since they are more rapidly assimilated in the body.

Kaufmann *et al.* (1982) have examined the 'time-controlled' fractionation of butterfat in an SCF extraction process at 80°C, 200 bar and the two fractions taken showed clearly different compositions of component oils. The first fraction was found to be 81% enriched in the short-chain (C_4–C_{10}) acids with a simultaneous 44% reduction in the oleic acid content. Accordingly the second fraction showed a higher proportion of longer-chain acids and contained 99% of the C_{24}–C_{44} triglycerides.

More recently Kanare *et al.* (1989) have reported the composition of extracts taken using a 'pressure-controlled' SCF extraction process in which the pressure was progressively increased from 100 to 400 bar at 50°C. Analysis revealed a similar pattern to

Kaufmann with the short-chain acids being selectively extracted in the early stages of the extraction. Both cholesterol and lactones were also preferentially removed in the initial extracts.

Fish oils

Certain fish oils, which are rich in the highly unsaturated eicosapentaenoic (EPA, C20:5) and docosahexaenoic (DHA, C22:6) acids, are now believed to be of great dietary value. There is therefore considerable interest in the isolation of fractions of fish oils enriched in these component acids. To effect a realistic separation by any technique the acids in the oil must firstly be removed from the triglyceride structure by hydrolysis and esterified, usually as the ethyl ester. The conventional fractionation process then involves distillation techniques which are limited in terms of both thermal degradation of the oils and the separation factors obtainable.

One of the first reports of the application of SCF extraction in the fractionation of fish oils was by Eisenbach (1984) who demonstrated enrichment of the component acids of cod liver oil by carbon number. Further separation of acids differing only by number and position of double bonds could not be achieved. However, the technique was successful in producing a C_{20} fraction containing about five times the concentration of EPA of the starting material.

Nilsson *et al.* (1988) solved the problem of limited separation of EPA in a combined SCF extraction/urea crystallisation process in which less saturated acids (e.g. C20:0, C20:1) can be selectively removed. In this combined process, fractions containing 96% pure EPA were obtained from menhaden (large herrings) oil. In the course of this work distribution coefficients of more than twenty individual acid components were determined in SCF CO_2. The feasibility of this fractionation process has recently been established in a series of pilot scale tests by Krukonis (1988).

2.6.6 Extraction of flavours and fragrances

Flavours and fragrances are conventionally isolated from botanical sources either as an absolute, using solvent extraction, or by steam distillation. The main drawbacks to these methods are thermal degradation (e.g. of sesquiterpenes), loss of volatile 'top notes' and indiscriminate separation of high molecular weight components. It was realised at an early stage in the development of NCF extraction that these problems could be largely overcome by using CO_2.

The main advantages in using NCF CO_2 for the isolation of flavours and fragrances are:

(1) The extraction and separation can be carried out at low temperature in an inert environment, thereby avoiding thermal damage and chemical degradation.
(2) The extract has improved solubility in formulations since less terpenes are extracted.
(3) The high vapour pressure of CO_2 enables it to be removed without losses in the highly volatile 'top notes'.
(4) Undesirable components such as proteins, waxes, sugars, chlorophyll are not extracted.

Most of the early work on the use of liquid CO_2 for flavour extraction was carried out in the Soviet Union in the 1960s and reported in Russian. Since then the variety of flavours and fragrances examined has grown enormously and now represents probably the largest class of researched applications for CO_2 extraction in the food industry. Of all examples, that of hop oil extraction deserves special mention since it is one of the few applications regularly operating on a large commercial scale. In hop oils it is the soft resins consisting of alpha acids (humulones) and beta acids (lupulones) that are the important components for flavour production. In the brewing process alpha acids are isomerised to give beer its characteristic taste. Conventional hop extraction processes use either hexane or methylene chloride and, although residual levels of these solvents are within allowed limits, CO_2 offers many of the above advantages ((1)–(4)) and, being non-toxic, has a far higher level of acceptability.

A process for hop extraction using SCF CO_2 has been patented by Vitzthum *et al.* (1972) and a complementary process using NCL CO_2 was later developed by Laws *et al.* (1980). The latter process uses NCL CO_2 at 7–10°C, where it is claimed that a solubility maximum exists. The hop extract is rich in the desirable acids and essential oils and contains no 'hard' resins. The process was implemented on a commercial scale at Reigate (UK). Similar plants are operating in West Germany at supercritical conditions (Münchsmünster). Although the higher temperature of the SCF process facilitates a more rapid extraction, some selectivity is lost and more chlorophyll is also co-extracted.

Hop extraction is a highly seasonal process and can be carried out only over a limited period. Most commercial hop extraction plants are therefore also used for other applications out of the hop season, e.g. in the extraction of flavours, herbs and spices, and are also often offered for hire on a 'contract' basis.

Judging by the number of publications appearing over the last decade there must be few botanical plants providing flavour and fragrance extracts that have not been subjected to CO_2 extraction. Most tests show an improved quality extract resembling more closely the botanical source for which quantitative evidence is often presented in the form of chromatograms. However, specific information regarding efficiency and mass transfer is often lacking. As in the case of hop extraction liquid CO_2 is found to give a higher quality extract, but this is probably offset by a less efficient extraction. A selection of relevant literature is given in Table 2.4. NCF extraction has also been used to extract flavours and aromas from a variety of other sources including fruit juices (Schultz and Randall, 1970), wine (Jolly, 1981), tea (Vitzthum *et al.*, 1975a), cocoa (Vitzthum *et al.*, 1975b) and tobacco (Luganskaya *et al.*, 1967).

Inexperience of high-pressure technology, lack of design data and high capital costs have all contributed to limit the application of SCF extraction in the food industry. However, selected applications making use of the unique properties of SCFs have been, and are being, profitably exploited. It is noteworthy that the products of such processes, although sometimes more expensive, have found a place in the market by virtue of their improved quality. This reflects the current trends of the consumer towards purer and more natural processed foods. It seems likely that these consumer demands, combined with increasing legislative restrictions, will dictate greater implementation of SCF extraction technology in the future.

Table 2.4. Examples of SCF extraction processes

Plant	Reference
Almonds	Calame and Steiner (1982)
Anise	Stahl and Gerard (1982)
Basil	Pekhov *et al.* (1975)
Bergamot*	
Black pepper*	Moyler (1986), Sankar and Manohar (1988), Calame and Steiner (1982), Vitzthum and Hubert (1971)
Camomile*	Stahl and Schilz (1976), Weust *et al.* (1981)
Caraway	Stahl and Hubert (1976), Stahl and Gerard (1982)
Cardamon*	Meerov *et al.* (1971)
Cinnamon	Stahl and Gerard (1982), Tateo and Gerard (1989)
Clove*	Stahl and Chizzini (1982), Moyler (1986), Gopalakrishnan *et al.* (1990)
Coriander*	Volodicheva and Lybarskii (1974)
Cumin	Gangadhara and Mukhopadhyay (1988)
Fennel	Volodicheva and Lybarskii (1974)
Ginger*	Stahl and Schilz (1976), Pellerin (1988), Sankar and Manohar (1988), Moyler (1986), Chen *et al.* (1986)
Hop*	Moyler (1986)
Horseradish	Stahl *et al.* (1983a or b)
Juniper*	
Lemon*	Calame and Steiner (1982), Temelli *et al.* (1988)
Lime*	
Lovage*	
Mace*	
Marjoram*	Bouclier and Koller (1986)
Nutmeg*	Hubert and Vitzthum (1978), Moyler (1986)
Orange*	
Paprika*	Coenen and Hagen (1983)
Parsley*	
Patchouli*	
Peppermint	Shaftan *et al.* (1973)
Pimento*	Moyler (1986)
Red pepper	Hubert and Vitzthum (1978)
Rosemary*	Pellerin (1988), Tateo and Fellin (1988)
Sage*	Stahl and Shilz (1976), Pellerin (1988)
Sandalwood	Sankar and Manohar (1988)
Thyme*	Bestmann *et al.* (1985)
Vanilla*	Shuetz (1981), Vitzthum and Hubert (1971)

* Commercially available extracts.

Symbols used in text

a = constant in attractive term of EOS
b = constant in repulsive term of EOS
C = concentration
D = diffusion coefficient
D^{eff} = effective diffusion coefficient
f = fugacity
k = binary interaction parameter
K = equilibrium constant
L = half thickness of slab
M_t = mass of solute extracted at time t
M_∞ = mass of solute extracted at infinite time
P_c = critical pressure
R = gas constant
S = solubility
T_c = critical temperature
V = molar volume
x = mole fraction (in liquid or solid phase)
y = mole fraction (in gas or NCF phase)
η = viscosity
τ = tortuosity factor
ϕ = fugacity coefficient
φ = volume fraction of pores

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Chapter 3

Pressure-activated membrane processes

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3.1 INTRODUCTION

Over the last 30 years, a number of membrane processes have evolved, which make use of a pressure driving force and a semi-permeable membrane in order to effect a separation of components in a solution or colloidal dispersion. The separation is based mainly on molecular size, but to a lesser extent on shape and charge. The three main processes are reverse osmosis (hyperfiltration), ultrafiltration and microfiltration. The dimensions of the components involved in these separations are given in Fig. 3.1, and are typically in the range of less than 1 nm to over 1000 nm. A brief summary of the main differences between them, in terms of the components which are rejected by the membranes, is also illustrated. More recently the process term 'nanofiltration' has been introduced, which is somewhere between reverse osmosis (RO) and ultrafiltration, bringing about a separation of low molecular weight components such as monovalent ions and salts from organic

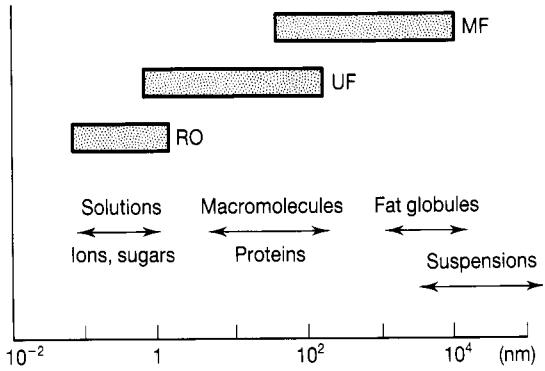


Fig. 3.1. Size ranges for different membrane processes.

compounds such as sugars. These pressure-activated processes can also be regarded as a continuous spectrum of processes, with no obvious distinct boundaries between them. However, it should be noted that the sizes of the components being separated range over several orders of magnitude, so it is highly likely that the separation mechanisms and hence the operating strategies may change as we move through the spectrum.

3.2 TERMINOLOGY

The feed material is applied to one side of a membrane. The feed is usually a low-viscosity fluid, which may sometimes contain suspended matter and which is subjected to a pressure. In most cases the feed flows in a direction parallel to the membrane surface and the term cross-flow filtration is used to describe such applications. Dead-end systems are used, but mainly for laboratory scale separations. The stream which passes through the membrane under the influence of this pressure is termed the permeate (filtrate). After removal of the required amount of permeate, the remaining material is termed the concentrate or retentate. The extent of the concentration is characterised by the concentration factor (f), which is the ratio of the feed volume to the final concentrate volume (see equation (3.5)).

The process can be illustrated simply in Fig. 3.2(a). From a single membrane processing stage, two fractions are produced, named the concentrate and permeate. The required extent of concentration may not be achieved in one stage, so the concentrate may be returned to the same module for further concentration or taken to other modules in a cascade, or multistage process. The permeate may also be further treated in a separate process.

In terms of size considerations alone, one extreme is a membrane with very small pore diameters (tight pores). In this case the permeate will be pure water because even small molecular weight solutes will be rejected by the membrane; high-pressure driving forces are required to overcome frictional resistance and osmotic pressure gradients. If the permeate is predominantly water, then the process is known as reverse osmosis or hyperfiltration; it is similar in its effects to evaporation or freeze-concentration. A concentrate will be produced, in which there is virtually no alteration in the proportion of the

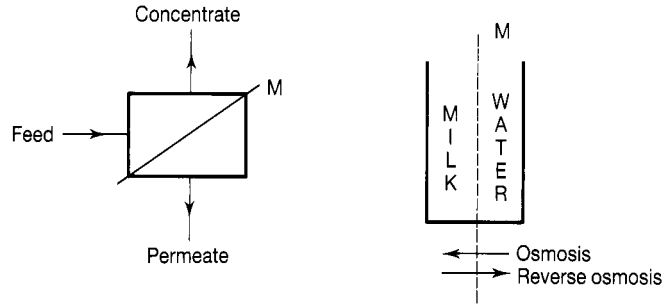


Fig. 3.2. Separation of feed into a concentrate and permeate stream.

solid constituents. In some applications it is the permeate which is the required material; for example the production of 'drinking water' from sea-water or 'pure water' from brackish water. The best processes are those where both the concentrate and the permeate are fully utilised.

There have been several comparisons made between evaporation and reverse osmosis, in terms of capital costs, energy costs and product quality (Renner, 1991). In general terms RO is less energy intensive and can improve product quality. Some limitations are the high capital costs, membrane replacement costs and extent of concentration, which is not as high as that obtainable by evaporation.

If a fluid, for example milk, is separated from water by a semi-permeable membrane (see Fig. 3.2(b)), there will be a flow of water from the water to the milk, in order to equalise the chemical potential of the two fluids; this is termed osmosis. This flow of water can be stopped by applying a pressure to the milk. This pressure that stops the flow is termed the osmotic pressure. If a pressure greater than the osmotic pressure is applied, the water will flow from the milk to the water, thereby reversing the natural process of osmosis and achieving a concentration of the milk. Therefore in reverse osmosis, the pressure applied needs to be in excess of the osmotic pressure. Osmotic pressure (π) is a colligative property, the pressure being dependent upon the number of particles and their molecular weight. In classical terms it is determined from the Gibb's free energy equation:

$$\pi = \frac{RT}{V_m} \ln \gamma X \tag{3.1}$$

where R = gas constant, T = absolute temperature, γ= activity coefficient, X = mole fraction, and V_m = partial molar volume.

For dilute solutions of non-ionisable materials, the Van't Hoff equation can be used

$$\pi = RT(c/M) \tag{3.2}$$

where c = concentration (kg m^{-3}) and M = molecular weight.

For ionisable salts this becomes

$$\pi = iRT(c/M) \tag{3.3}$$

where i = the degree of ionisation, e.g. for NaCl, $i = 2$; for $FeCl_2$, $i = 3$.

This equation predicts a linear increase in osmotic pressure with concentration. However, this relationship breaks down, even at relatively low concentrations, with the relationship between osmotic pressure and concentration becoming non-linear. For example, the osmotic pressure of a 25% serum albumin solution was 300 τ, which is about six times higher than predicted from the Van't Hoff equation. It is also affected by pH.

This non-linear relationship can be represented by Virial type equations:

$$\pi = Ac + Bc^2 + Dc^3 \tag{3.4}$$

where c = concentration and A, B and D are constants. The constants are presented for dextran and whey by Cheryan (1986).

Osmotic pressures are highest for low molecular weight solutes, so the highest osmotic pressures arise for salt and sugar solutions. Concentration of such solutions results in a large increase in their osmotic pressure. On the other hand, proteins and other macromolecules do not produce high osmotic pressures. There will only be small increases during their concentration as well as small differences in osmotic pressure between the feed and permeate in ultrafiltration. Values for osmotic pressures are not easy to find in the literature and a selection of values is given in Table 3.1. A further complication with foods and other biological systems is their complexity, with not just one but many components. *In reverse osmosis the applied pressure must exceed the osmotic pressure, and the driving-force term in reverse osmosis is normally the difference between the applied pressure and the osmotic pressure.* It could be that osmotic pressure is one of the factors that limits the extent of concentration. One suggested experimental method for measuring osmotic pressure is to determine the pressure that would give zero flux, by extrapolation. In ultrafiltration and microfiltration, there is little osmotic pressure difference over the membrane as the low molecular weight components are almost freely permeating (see equation (3.8)).

Table 3.1. Osmotic pressures of some solutions

Solution		Osmotic pressure (bar)
Sugar beet	20° Brix	34.1
Tomato paste	33° Brix	69.0
Apple juice	15° Brix	20.4
Citrus juice	10° Brix	14.8
	34° Brix	69.0
Sucrose	44° Brix	69.0
Coffee extract	28% TS	34.0
Sea-water	3.5% salt	23.2
	15.0% salt	138.0
Milk		6.9
Whey		6.9
Lactose	1% w/v	3.7

Compiled from data in Cheryan (1986) and Lewis (1982).
Some equations for osmotic pressure are given by Cheryan (1986).

As the membrane pore size increases, the membrane becomes permeable to low molecular weight solutes in the feed; even the transport mechanisms are likely to change. Lower pressure driving forces are required as osmotic pressure differences between the feed and permeate are reduced. However, molecules of a larger molecular weight are still rejected by the membrane. Therefore some separation of the solids present in the feed takes place; the permeate contains low molecular weight components at approximately the same concentration as they are in the feed, and the concentrate contains large

molecular weight components at an increased concentration, compared to the feed. Note that some of the low molecular weight components will be retained in the concentrate. It is this fractionation and concentration process that makes the ultrafiltration process more interesting than reverse osmosis, although, as mentioned earlier, there is no sharp demarcation between the processes. More porous membranes still allow not only sugars and salts, but also macromolecules, to pass through, but retain particular matter and fat globules, i.e. greater than 100 nm (see Fig. 3.1); this is termed microfiltration. Because of their increased potential for separating components in mixed feeds, ultrafiltration and microfiltration are covered in more detail in Chapters 4 and 5. However, much of the discussion, particularly that on membrane performance and rejection, will also be pertinent to all three pressure-activated processes. Major points of difference are discussed later in this chapter.

3.3 CONCENTRATION FACTOR AND REJECTION

Two important processing parameters for all pressure activated processes are the concentration factor (f) and the membrane rejection characteristics. The concentration factor is defined as follows:

$$\textit{Concentration factor} \quad (f) = V_F / V_c \tag{3.5}$$

where V_F = feed volume and V_c = final concentrate volume.

The term volume reduction factor (VRF) is sometimes used:

$$\text{VRF} = 100\,(V_F - V_c)/V_F = 100(1 - 1/f) \tag{3.6}$$

Thus a process with a concentration factor of 10 would have a volume reduction factor of 90%.

The permeate volume (V_p) equals the feed volume minus the concentrate volume (assuming no losses)

$$V_p = V_f - V_c \tag{3.7}$$

As soon as the concentration factor exceeds 1, the volume of permeate will exceed that of the concentrate. Concentration factors may range from as low as 1.5 for some viscous materials, to up to 50 for dilute protein solutions, e.g. chhana whey (Jindal and Grandison, 1992). Generally higher concentration factors are used for ultrafiltration than for reverse osmosis, e.g. up to 25–30 for UF of cheese-whey, compared to 5 for RO of cheese-whey.

A mass balance for the process can be applied and is useful for estimating the distribution of components between the permeate and concentrate, or for estimating the losses that are incurred in practical situations.

The *rejection* or *retention factor* (R) of any component is defined as

$$R = c_F - c_p / c_F \tag{3.8}$$

where c_F is the concentration of component in the feed and c_p is the concentration in the permeate.

It can be determined experimentally for each and every component in the feed, by sampling the feed and permeate at the same time and analysing the component in question. It is a very important property of a membrane, as it will influence the extent (quality) of the separation that can be achieved.

Rejection values normally range between 0 and 1; sometimes they are expressed as percentages (0–100%).

when $c_p = 0$; $R = 1$; all the component is retained in the feed

when $c_p = c_F$ $R = 0$; the component is freely permeating.

An ideal RO membrane would give a rejection value for all components of 1, whilst an ideal UF membrane, being used to concentrate a high molecular weight component or remove a low molecular weight component would give respective rejection values of 1 and 0. If the concentration factor and rejection value are known, the yield of any component, which is defined as the fraction of that component present in the feed, which is recovered in the concentrate, can be estimated. Obviously for reverse osmosis, the yield for an ideal membrane is 1.0. Rejection data for membranes and their effects on yield and separation performance will be discussed in greater detail in Chapter 4.

3.4 MEMBRANE CHARACTERISTICS

The membrane itself is crucial to the process. The first commercial membranes were made of cellulose acetate and these are termed first-generation membranes. For food-processing applications, they had some limitations, with temperatures below 30°C and a pH range of 3–6. These were followed in the mid-1970s by other polymeric membranes (second-generation membranes), with polyamides and, in particular, polysulphones being widely used for foods. The resulting improvements in cleaning and hygiene are covered in Section 3.8. It is estimated that over 150 organic polymers have now been investigated for membrane applications. Inorganic membranes based on sintered and ceramic materials are also now available. The physical structure of these membranes is complex, and as most of them are used for microfiltration, their structure is described in more detail in Chapter 5.

The main terms used to describe membranes are microporous or asymmetric. Microporous membranes have a uniform porous structure throughout, although the pore size may not be uniform across the thickness of the membrane. They are usually characterised by a nominal pore size and no particle larger than this will pass through the membrane. In contrast to this, most membranes used for ultrafiltration are of asymmetric type, having a dense active layer or skin of 0.5–1 μm in thickness, and a further support layer which is much more porous and of greater thickness (Fig. 3.3). Overall the porosity of these membranes is high, although the surface porosity may be low, with quoted values in the range 0.3–15% (Fane and Fell, 1987). Often the porous path may be quite tortuous, the distance covered by the solvent or solute being much greater than the thickness of the membrane; the term tortuosity has been used as a measure of this property. The pores are not of a uniform size, as can be seen when viewed under the electron microscope, and are best characterised by a pore size distribution. This

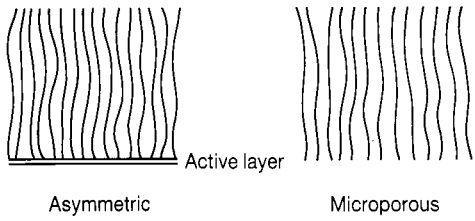


Fig. 3.3. Closer examination of the membrane structure.

distribution can be measured by electron microscope techniques, or by combined bubble size and solvent permeability methods (Munari *et al.*, 1985). It is claimed that this method is capable of measuring the pore-size distribution in the thin skin. Another technique mentioned by Fane and Fell is the capillary condensation/permeability method. Pore sizes may range from 1 to 100 nm. This distribution of pore size is one of the main factors preventing a sharp separation of components of almost similar size, e.g. mono- and di-saccharides (see Rejection). An indirect measurement of pore size can be made by measuring the permeability of solutes, such as dextrans with a range of molecular weights.

It can be seen that in physical terms alone, there are a number of membrane structures available. The membrane also has a chemical nature, and many materials have been evaluated. It may be hydrophilic or hydrophobic in nature. Fane and Fell (1987) stated that the hydrophobic nature can be characterised by measuring its contact angle (θ). The higher the contact angle the more hydrophobic is the surface. Polysulphones are generally much more hydrophobic than cellulosic membranes. There was shown to be a good correlation between flux decline and hydrophobicity, with the least hydrophobic membrane showing the least flux loss over a period of 150 min. The surface may also be charged. All these factors will give rise to interactions between the membrane and the components in the feed and influence the components passing through the membrane, as well as the fouling of the membrane.

The physical chemistry of membranes has been described in more detail by Cheryan (1986), Gutman (1987) and Tsujita (1992).

3.5 PERMEATE RATE

Two other important processing parameters are the flux or permeate rate and the power consumption.

The flux is usually expressed in terms of volume per unit time per unit area ($l\ m^{-2}\ h^{-1}$). Expressed in this way it permits a ready comparison of different membrane configurations of different surface areas. It can also be expressed as a permeate velocity. If energy is to be taken into account, it may be relevant to measure and maximise the flux to energy consumption ratio. Flux values may range from higher than $500\ l\ m^{-2}\ h^{-1}$ to less than $5\ l\ m^{-2}\ h^{-1}$.

(*Note*: Imperial units are still sometimes used, where 1 gal $ft^{-2}\ d^{-1} = 2.036\ l\ m^{-2}\ h^{-1}$.)

Factors affecting the flux rate are the applied pressure: the flow rate and viscosity, both of which affect turbulence, and the processing temperature. Increasing the

temperature and inducing more turbulence increases the flux. However, the flux is only affected by the applied pressure in the pressure-dependent region. These factors are discussed in more detail in Chapter 4.

The power utilisation (W) is related to the pressure (head) developed and the mass flow rate as follows:

$$W = m'hg \tag{3.9}$$

where m' = mass flow rate (kg s^{-1}), h = head developed (m) and g = acceleration due to gravity (9.81 m s^{-2}).

This energy is largely dissipated within the fluid as heat and will result in a temperature rise. Cooling may be necessary if a constant processing temperature is required.

The membrane offers a resistance to the transfer of both solvent (normally water) and solute. The permeate flux is a measure of the flow rate of solvent through the membrane, whereas the rejection describes the amount of solute which passes through (see eq. (3.8)). From a process engineering standpoint, it is highly desirable to be able to predict the flux and rejection from the physical properties of the solution, the membrane characteristics and the hydrodynamics of the flow situation, in order to optimise the performance of the system. Membrane operations have been subject to a number of modelling processes, in order to achieve these objectives. However, before these models are discussed in more detail, it is important to consider the phenomena of concentration polarisation and fouling.

3.6 TRANSPORT PHENOMENA AND CONCENTRATION POLARISATION

A very important consideration for pressure-driven membrane processes is that the separation takes place not in the bulk of solution, but in a very small region close to the membrane, known as the boundary layer, as well as over the membrane itself. This gives rise to the phenomenon of concentration polarisation over the boundary layer. (Note that in streamline flow the whole of the fluid will behave as a boundary layer.) It is manifested by a quick and significant reduction (2–10 fold) in flux when water is replaced by the feed solution, for example in a dynamic start.

Concentration polarisation occurs whenever a component is rejected by the membrane. As a result, there is an increase in the concentration of that component at the membrane surface, together with a concentration gradient over the boundary layer. Eventually a dynamic equilibrium is established, where the convective flow of the component to the membrane surface equals the flow of material away from the surface, either in the permeate or back into the bulk of the solution by diffusion, due to the concentration gradient established. This increase in concentration, especially of large molecular weight components, offers a very significant additional resistance. It may also give rise to the formation of a gelled or fouling layer on the surface of the membrane (see Fig. 3.4). Whether this occurs will depend upon the initial concentration of the component and the physical properties of the solution; it could be very important as it may affect the subsequent separation performance. Concentration polarisation itself is a reversible phenomenon; thus if the solution is then replaced by water, the original water flux should be restored. However, this rarely occurs in practice due to the occurrence of fouling,

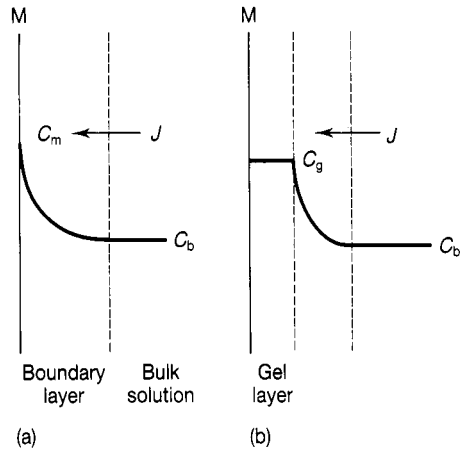


Fig. 3.4. (a) Concentration polarisation in the boundary layer; (b) concentration polarisation with a gel layer.

which is detected by a decline of flux rate at constant composition. Fouling is caused by the deposition of material on the surface of the membrane or within the pores of the membrane. Fouling is irreversible and the flux needs to be restored by cleaning. Therefore, during a concentration process, flux declines due to a combination of these two phenomena.

A number of mechanisms have been proposed to explain the transport of solvent and solute in reverse osmosis. These have been reviewed by Cheryan (1989), and Cheryan and Nichols (1992).

The simplest mechanism to visualise conceptually is a simple sieving mechanism, based on size. However, for reverse osmosis, this does not explain the high rejection of salt and the permeation of water, as the molecules are about the same size. Other physico-chemical factors concerned with the structure of the membrane and interaction of solvent and solutes with the membrane influence the performance. It is believed that for reverse osmosis, the phenomena are more complex than those occurring with ultrafiltration, which is generally regarded as a sieving process. Therefore the models proposed for reverse osmosis and ultrafiltration are different in nature.

One of the most common models used for reverse osmosis is the solute diffusion model, in which the solvent flux (J_w) is influenced in the main by the pressure driving force and the solute flux (J_s) by diffusion. All the resistance to mass transfer takes place in the active layer, which is between 0.5 and 1.0 μm in thickness. In this case the pressure driving force is the difference between the applied pressure and the osmotic pressure. The transport of solvent and solute are not connected. It is assumed that the solute dissolves in the skin instantaneously and then passes through the pores of membrane by diffusion, whereas the flow of solvent (which also dissolves) is influenced by the pressure differential. The relationship between concentration dissolved in the membrane at its surface and that in the solution depends upon the partition coefficient for that component.

Thus the diffusivity and membrane thickness and its partition coefficient all influence the transport of solute. (Convective flow of solute is ignored at high solute rejections.)

According to this model, increasing the pressure will have a preferential effect on solvent flow, thereby reducing solute concentration in the permeate and increasing rejection. It also predicts that increasing the solute concentration in the feed preferentially increases solute transport and increases the concentration in the permeate, thereby decreasing rejection. Increasing the temperature increases both solvent flux and solute flux by about the same amounts, thereby leaving the rejection unchanged. A 1°C change in temperature changes the flux (solute and solvent) by approximately 3%.

Although many of these trends are observed with simple solutions, the theory does not account for all the observed facts for multi-component solutions, whereby the presence of one component increases the permeability of other components. Glover (1985) pointed out that this theory does not describe the behaviour of complex systems precisely, but it gives a background of understanding. In multicomponent systems, differences in the partition coefficients could explain the difference in permeabilities between different components.

A second proposed model is the preferential adsorption, capillary flow model, which predicts that the component concentrated preferentially in the permeate will be that component which is adsorbed most strongly on the membrane surface. The component that is preferentially adsorbed onto the membrane surface provides a thin layer of that component adjacent to the surface. This thin layer then moves through the pores of the membrane by capillary flow, under the influence of a pressure gradient, and in this way is thus preferentially transported through the membrane. For membranes of a hydrophilic nature, the component preferentially absorbed and transported is water. Thus it is postulated that there is a thin film of water adjacent to the membrane surface. This theory also explains the low rejections and sometimes negative rejections for highly polar organic solutes, found with cellulose acetate membranes, due to their preferential adsorption on the surface. Other situations, where rejection of organic solutes decreases with increasing pressure, are explained by their adsorption onto the more hydrophobic regions of the membrane.

A third model is based upon the wetted surface mechanism, whereby water adsorbs onto the surface of the membrane by hydrogen bonding. It is postulated that these clusters of water prevent solute entering the pores and that the water passes through the membrane from one adsorbed site to the next. The energy requirements for water migration are much less than salt migration, thereby promoting separation of the salt and water. All these models are based on knowing the transport mechanisms involved. The physical chemistry of a wide variety of membrane materials, including permeability data, diffusion data and sorption data which are required for the models described earlier, have been reviewed by Tsujita (1992). Also reviewed are transport properties related to non-equilibrium thermodynamics with uncharged and charged membranes.

Other models are based upon irreversible thermodynamics, where the driving force for transport of solvent and solute is expressed in terms of differences in their chemical potential over the membrane. However, the fluxes for solvent and solute are coupled and the flux for each component is influenced by the chemical potential difference for both components. With such models, the exact mechanisms are not known, but the

relationship between the fluxes and driving forces are described by phenomenological coefficients, which are nevertheless connected with the mechanism. Such phenomenological models provide an empirical description of the transport of molecules through membranes, but they do not give an explanation for the separation mechanism. Gekas (1992) discusses these types of model in more detail for membrane processes.

Cheryan (1989) has concluded that none of the proposed models can satisfactorily explain the rejection characteristics of all solvent–solute–membrane reverse osmosis systems.

On the other hand, *ultrafiltration* is regarded as a sieving process and is often modelled as a bundle of capillaries, with flow through the capillaries being described by the Hagen–Poiseuille equation:

$$Q = d^4 \Delta P \ \pi/128\mu L \tag{3.10}$$

where Q = volumetric flow rate, ΔP = pressure drop; μ = dynamic viscosity, L = capillary length and d = capillary diameter. (Note that this equation is often expressed as flux rate/unit area ($d^2 \Delta P/32\mu L$).)

Additional terms may be introduced to account for porosity and tortuosity effects in the membrane.

These equations predict that the flow will be strongly biased toward the largest pores. Such pores may plug quicker because of the higher degree of polarisation encountered due to the higher flux. (See concentration polarisation.) Also, as pore size increases, the flow rate through the membrane will also increase.

The flux for most food solutions is lower than that for water and other pure solvents for a number of reasons. The viscosity of the feed solution may be higher and this further increases throughout the concentration process. Also concentration polarisation and fouling occur. Factors which influence concentration polarisation and fouling are the prevailing flow conditions and the composition of the material being processed. Some of these factors are discussed in more detail in Chapter 4.

3.7 MEMBRANE EQUIPMENT

Membrane suppliers now provide a range of membranes, each with different rejection characteristics; for reverse osmosis salt rejection values are quoted, e.g. from 99% rejection of sodium chloride down to 60% rejection of sucrose; and for ultrafiltration, different molecular weight cut-offs in the range 1000 to 500 000. For example, it is implied that a membrane with a molecular weight cut-off of 20 000 will reject components with a molecular weight greater than that value. Such figures are provided for guidelines only (see Section 4.2.1). Tight ultrafiltration membranes have a molecular weight cut-off value of around 1000–5000, whereas the more 'open' or 'loose' membranes have a value in excess of 100 000. Thus molecular weight cut-off value is related to porosity and rejection characteristics; as membranes become more permeable to solutes, their molecular weight cut-off values increase. However, because there are many other factors that affect rejection, molecular weight cut-off should only be regarded as giving a relative guide to its pore size and true rejection behaviour. Experimental determinations should always be made on the system to be validated, at the operating conditions to be used.

Other desirable features for membranes to ensure commercial success with food components are listed as

- Reproducible pore size from batch to batch, offering uniformity in terms of both their permeate rate and their rejection characteristics.
- High flux rates and sharp rejection characteristics.
- Compatibility with processing, cleaning and sanitising fluids.
- Resistance to fouling.
- An ability to withstand temperatures required for disinfecting and sterilising surfaces, which is an important part of the safety and hygiene considerations. Extra demands placed upon membranes used for food processing include: the ability to withstand hot acid and alkali detergents (low and high pH), temperatures of 90°C for disinfecting or 120°C for sterilising and/or widely used chemical disinfectants, such as sodium hypochlorite, hydrogen peroxide or sodium metabisulphite. The membrane should be designed to allow cleaning both on the feed/concentrate side and the permeate side. See Section 3.8.

Membrane processing operations can range in their scale of operation, from laboratory bench-top units, with samples less than 10 ml, through to large commercial scale operations, processing more than 50 $m^3 h^{-1}$. Furthermore, the process can be performed at ambient temperatures, which allows concentration without any thermal damage to the feed components.

The process can be batch or continuous; the fluid can be static or in motion, either agitated in a stirred cell or moving across the surface of the membrane. The membrane itself can be configured in a variety of forms.

3.7.1 Membrane configuration

The membranes themselves are thin and in most cases require support against the high pressure. The support material itself should be porous. The membrane and its support, together, are normally known as the module.

There are a number of criteria that have to be satisfied in the design of a pressure-activated membrane module. It must provide a large surface area in a compact volume; it must be able to support the membrane and the configuration must allow suitable conditions to be established, with respect to turbulence, high wall shear stresses, pressure losses, volumetric flow rates and energy requirements, thereby minimising concentration polarisation. Hygienic considerations are important; there should be no dead spaces and the module should be capable of being in-place-cleaned on both the concentrate and the permeate side. The membranes should be readily accessible, for both cleaning and replacement. It may also be an advantage to be able to collect permeate from individual membranes in the module to be able to assess the performance of each one individually.

The two major configurations which have withstood the test of time are the *tubular* and the *flat-plate* configurations. The main features of these configurations will be further discussed.

Tubular membranes come in a range of diameters. In general tubes offer no dead spaces, do not block easily and are easy to clean. However, as the tube diameter increases, they occupy a larger space, have a higher hold-up volume and incur higher

pumping costs. The two major types are the *hollow fibre*, with a fibre diameter of 0.001–1.2 mm and the wider tube with diameters up to 25 mm, although about 12 mm is a popular size. Some of the configurations are shown in Fig. 3.5 and 3.6.

For the hollow fibre system, the membrane wall thickness is about 250 μm and the tubes are self-supporting. The number of fibres in a module can be as little as 50 but sometimes greater than 1000. The fibres are attached at each end to a tube sheet, to ensure that the feed is properly distributed to all the tubes. This may give rise to pore plugging at the tube entry point. Prefiltration is recommended to reduce this. The feed is usually pumped through the tubes but it can also be pumped through the shell-side. In general the length of the fibres and modules is not more than 1 m. The geometry of the system ensures that there is a large surface area to volume ratio and that hold-up volumes are low. Pressure drops required are also not excessive because of the short tube lengths. In ultrafiltration applications, the applied pressures are fairly low (usually less than 2 bar). This permits only low inlet pressures, which limits the flow rate that can be achieved as well as restricting its application to low viscosity fluids. However, they are widely used for desalination and in these RO applications are capable of withstanding high pressures. For these applications the outer diameter should be 2–4 times higher than the inner diameter. It is the ratio of the external to internal diameter, rather than the membrane wall thickness, which determines the pressure that can be tolerated (Nishimura and Koyama, 1992).

Hollow fibre systems usually operate in the streamline flow regime. However, the wall shear rates are high. They may not perform well with viscous feeds and those containing particulate matter.

One big advantage is their backflushing capability, which helps improve their cleanability. This is due to the self-supporting nature of the membrane.

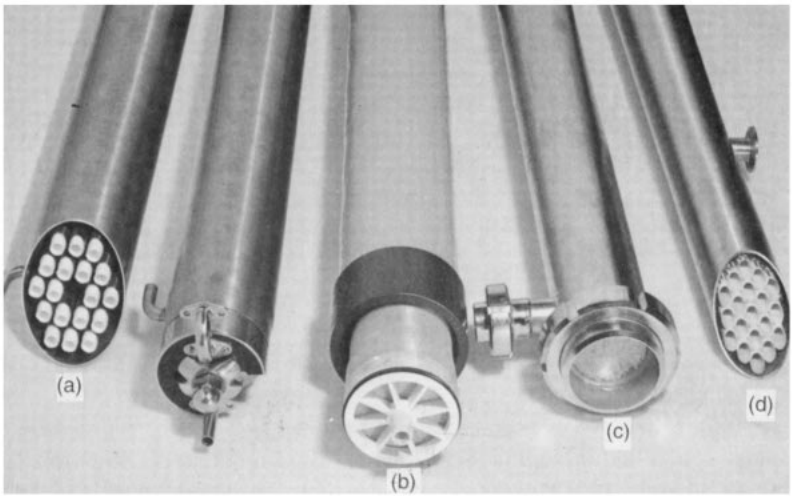


Fig. 3.5. Plate showing different configurations: (a) and (d) tubular; (b) spiral wound; (c) hollow fibre (with courtesy of PCI).

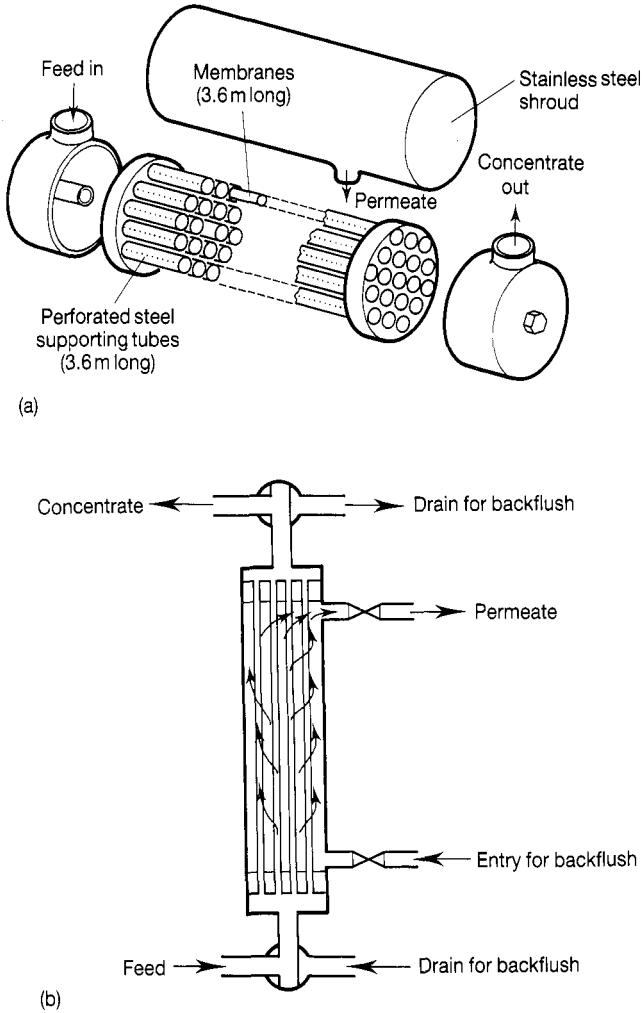


Fig. 3.6. Tubular and hollow fibre configurations: (a) tubular, 12.5 mm diameter, 3.6 m long. Module has 18 tubes, total area 2.5 m^2 (with courtesy of PCI); (b) hollow fibre, length 1.1 m, diameter 7.6 cm, fibres 0.5 mm or 1.0 mm; total area 1.4 or 2.5 m^2. Each module has 1000 or more fibres (with courtesy of Romicon).

Hollow fibre systems tend to be expensive, because if one or several fibres burst, the whole cartridge needs to be replaced.

For *wider tubes*, the feed is normally pumped through the tube, which may be up to 25 mm diameter, although a popular size is about 12 mm diameter. There may be up to 20 tubes in one module; tube lengths may be between 1.2 m and 3.6 m (4 and 12 ft) and tubes within the module may be connected in series or parallel. The membrane is cast or inserted into a porous tube which provides support against the applied pressure. Cheryan

(1986) describes different systems in more detail. Therefore they are capable of handling more viscous fluids and even materials with small suspended particles, up to one-tenth the tube diameter. They normally operate under turbulent flow conditions with flow velocities greater than 2 m s^{-1}. The corresponding flux rates are high, but pumping costs are also high, in order to generate the high volumetric flow rate required at the operating pressure. They are less susceptible to fouling and relatively easy to clean and disinfect by clean-in-place (CIP) methods.

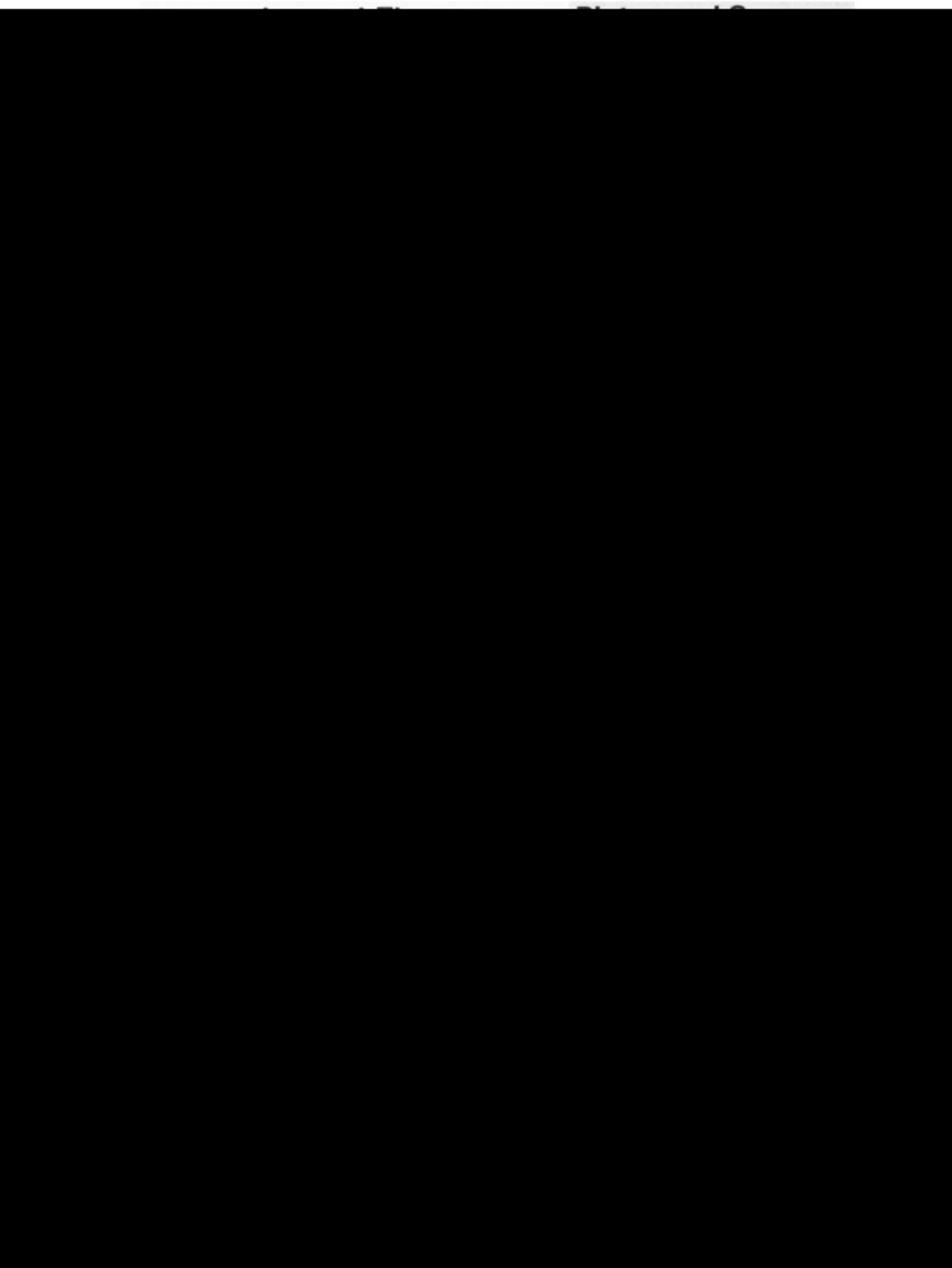
The *flat plate* module can take the form of a plate and frame-type geometry or a spirally wound geometry.

The plate-and-frame system employs membranes stacked together, with appropriate spacers and collection plates for permeate removal, somewhat analogous to plate heat exchangers. The channel height can be between 0.4 and 2.5 mm. Flow may be either streamline or turbulent and the feed may be directed over the plates in a parallel or series configuration. This design permits a large surface area to be incorporated into a compact unit. These systems have been developed for laboratory, pilot plant and large-scale processes, and their performance depends upon the hydrodynamics of the system.

One commonly used system commercially is the DDS plate-and-frame system. Here the channel height is between 0.4 and 1 mm. One membrane is fixed on both sides of a permeate collection plate to form a sandwich. Between each sandwich is a spacer plate (see Fig. 3.7(a)). This system usually operates under streamline flow conditions and the energy required is somewhere between that of a tubular and spirally wound system. Membranes are easily replaced and it is easy to isolate any damaged membrane sandwich. Considerable attention has been devoted to the design of the plate to improve performance. This has been achieved by ensuring a more uniform distribution of fluid over the plate, by increasing the channel width of the longer channels and reducing the ratio of the longest to the shortest channel length. More details are provided by Glover (1985). Also shown in Fig. 3.7(b) is a Pellicon system, used for filtering between 2 and 200 litres.

The *spiral wound system* is now widely used and costs for membranes are quite low. In this case a sandwich is made from two sheet membranes, which enclose a permeate spacer mesh. This is attached at one end to a permeate removal tube and the other three sides of the sandwich are sealed. Next to this is placed a feed spacer mesh and the two together are rolled round the permeate collection tube in the form of a Swiss roll (see Fig. 3.8). The channel height is dictated by the thickness of the feed spacer. Wider channel heights reduce the surface area to volume ratio, but reduce the pressure drop.

The typical dimensions of one spiral membrane unit would be about 12 cm in diameter and about 1 m in length. Up to three units may be placed in one housing, with appropriate spacers to prevent telescoping, which may occur in the direction of flow and could damage the sandwich. Another practical aspect which has received attention is excessive by-passing of feed between the periphery of the membrane sandwich and the housing. Reducing this completely by use of a seal at the inlet improves the performance, but causes a dead space, which may be difficult to clean. Meshes or seals which allow limited by-pass provide a reasonable compromise. This configuration is becoming very popular and relatively cheap. Again the flow may be streamline or turbulent. Pressure drop/flow rate relationships suggest that flow conditions are usually turbulent.



this is often referred to as the transmembrane pressure. Pressure may also be applied to the permeate side, in order to reduce the initial flux and concentration polarisation during the transition from water to product. The pressure difference $(P_1 - P_2)$ will depend upon the flow rate and viscosity of the fluid.

The processing variables which have been found to be important are as follows:

(1) *Pressure.* An increase in the pressure increases the permeate flux, usually up to a limiting value, above which the permeate rate becomes independent of pressure: for these reasons we talk of a pressure-dependent and pressure-independent region. The pressure-independent region has been linked with the formation of a gel-permeation layer, adjacent to the membrane. Higher pressures are said to cause compaction of the gelled layer near the membrane surface and perhaps the membrane itself.

(2) *Flow rate.* Increasing the flow rate increases turbulence and decreases the boundary layer thickness; this increases the permeate rate; the relationships between permeate rate, pressure and flow rate are shown in Fig. 4.6. An increase in the flow rate leads to an increase in the pressure, at which the onset of pressure-independence begins.

Higher flow rates give rise to higher frictional losses and higher energy requirements:

- for streamline flow conditions: pressure drop $\propto$ (flow rate)$^{1.0}$
- for turbulent flow conditions: pressure drop $\propto$ (flow rate)$^{1.8}$.

(3) *Temperature.* All investigators have found that increasing the temperature increases the permeate flux; the relationship between permeate flow rate and temperature is almost linear over the relatively narrow range of temperatures investigated. An increase of about 2% per °C rise in temperature is quite normal. Suggestions for the increase in permeate rate are: a change of porosity of the membrane, decrease in viscosity of the feed, higher diffusion rates or increased solubility of the diffusing material in the membrane.

In practice, for foodstuffs, chemical and microbiological considerations have a big influence on the operating temperatures used. Temperatures between 5 and 45°C are often avoided, as microorganisms may grow. With non-cellulosic membranes, the highest temperatures which cause no loss of activity or functionality are used; often in the temperature range 50 to 60°C. Higher temperatures (70–90°C) are used for cleaning and disinfecting the equipment.

As well as the operating conditions, the nature and composition of the feed affect the permeate rate; particularly its chemical nature and its viscosity. More detail will be provided in the applications section. There is some interdependence; for example, temperature affects viscosity and all the latter three factors influence the degree of turbulence.

(4) *Turbulence.* The role of turbulence has been described in an earlier section. The extent of turbulence is also influenced by temperature and composition and is characterised by the Reynolds number (eq. (4.16)).

The two extremes are laminar flow conditions, usually combined with high shear rates, through to fully developed turbulent flow.

Other factors such as pH and osmotic pressure may also influence the permeate flux, as well as membrane configuration (see Chapter 3).

4.4 DIAFILTRATION

Introduction

An extension of ultrafiltration employs the addition of water at some stage during the concentration process. It should be remembered that during ultrafiltration, the concentration of any component in the retentate can never decrease (unless there is a true negative rejection). At best, for zero rejection, it will remain constant. However the amount (yield) of a component of low rejection value is significantly reduced, as is the dry weight composition, compared to a substance with a much higher rejection.

To effect a reduction in concentration, the retentate needs to be diluted with water; such a process is known as diafiltration. The net effect of diafiltration is to wash out more of the lower molecular weight components. The two main modes of operation are *discontinuous diafiltration* (DDF) and *continuous diafiltration* (CDF).

DDF is where the water is added back to the retentate intermittently. For example a dilute extract may be concentrated by a factor of 10, by normal ultrafiltration. Water could then be added back to the extract, say to restore the original volume. The diluted retentate could then be further processed (diafiltered) to further reduce the volume of the retentate. Further water can be added as required. Extensive diafiltration should eventually reduce the concentration of low molecular weight compounds to very low levels. DDF can be analysed in the same way as a batch ultrafiltration process, with account being taken of dilution effects. Lewis and Finnigan (1989) found DDF useful for removing glucosinolates from rapeseed meal extracts. After a fivefold concentration, followed by dilution back to the starting volume and a further fivefold concentration, the protein content had increased from 45.3 to 83.0% (dwb) and the glucosinolate concentration was reduced from 2.5% to 0.3%. However glucosinolate rejection values were greater than zero and there were considerable losses of protein in the permeate.

There are many similar examples of diafiltration processes in the literature.

4.4.1 Washing out at constant volume

CDF involves the continuous addition of water, at a rate equal to the rate at which permeate is removed. Normally this would be termed washing out. This is probably the most common form of diafiltration and can be subjected to mathematical analysis (see Fig. 4.8).

Let

system volume = V_0

permeate flow rate = P

water flow rate = W

At steady state $W = P$

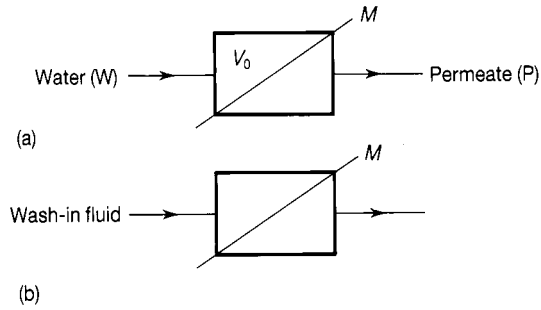


Fig. 4.8. Diafiltration: (a) washing out; (b) washing in.

Solute balance:

$$V_0 c = V_0 (c + \mathrm{d}c) + P c_P \, \mathrm{d}t$$

$$-V_0 \, \mathrm{d}c = P c_P \, \mathrm{d}t$$

For $R = 0$, $c = c_P$

$$-V_0 \, \mathrm{d}c = P c \, \mathrm{d}t$$

$$-V_0 \int_{c_0}^{c} \frac{\mathrm{d}c}{c} = P \, \mathrm{d}t$$

$$-V_0 \ln\left(\frac{c}{c_0}\right) = Pt$$

$$\ln\left(\frac{c_0}{c}\right) = \frac{Pt}{V_0} = \frac{V}{V_0} \tag{4.21}$$

where V = cumulative permeate volume.

The ratio V/V_0 is termed the number of diavolumes removed. Equation (4.21) gives the number of diavolumes required to bring about a certain removal of the freely permeating species ($R = 0$). Thus, a tenfold reduction in concentration would require the removal of 2.303 diavolumes.

When $R \neq 0$; $c_P = c\,(1 - R)$

$$\frac{-V_0 \, \mathrm{d}c}{c(1-R)} = P \, \mathrm{d}t$$

$$\frac{-V_0}{(1-R)} \int_{c_0}^{c} \frac{\mathrm{d}c}{c} = \int P \, \mathrm{d}t$$

$$\frac{V_0}{(1-R)} \ln\left(\frac{c_0}{c}\right) = Pt$$

Therefore

$$\ln\left(\frac{c_0}{c}\right) = (1-R)\left(\frac{V}{V_0}\right) \tag{4.22}$$

Again, this equation is only applicable if R remains constant.

Thus a component with a rejection of 0.1 would require the removal of 2.59 diavolumes to reduce its concentration ten times. (cf. 2.3 diavolumes for $R = 0$.) Glover (1985) stated that in milk, the most efficient diafiltration was to concentrate to about 2, followed by continuous diafiltration.

If the analysis is done on the permeate, the equation becomes

$$\ln\left(\frac{c_0(1-R)}{c_P}\right) = (1-R)\left(\frac{V}{V_0}\right) \tag{4.23}$$

This shows how the permeate concentration changes with the number of diavolumes, or with time.

All these analyses have ignored the void volume V', which can be accounted for by an additional term in eq. (4.21), as follows:

$$\ln(c/c_0) = V/V_0 - V'/V_0$$

Analysis shows that the optimum concentration for the retained species ($R = 1$), to minimise the processing time, is given by c_g/e, where c_g is the gel layer concentration (see Table 4.5). Further analyses for optimising the processing time for diafiltration of soya bean extracts and cheese whey are performed by Ali Asbi and Cheryan (1992). The optimum appeared to be a combined process where the feed is first concentrated by ultrafiltration, followed by continuous diafiltration.

Washing-in

Washing-in applications can be analysed in a similar way:

$$\text{for } R = 0 \qquad \ln\left(\frac{c_f}{c_f - c}\right) = \frac{V}{V_0} \tag{4.24}$$

$$\text{for } R \neq 0 \qquad \ln\left(\frac{c_f}{c_f - c}\right) = \left(\frac{V}{V_0}\right)(1-R) \tag{4.25}$$

where c_f = concentration of component in the wash-in fluid.

Washing-in procedures have been used in binding studies, for example protein solution is placed in the cell and the binding component is added in the wash-in solution. Analysis of the component in the permeate together with a knowledge of its rejection characteristics and a mass balance will permit the total quantity of the component bound to the protein to be calculated. Further examples and details are provided by Cheryan (1986).

4.4.2 Diafiltration applications

Diafiltration is widely employed to remove low molecular weight components, whether they be anions, cations, sugars, alcohol or antinutritional compounds. It therefore offers an alternative process to ion exchange, electrodialysis and dialysis. One interesting medical application involves haemodialysis, for kidney patients.

Another interesting application involves the use of membranes to process alcoholic drinks. The membranes allow alcohol and water to permeate but retain other components within the drink. If water is added back to the concentrate, the net result is that alcohol is reduced. Here also is an example where there is an interest in the permeate.

4.4.3 Protein fractionation

There is potential for using ultrafiltration, combined with diafiltration, to separate individual proteins from a mixture to produce reasonably pure fractions, provided that the molecular weights of the proteins are well distributed and the membranes have a narrow pore size distribution. Lewis (1982) illustrates a scheme for separating three proteins of molecular weight 5000 (A), 50 000 (B) and 500 000 (C), present in equal concentrations. Fractionation is achieved in a two-stage process, using membranes *X* and *Y*, with molecular weight cut-off (MWCO) values of 10 000 and 100 000 respectively, in the first and second stages (see Fig. 4.9). Two situations are illustrated, using a membrane with ideal rejection values and those with rejection values found in practice (real membrane). Rejection values used for these calculations are given in Table 4.7.

The feed is concentrated in the first stage by a factor of 10 and the permeate (P_1) consists mainly of A. The concentrate is diluted back to its original volume with water and ultrafiltered by a factor of 10, using membrane *Y*. The concentrate (C_2) and permeate (P_2) from the second ultrafiltration process comprise predominantly C and B respectively. The compositions of the three streams (dry weight basis) are given in Table 4.8, for the ideal membrane and the real membrane; eqs. (4.6) and (4.8) can be used to perform such calculations.

Better separations would also be achieved by using higher concentration factors.

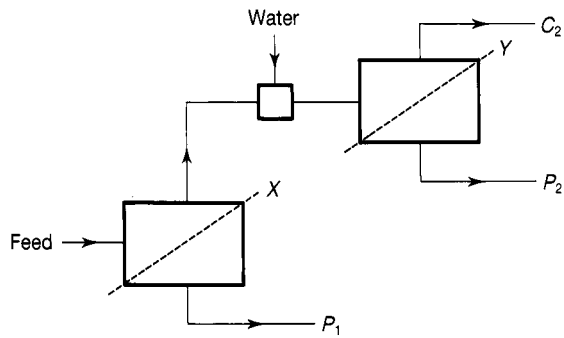


Fig. 4.9. Fractionation of proteins A, B and C, with membranes *X* and *Y*; P_1, permeate from first stage; P_2, permeate from second stage; C_2, concentrate from second stage.

Table 4.7. Membrane data for protein fractionation

Membrane	MWCO	Rejection (ideal)			Rejection (real)		
		A	B	C	A	B	C
X	10 000	0	1	1	0.1	0.99	1.00
Y	100 000	0	0	1	0.05	0.10	0.99

Table 4.8. Composition of different streams

	Composition (dry weight basis)					
	Ideal			Real		
	A	B	C	A	B	C
Feed	33.3	33.3	33.3	33.3	33.3	33.3
P_1	100	0	0	97.4	2.6	0
C_2	0.9	9.0	90.1	1.3	11.0	87.7
P_2	9.1	90.9	0	11.3	86.4	2.3

Further purification of each stream could be accomplished by more extensive diafiltration. It can be seen that although there is some scope for enrichment of proteins, true fractionation is difficult. The major limitation would appear to be due to the diffuse nature of the membrane and the wide distribution of pore sizes. Cheryan (1986), using a similar type of analysis, came up with a similar set of data and arrived at similar conclusions.

An alternative procedure would be to employ continuous diafiltration to wash out each of the proteins in turn, starting with the smallest.

McGregor (1986) has examined a wide range of membranes for separating complex mixtures of proteins. Some membranes showed reasonably sharp characteristics in terms of the permeate, i.e. it was possible to produce permeates with very little protein above a certain molecular weight, but with a mixture of proteins in the concentrate.

A novel approach to enrichment, in situations where the fraction of interest represents a small proportion of the total protein, has involved depositing inorganic or organic compounds within the matrix of a porous stainless steel tube. These components separate on the basis of charge or size exclusion. Immunoglobulin G has been enriched from 8% to 20% in cheese whey by this method (Thomas *et al.*, 1992).

4.5 ULTRAFILTRATION APPLICATIONS

Some applications of ultrafiltration will now be discussed, which take advantage of the opportunity to concentrate macromolecules or to remove small molecular weight components, at ambient temperature, without changing pH or ionic environment.

A wide range of membrane materials is available, the most common being cellulose acetate, polyamides, polysulphones and polyethersulphones; each with different flux characteristics, rejection values, and other physicochemical characteristics, such as charge and extent of hydrophobicity.

Molecular weight cut-offs range from 2000 to 300 000, with operating temperatures up to 80°C, over the pH range 1 to 14.

Ultrafiltration is also very useful for recovering valuable components from food-processing waste stream and fermentation broths. Probably the greatest impetus has come from the dairy industry and dairying applications. However, in all applications, flux decline due to concentration polarisation and fouling are probably the two most important practical aspects.

4.5.1 Dairy applications

Milk is chemically complex, containing components of a wide range of molecular weights, such as protein, fat, lactose, minerals and vitamins. It also contains micro-organisms, enzymes and perhaps antibiotics and other contaminants. An idea of the complexity of milk is given by Walstra and Jenness (1984), who list well over 50 components. Whole milk contains about 30–35% protein and about the same amount of fat (dry weight basis). Therefore it is an ideal fluid for membrane separation processes, in order to manipulate its composition, thereby providing a variety of products or improving the stability of a colloidal system. The same applies to skim-milk, standardised milk and some of its by-products such as cheese whey. Skim milk can be concentrated up to seven times and full-cream milk up to about five times (Kosikowski, 1986). Milk can be derived from a number of different species, cows' milk being the most common, with milk from buffalo, goats and sheep being drunk in substantial quantities throughout the world. Milk is either consumed in its liquid form or converted to a wide variety of products. Surplus milk is usually preserved as skim milk powder and butter or anhydrous butterfat. At one time the most valuable component was the fat, with cream products and butter fetching high returns. Skim milk was therefore a by-product of cream and butter manufacture, along with lesser quantities of buttermilk. Most of the skim milk was dried, and in some situations fed to animals. However, a further important trend over the last ten years has been the move to a more health-conscious diet, and in this sense skim milk is more widely used as the starting material for yoghurts, low-fat cheeses and other desserts. Ultrafiltered milk also forms the starting material for some of these types of product (de Boer and Koenraads, 1991).

Cheese is a very important product derived from milk. In cheese production, most of the fat and the casein fractions are incorporated into the curd. However the by-product of cheese manufacture is whey, which incorporates the whey proteins (about 20% of milk protein). The compositions of whey and skim milk are given in Table 4.6.

An IDF publication (International Dairy Federation, 1979) gives a summary of the rejection values obtained during the ultrafiltration of sweet whey, acid whey, skim milk and whole milk, using a series of industrial membranes. These results are summarised in Table 4.1. It should be noted that protein rejections are based on Kjeldahl nitrogen × 6.38. Where this is the case, it is also measuring non-protein nitrogen (NPN). Rejection values obtained may not be a true reflection of the behaviour of the protein. Such

rejection values could be lower than expected for materials containing substantial amounts of NPN. One example is chhana whey, which is produced from a heat-coagulated cheese and contains substantial NPN (Jindal and Grandison, 1992).

However, low rejection values could also be indicative of excessive protein leakage, and may well warrant investigation if not expected. Rejection values based on true protein can be determined by Kjeldahl or by using electrophoresis or high performance liquid chromatography (HPLC) for analysis and will give a clearer picture of the behaviour of the proteins.

Rejection values for ash are interesting and are higher than would be expected from their molecular weight. This would suggest that binding of minerals to the protein is occurring. During ultrafiltration of whey and buttermilk, Hiddink *et al.* (1978) observed that at pH 6.6, anions such as Cl^- and NO_3^- were preferentially removed. However, at pH 3.2, cations such as Na^+, K^+, and Ca^{2+} were preferentially removed. This is an example of the Donnan effect. Maximum ash removal was obtained by ultrafiltration at pH 6.6, followed by diafiltration at pH 3 to 3.5.

Bastian *et al.* (1991) compared the rejection values during ultrafiltration and diafiltration of whole milk. They found that the rejection of lactose, riboflavin, calcium, sodium and phosphorus was higher during diafiltration than ultrafiltration. Diafiltration of acidified milk gave rise to lower rejections of calcium, phosphorus and sodium. Premaratne and Cousin (1991) have performed a detailed study on the rejection of vitamins and minerals during ultrafiltration of skimmed milk. During a five-fold concentration the following minerals were concentrated by the following factors: Zn (4.9), Fe (4.9), Cu (4.7), Ca (4.3), Mg (4.0) and Mn (3.0), indicating high rejection values. On the other hand, most of the B vitamins examined were almost freely permeating.

The use of nanofiltration or partial demineralisation has been discussed by Kelly *et al.* (1991) and its effects on lactose crystallisation, which was improved by about 8% at a concentration factor of 3, by Guu and Zall (1992).

Both cheese whey and skim milk contain substantial protein and other nutrients. A great deal of attention has been paid to ultrafiltration of these products to increase their functionality and hence profitability.

Cheese whey contains only about 10–12% protein on a dry weight basis. However the proteins are soluble and have excellent functional properties. The main thrust has been toward using ultrafiltration to increase the protein content, in order to produce concentrates, which could then be dried to produce high protein powders (concentrates and isolates) with useful functional properties. Some typical concentration factors (f) used are as follows:

$f = 5$	protein content (dwb) about 35% (similar to skimmed milk)
$f = 20$	protein content about 65%
$f = 20$; plus diafiltration	protein content about 80%

The product starts to become very viscous at a concentration factor of about 20. Therefore if a protein concentrate with a higher total solids is required, diafiltration is required.

However, some of the minerals are difficult to remove. In order to produce protein isolates, containing over 90% protein, ion exchange is often used. Whey from some cheese varieties, e.g. chhana, may contain substantially less true protein; because of this, concentration factors up to 40 are possible (Jindal and Grandison, 1992).

The most effective concentration is achieved when the rejection for protein is 1.0 and that for lactose and minerals is 0. Figure 4.10 shows how the protein content of the whey concentrate, measured on a dry weight basis, is likely to change with concentration factor for this ideal situation and also for a practical situation, where protein, lactose and mineral rejections would be about 0.98 and 0.1 and 0.2 respectively. For the practical situation, the protein content is always lower, at any concentration factor. There are also slight losses of protein in the permeate.

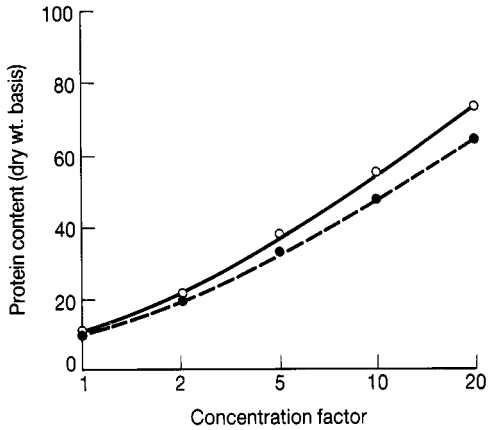


Fig. 4.10. Illustration of change of protein content (dry weight basis) during ultrafiltration of whey; o ideal situation; • practical situation.

More recently inorganic membranes have been used for whey protein concentration and a study of the fouling of these membranes has been made by Daufin *et al.* (1991). Calcium phosphate was found to be partly responsible for flux decline and the role of protein increased as its concentration increased and its pH fell. Coolbear *et al.* (1992) have examined the effects of some thermophilic proteinases for cleaning ultrafiltration membranes fouled during whey processing. Performance was synergistically enhanced using the anionic detergent SDS in combination with the enzyme. Bohner and Bradley (1992) have described an effective cleaning and disinfecting procedure for polysulphone membranes, fouled by whey solids.

The permeate from ultrafiltration of whey contains about 5% total solids, the predominant component being lactose. Since this is produced in substantial quantities, the economics of the process are dependent upon its utilisation. It can be concentrated by reverse osmosis and hydrolysed to glucose and galactose to produce sweeteners or fermented to produce alcohol or microbial protein.

Skim milk has been investigated also. However because of its original higher protein content, concentration factors of about 7 are the maximum achievable. Protein concentrates based on skim milk have not received the same amount of commercial interest as those based on whey proteins. However, it has been suggested that the concentrates can be further modified to produce an interesting range of products with good whipping and foaming characteristics. Rajagopalan and Cheryan (1991) reported that it was not possible to produce a pure protein isolate by ultrafiltration and diafiltration of skim milk, due to a high mineral rejection. An isolate containing about 90% protein and 8% ash was obtained.

Yoghurt and other fermented products have been made from skim milk and whole milk concentrated by ultrafiltration (Renner and El-Salam, 1991). Whey protein concentrates have also been incorporated (de Boer and Koenraads, 1991). Production of labneh, which is a strained or concentrated yoghurt at about 21% total solids, has been described by Tamime *et al.* (1991), by preconcentrating milk to 21% TS. Inorganic membranes have also been used for skim milk, and Daufin *et al.* (1992) have investigated the cleanability of these membranes using different detergents and sequestering agents.

As well as exploiting the functional properties of whey proteins, full cream milk has been concentrated by UF, prior to cheesemaking. The UF concentrate has been incorporated directly into the cheese vats. Some advantages of this process include: increased yield, particularly of whey protein, lower rennet and starter utilisation, smaller vats, or even complete elimination of vats, little or no whey drainage and better control of cheese weights. Lawrence (International Dairy Federation, 1989), suggests that concentration below a factor of 2 gives protein standardisation, reduced rennet and vat space, but no increased yield. At concentration factors greater than 2, an increased yield is found.

Some problems result from considerable differences in the way the cheese matures and hence its final texture and flavour. The types of cheese that can be made in this way include: Camembert type cheese, mozzarella, feta and many soft cheeses. Those which are difficult include the hard cheeses such as Cheddar and also cottage cheese; the problems are mainly concerned with poor texture. Some debate about compositional standards for cheeses produced by such technology exists. More discussion is given by Kosikowski (1986).

Further reviews on the technological problems arising during the conversion of retentate into cheeses are discussed by Lelievre and Lawrence (1988). Spangler *et al.* (1990a) have investigated how the quality of Gouda cheese was affected by concentration factor (3.6–5), rennet concentration and coagulation temperature. The best cheeses were made from the five-fold concentration and a rennet concentration of 0.14%. Higher rennet concentrations lead to the development of very bitter flavoured cheese. Spangler *et al.* (1990b), for example, found that Gouda production from ultrafiltered milk could be improved by attention to detail; such as preacidification of milk prior to ultrafiltration; the amount of rennet was also very important. Sachdeva and Reuter (1991) produced acceptable chhana by ultrafiltration; it had a lower moisture content, giving a yield improvement of 31.4% (product basis) and 16.4% on a dry matter basis. Quarg is also produced from ultrafiltered milk.

Whole milk has been concentrated five times by ultrafiltration and the retentate heated at 120°C for 5 min before being recombined with the permeate prior to pasteurisation (Kosikowski and Mistry, 1990). This procedure is claimed to produce an extended shelf-life product with a superior flavour to a conventionally pasteurised product.

Sweetsuir and Muir (1985) have investigated the production of sterile concentrates, produced by the ultrafiltration of skim milk, with and without the addition of fat. Such concentrates were able to withstand sterilisation at 120°C for 7 min. The organoleptic qualities were improved by the presence of fat and the heat stability was improved by procedures which reduced the levels of salts in the concentrates.

Protein hydrolysates produced from milk proteins have been discussed by Donnelly (1991).

Ultrafiltration is an extremely valuable method of concentrating and recovering many of the minor components, particularly enzymes from raw milk, many of which would be inactivated by pasteurisation. Such enzymes are discussed in more detail by Kosikowski (1988). On-farm ultrafiltration processing of milk has been suggested as a way of reducing transportation costs, but is only viable for long distances, i.e. greater than 100 km, and is not likely in the UK; the permeate can be fed to pigs.

Further reviews on membrane processing of milk are provided by Glover (1985, 1986), Renner and El-Salam (1991), El-Gazzar and Marth (1991) and International Dairy Federation (1991).

4.5.2 Oilseed and vegetable proteins

Crops such as soyabean, cottonseed, sunflower and maize are grown primarily for oil. After its extraction a solid residue remains which is often rich in protein. However, certain toxic components may also be associated with the residues. There have been many laboratory investigations into the use of ultrafiltration for extracting proteins from these residues, or removing the toxic residues. Lewis (1982) and Cheryan (1986) have reviewed the more important of these. For soya, the separation of low molecular weight peptides from soy hydrolysates, with the aim of improving quality; the dissociation of phytate from protein, followed by its removal by ultrafiltration; the removal of oligosaccarides; the removal of trypsin inhibitor and comparisons of performance of different membrane configurations. For cottonseed, the use of different extraction conditions has been evaluated, as have the functional properties of the isolates produced by ultrafiltration. Investigations were performed with sunflower and alfalfa to remove the phenolic compounds responsible for the colour and bitter flavour.

Many have been very successful in their objectives and good quality protein concentrates and isolates have been produced, particularly with soyabean. For example, Nichols and Cheryan (1981) showed an 86% recovery of soya protein by a process involving multiple extraction, centrifugation, ultrafiltration and diafiltration. Diafiltration was necessary to produce an isolate, but there was still a relatively high amount of minerals, due to a higher than expected mineral rejection.

However, in comparison to dairy processing, few have come to commercial fruition, mainly because of the economics of the process, dictated by the relatively low value of the products and the fact that acceptable food products can be obtained by simple

technology, such as isoelectric precipitation. A further problem arises from the fact that the starting residue is in the solid form, thereby imparting an additional extraction procedure. Extraction conditions may need to be optimised, with respect to time, temperature, pH and antinutritional factors. Most vegetable proteins have their minimum solubility somewhere between pH 4 and 7. They are much more soluble at high and low pH values, so the use of second-generation membranes has helped enormously with processing such extracts and cleaning of such membranes. However, ultrafiltration at high pH and high temperatures may cause protein hydrolysis and excessive loss of nitrogen in the permeate. This has been observed by Lawhon *et al.* (1978). It is usually advisable to prefilter the extracts down to below 50 μm to avoid excessive flux decline due to fouling.

A further problem arises from the complexity of oilseed and vegetable proteins, compared to milk products, evidenced by electrophoretic measurement. It is likely that many of these proteins are near their solubility limits after extraction and further concentration will cause them to come out of solution and promote further fouling. Fouling and cleaning of membranes was found to be a serious problem during ultrafiltration of rapeseed meal extracts (Lewis and Finnigan, 1989). Again it was possible by extensive diafiltration to produce an extract with a high protein content and virtually no glucosinolate (dry weight basis). The rejection of glucosinolate was higher than expected, suggesting some form of binding with the proteins.

Work has also been performed on potatoes and potato-processing wastes as well as bean and pea extracts. Waste streams from the vegetable industry have received attention, in order to recover starch and protein, as well as reducing biological oxygen demand (BOD). Some examples are given by Walters and Elliott (1983) and Koseoglu *et al.* (1991).

Kosikowski (1986) has reviewed some of the work on clarification of alcoholic beverages and sugar juices from cane, beet and fruit rind.

Fruit juices have been clarified successfully by membrane techniques; these are covered in more detail in Chapter 5. Vegetable juice processing is also important and is covered in more detail in the section on reverse osmosis (Chapter 3).

Applications in cereal processing have been reviewed by Koseoglu (1991); these include the recovery of solids from thin stillage, protein from wet-milling processes and foam proteins from fermentation broths. A further important aspect is the use of enzyme reactors for the breakdown of starch to sugars. This is discussed in more detail later in this chapter under membrane reactors (Section 4.5.4). Oil-processing applications are very much in their infancy. Potential applications in vegetable oil production and refining are reviewed by Koseoglu (1991). One application is in the solvent-oil extraction process to recover the solvent from the oil, thereby reducing the amount of distillation required. This is possible using reverse osmosis and ultrafiltration membranes but the main problem is compatibility of the membrane with the hydrocarbon solvent, which can lead to collapse of the structure. For polysulphone membranes, this problem can be partially overcome by presoaking the membrane with solvents of decreasing polarity; 50/50 solution of isopropanol/water, followed by 50/50 solution of isopropanol/pentane, followed finally by pentane. Research and development has led to a new generation of membranes, which are more resistant to hydrocarbons, which not only remove solvent but also refine the oil by removing lecithin, free fatty acids, and the majority of the

coloured pigments. The membrane processing is claimed to eliminate degumming, alkali refining, water washing and drying.

4.5.3 Animal products

Slaughterhouse wastes contain substantial amounts of protein. Two important streams that could be concentrated by ultrafiltration are blood and waste water.

Blood contains about 17% protein. It can be easily separated by centrifugation into plasma (70%) and the heavier erythrocytes (red blood corpuscles or cells – 30%). Plasma contains about 7% protein, whereas the blood corpuscles contain between 28 and 38% protein. The proteins in plasma possess useful functional properties, particularly gelation, emulsification and foaming. They have been incorporated into meat products and have shown potential for bakery products, as replacer for egg white.

Whole blood, plasma and erythrocytes have all been subjected to ultrafiltration processes (Ockerman and Hansen, 1988; Cheryan, 1986). The process is concentration polarisation controlled and flux rates are low. High flow rate and low pressure regimes are best. Gel concentrations were approximately 45% for plasma protein and 35% for red blood cells. The fouling characteristics of different blood fractions have been investigated by Wong *et al.* (1984). Whole blood was found to be the worst foulant, when compared to lysed blood and blood plasma.

One significant problem with the erythrocyte fraction is the dark colour, due to haemoglobin, which affects its acceptability in most foods. Methods to reduce this include oxidation using hydrogen peroxide or separation of the haem group from the haemoglobin. However, this latter procedure reduces the stability of the protein. A third procedure involves digestion of the haemoglobin with proteolytic enzymes and the use of membranes to recover the breakdown products.

Blood and other parts of the animal are used as raw materials for a wide range of active components.

Waste water, like many other factory wastes, has a high BOD, as well as containing up to 10% of the incoming protein (Cheryan, 1986), and processes for removal of BOD and recovery of fat and protein are described by him.

Another important material is gelatin, which can be concentrated from very dilute solutions by ultrafiltration. As well as concentration, ash is removed which improves its gelling characteristics. This is one example where there have been some high negative rejections recorded for calcium, when ultrafiltered at low pH.

Eggs have also been processed by UF. Egg white contains 11–13% total solids (about 10% protein, 0.5% salts and 0.5% glucose). Large amounts of egg white are used in the baking industry. The glucose can cause problems during storage and causes excessive browning during baking. Whole egg contains about 25% solids and about 11% fat, whereas egg yolk contains about 50% solids. It is unusual to evaporate eggs prior to drying, because of the damage caused. Both RO and UF have been investigated as a means of concentrating egg prior to drying; UF also results in the partial removal of glucose; further removal can be achieved by diafiltration. Flux values during UF are much lower than for many other food materials, most probably due to the very high initial protein concentration; rates are also highly velocity dependent and temperature dependent (Cheryan, 1986). It is possible to remove any remaining glucose by fermentation, prior to

spray drying. Egg white can also be concentrated to about 20% total solids by RO; again, the remainder of the glucose can be removed by fermentation before being spray dried. There are various processes to remove proteins from eggs (lysosyme by ion exchange) after they have initially been concentrated by RO and UF. There is a great deal of interest in removing cholesterol from yolk, with strong marketing advantages for the resulting product. However, despite its relatively low molecular weight, this has not been achieved by UF.

Extracts from cod offal have been subjected to ultrafiltration processes and three protein fractions were produced by a combination of ultrafiltration and pH precipitation. The extracts were hydrolysed but it was not found possible to further fractionate the hydrolysates by ultrafiltration (Vega and Brennan, 1987).

4.5.4 Biotechnology applications

There is a range of applications within the field of biotechnology, concerned with the transformation of components by biocatalysts. These include microorganisms, enzymes and plant or animal cells. The range of substrates and products is vast and the desired product may be either the cells themselves or their metabolic by-products, which may be intracellular or extracellular. UF membranes have a role, both in the primary production process, for example as a reactor for fermentation processes or enzymatic reactions, or in downstream processing for recovering the products.

Membrane-based bioreactors

Membrane-based bioreactors appear to be a very promising application for the production of ethanol, lactic acid, acetone and butanol, starch hydrolysates and protein hydrolysates; further refinements include the introduction of electrodialysis and pervaporation, into selective processes. The two main types of membrane reactor are the enzyme reactor and the fermentation reactor.

Enzyme reactors

Many industrial processes involve the use of enzymes to break down molecules, for example the hydrolysis of starch or proteins, or of simpler molecules such as sucrose or lactose. Most such reactions involving enzymes are of a batch nature and the enzyme needs to be inactivated or is lost at the end of the process because it is not easy to separate it from the product. The processing costs of batch processes are also high. There is considerable interest in continuous processes, where the enzyme is either immobilised or retained within the reaction vessel and reutilised, whereas the products are removed and reutilised. Ultrathin membranes have been investigated for both these types of application, in the form of enzymatic reactors. Immobilisation of enzymes provides an alternative method to overcome some of the disadvantages of using free enzymes. One system which has been much investigated is the immobilisation of the enzyme by adsorption and entrapment onto the outside of tubular membranes. Hollow fibre systems have been found useful for this purpose, with the enzyme being immobilised on the shell side of the reactor (see Fig. 4.11(a)). This is useful in situations where the product and reactant are normally much smaller than the enzyme and both diffuse into the membrane from the feed. One example is the hydrolysis of lactose. These types of reactor are

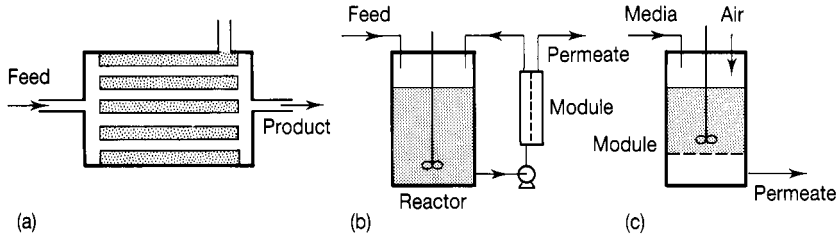


Fig. 4.11. Some immobilisation systems: (a) immobilisation within the shell space of a hollow fibre system; (b) continuous reactor linked to ultrafiltration module, for either enzymatic or fermentation reactions; (c) stirred cell or dead end reactor for enzymatic or fermentation reactions.

usually operated in plug flow mode. A variation on this theme is to immobilise the enzyme in the tubes and to pass the reactant through the shell; this is less common. The enzyme may also be immobilised within the membrane by incorporating it into the casting solution.

The second variant is where the enzyme is incorporated into the reaction vessel and the membrane not only physically retains the enzyme, but also allows a continuous removal of the product. This is not immobilisation in the strict sense of the word, but it does provide a mechanism for long-term usage. This arrangement is useful for hydrolysis of macromolecules, where the required end-product is often much smaller in molecular weight than the reactant or the enzyme. Dead-end or flow-through systems have been used (see Fig. 4.11(b, c)). The dead-end system uses the membrane unit both as the reaction unit and for separation and has been more widely investigated. Flow-through systems use a cross-flow filtration unit connected in series with the reaction vessel. This arrangement has two major advantages over the dead-end system, namely better induced turbulence to reduce concentration polarisation, which leads to poor activities in dead-end reactors after only a few hours' operation and a separate location for the reaction and separation vessels, thereby permitting each to be operated at its optimum set of working conditions, rather than at a compromise set of conditions.

Both systems operate essentially as a continuous stirred tank reactor (CSTR) and feed is supplied to replace the permeate removed. CSTRs are characterised by a wide distribution of residence times and are usually considered to be well mixed, i.e. the concentration of the feed to the membrane is the same as that in the reactor vessel. Thus at high conversions, the substrate concentration would be low; they are particularly useful for reactions which may be inhibited by substrate concentration; at first sight they may also appear useful because it helps reduce end-product inhibition. However, this is not necessarily the case because the end-product concentration in the permeate will be the same as that in the feed. A further feature for first- and second-order reactions is that a large volume is required for a CSTR compared to a plug flow reactor. The major advantages offered by this type of reactor are a high throughput and productivity, governed by a high enzyme concentration attainable and a constant and long-term use of the enzyme.

Cheryan (1986) has reviewed some of the potential applications for bioreactors for many types of protein and carbohydrate sources. Some other important aspects to be considered are the operating temperature and shear damage. Sometimes the optimum temperature for high activity may be one that also inactivates the enzyme quickly. Retained activity over a long period is an important function for membrane reactors.

Shear damage to enzymes was once thought to be important, although there is little experimental evidence to suggest that it is a major cause of reduced activity over the short term. Denis *et al.* (1990) examined the effects of pumping on the activity of pectate–lyase and found no loss of activity after 7 h pumping, but 36% loss of activity after 25 000 passes over a period of 6 days. Loss of activity due to adsorption could be reduced by selecting a membrane with the same charge as the enzyme, at the prevailing pH of the reaction. However, tissue culture and animal cells are much more sensitive to shear.

Vegetable proteins are more difficult to hydrolyse than animal proteins. The higher the levels of non-hydrolysable protein, the shorter will be the operational period for the reactor, due to the build-up of protein within the system. Cheryan and Deeslie (1983) reported run lengths of up to 90 h, during the hydrolysis of soya protein at 50°C, with yields between 85 and 94% and productivities (wt product/wt enzyme used) about 7 times higher than the equivalent batch process. There is scope for optimising the performance of such a reactor, in terms of temperature, enzyme concentration, substrate concentration, reactor volume and flow rate (also equivalent to permeate rate).

Protein hydrolysates have many uses, such as flavour enhancers, improvement of functional properties, and in special dietary or medical applications, some of which have been recently reviewed by Donnelly (1991).

Starch hydrolysis for the production of glucose syrups has also received some attention. There are two stages to the process, namely liquefaction and saccharification. Ideally, it would be desirable to perform both stages simultaneously. Liquefaction is difficult to achieve with enzymes at appreciable starch concentrations, because of the high viscosities involved (de Silva, 1991), although maltose has been produced by starch in a direct process (Nakajima *et al.*, 1990). Saccharification, which makes use of glucoamylase enzymes, is more suited to an enzyme membrane process as the batch process can take up to 48 h and the substrate is less susceptible to concentration polarisation and fouling, due to its lower viscosity.

Darnoko *et al.* (1989) examined the saccharification of cassava starch by glucoamylase in a dead-end stirred cell reactor. Essentially pure glucose was found in the permeate solids. Starch conversions were 64% at 25°C and 97% at 55°C. Productivity was 10–11 times better than a batch reactor over a 24 h period.

de Silva (1991) has studied the saccharification of starch in a CSTR-recycle membrane reactor, using maltodextrins (DE 17) as the feed material, at concentrations between 10 and 30% total solids. A tubular membrane was used with a molecular weight cut-off of 4000. The performance of the reactor was evaluated for different feed concentrations (10–30% TS), different enzyme concentrations and temperatures between 45 and 56°C. At 56°C and a feed concentration of 10% TS, a product containing 95% glucose was obtained, measured after 6 h operation. A mathematical model was developed to describe the process. This included the following terms: reaction kinetic parameters for

the enzyme, including product inhibition, and temperature effects, on both reaction rate and enzyme inactivation; membrane permeability, which was described by the film theory model; and the rejection characteristics of the membrane, which were determined experimentally. However, there were slight losses in enzyme activity during processing.

Other enzymic reactions which have been investigated include the hydrolysis of cellulose, soy protein, fish protein and casein (Cheryan, 1986). A semi-continuous process for lignocellulose hydrolysis has been described by Ishihara *et al.* (1991).

In these long-term continuous processes it is important to minimise enzyme losses and in some processes, to retain, recover or regenerate important enzyme cofactors. Cofactors are required by many enzymes for activity and are classified as prosthetic groups, coenzymes and metal ions. They include compounds such as NAD, FAD, ATP, Co-A. Prosthetic groups are distinguished from coenzymes by their tighter binding to the enzyme, although there is considerable overlap in binding affinities of these two groups (Parkin, 1993).

The cofactors may be immobilised or entrapped with the enzyme or retained within the system by selecting a membrane with a low molecular weight cut-off, which will totally reject them. Another approach is to use negatively charged membranes, which also give rise to higher rejections due to electrostatic interactions. Those which have mainly been investigated are the nicotinamide (NAD and NADP) cofactors.

Another approach is to regenerate cofactors by coupled reactions. Examples investigated include mannitol production from fructose, by mannitol dehydrogenase. Here the NADH produced is regenerated by a coupled reaction involving a glucose dehydrogenase reacting on glucose to produce gluconic acid. The coenzyme turnover number was reported to be 150 000 (Kulbe *et al.*, 1990).

Membrane fermenters

Fermentation processes can also be batch or continuous. Even with continuous processes, the conversion rate may be limited by end-product inhibition and the dilution rate must be less than the specific growth rate of cells, to avoid cell washout. Therefore, another application of membrane reactors is as part of a fermentation vessel, with containment of microorganisms and continuous removal of products. Again the microorganisms can be entrapped or immobilised, typically in the hollow fibre system, or simply allowed to circulate freely with the recycle broth. The freely circulating systems seem to perform more effectively than the immobilised systems. This system operates like a continuous system, with the additional feature that the cells are returned to the fermenter and retained within the system. It is possible to achieve high cell densities and high dilution rates, without the worry of washing out the cells. However, they have not been as thoroughly investigated as enzyme reactors.

One of the most popular applications is the production of alcohol. Cheryan (1986) compares the performance of batch fermentation processes, with continuous culture, immobilised cells and membrane recycle. The membrane recycle gives the biggest productivity in terms of mass of ethanol per unit volume and time. He has also described the use of a membrane reactor to produce alcohol from concentrated whey permeate, using a very high cell density which is retained by the membrane. Such a system has shown a high productivity and has been operated continuously for up to 10 days.

Long-term stability needs to be established. Teixeira *et al.* (1990) used lactose as the carbon source and found that ethanol productivity was 12.5 times higher than continuous fermentation and was directly proportional to the dilution rate. Biomass concentration showed a linear relationship with dilution rate and the largest concentration was eight times that obtained in a conventional continuous fermenter. Endo *et al.* (1990) describe a tubular bioreactor for high-density cultivation of microorganisms, used in the primary part of the circuit. They found improved cell density by over 10 times compared with a conventional fermenter and improved productivity compared to operating at constant dilution rate. Crespo *et al.* (1990) studied the effects of recirculation rate for a propionic acid bacterium fermentation process, linked to a tubular UF module, on permeate flux, fermenter homogeneity, energy consumption, temperature rise and cell damage. High circulation rates improved flux and the degree of homogeneity, but increased cell damage. Energy input was also estimated to determine the degree of cooling required. Propionic bacteria are useful for the production of propionic acid, cyanocobalamin, flavours and dairy starter cultures. Minier *et al.* (1990b) combined the membrane process with distillation, and found that the combination was useful to further concentrate the alcohols and to remove the low molecular weight inhibitors. A mineral ultrafiltration membrane has been used to separate lytic enzymes produced extracellularly during an acetonobutylic fermentation process. The membrane retained the cells but allowed the lytic enzymes to permeate. The permeate was heat treated to inactivate the enzymes, before being returned to the fermenter (Minier *et al.*, 1990a). However, although lytic enzyme activity was decreased by over 6 times, there was no marked increase in the yield of butanol compared to the control.

A novel method of immobilisation involves sandwiching the cells between a reverse osmosis and ultrafiltration membrane. The UF membrane allows free passage of the nutrients to the cells. The RO membrane helps immobilise the cells and permits separation of the alcohol from the sugars and salts. This system has been investigated experimentally and modelled by Jeong *et al.* (1991).

There also exists the possibility of cultivation of animal and plant cells in bioreactors. The performance of a hollow fibre immobilised system has been investigated for the production of antibodies (Piret and Cooney, 1990). Procedures which reduced cell and protein distribution on the shell side of the hollow fibre membrane, led to an improvement in productivity. Ultrafiltration membranes provide a number of advantages, including the retention of growth factors and the selective concentration of high molecular weight protein products.

Recovery of components and downstream processing

Ultrafiltration is now widely used for the recovery of cells, enzymes and other metabolic products. Ultrafiltration and microfiltration can be used for concentrating and harvesting cells but will be in direct competition with continuous centrifugation and to a lesser extent, conventional filtration. The cells may also be shear sensitive, so this factor needs some consideration. Factors such as membrane geometry, pore size, chemical nature of the membrane and ultrastructure should be investigated. Fouling is not necessarily related only to pore size, but to the distribution of pore size, the relationship between pore size and particle size, and whether fouling occurs on the surface or within the pores. Cheryan

(1986) concludes that the overall economics favour membrane processes for small-scale operations, for all particle sizes. However, for larger-scale operations, the economics may favour centrifuges with larger particles, e.g. yeast cells, but the membrane systems for smaller particles.

Ultrafiltration is widely used commercially for concentrating enzyme solutions. There are also many diafiltration operations, for washing out low molecular weight contaminants. For example, Olsen *et al.* (1990) describe the recovery of four enzymes, alkaline phosphatase, hyaluronoglucuronidase, chitinase and beta-N-acetylglucosaminidase, from shrimp waste. Flocculation was induced by ferric chloride at pH 7. The supernatant was then ultrafiltered to recover 99% of the enzyme activity. However, fractionation of the enzymes was not investigated. Pacheco-Oliver *et al.* (1990), used diafiltration to remove coloured impurities from a lipase enzyme, produced by fermentation.

There has been much less success for fractionating enzymes or other components, based solely on size difference, unless there is at least a tenfold difference in their molecular weights. Again the selection of membranes is very important, to minimise excessive enzyme loss in the permeate.

However, there has been some success at partial fractionation and purification, by using compounds which may either bind with one of the fractions and alter its molecular weight and hence its rejection characteristic, or by contacting it with a specific component with which it has a strong affinity. This other component could be introduced in the feed or immobilised within hollow fibre membrane systems. Thus, in conjunction with affinity or ion exchange gels or resins, a specific component in the crude extract can be bound and separated, and subsequently eluted. Examples are given for the purification of horse serum cholinesterase and cattle liver carboxylesterase by these types of method, using an affinity gel and DEAE sephadex, respectively (Molinari *et al.*, 1990). Trypsin has been purified by affinity chromatography and ultrafiltration, using a water-soluble ligand-bound polymer. This was incubated with the crude extract, forming a soluble polymer–trypsin complex. Unbound enzymes were removed by ultrafiltration, whereas the complex was retained. Trypsin was then eluted from the polymer and the eluant was subject to ultrafiltration to further purify the trypsin (US patent, 1990). A French patent (anon) (1991) describes the isolation and recovery of a wide range of enzymes from waste water treatment sludge. Ultrafiltration is used to concentrate the liquid phase, which results from the sludge treated by a number of methods.

Another development is the attachment of bacteria with outer layers rich in glycoproteins to microfiltration membranes. These are then used as a support for protein immobilisation, or as an ultrafiltration membrane (Sara *et al.*, 1990).

It is also possible to separate components such as dissolved organic materials or metal ions by micellar enhanced ultrafiltration (MEUF). The compounds are solubilised within micelles by the addition of surfactants and subsequently removed by ultrafiltration. The processes and factors affecting the flux and rejection of surfactants during ultrafiltration and reverse osmosis are reviewed by Akay and Wakeman (1993).

4.5.5 Medical applications: serum fractionation

One important medical application is for treatment of patients suffering kidney failure, which leads to an accumulation of toxic components, such as urea and creatine within the

blood. There are several treatment choices available: peritoneal dialysis, haemodialysis, haemofiltration and haemodiafiltration (de Burgh, 1992). Peritoneal dialysis makes use of the natural membrane in the peritoneum. A glucose solution is infused into the peritoneal cavity, and solutes and water from the blood diffuse through this membrane into the sugar solution. The sugar solution is infused and removed at hourly intervals. No direct access to the blood supply is required. The other three applications use a manufactured membrane to achieve the removal, with blood being supplied to the membrane. Haemodialysis is usually a short-term, intermittent process, of 2–4 h duration, which requires the availability of dialysis trained nursing staff, whereas haemofiltration and haemodiafiltration can be used continuously for up to several days. Haemodialysis involves the removal of plasma water, containing toxins, through the membrane filter, which is usually either a flat plate or hollow fibre configuration. This filtrate is replaced with a sterile infusion fluid to maintain the patient's fluid status. Where sufficient blood pressure is present, the blood flow can be driven by the patient's own arterial pressure, using either an arterio-venous shunt or an arterial groin line. Where blood pressure is insufficient, a blood pump may be incorporated. Whichever technique is used, the utmost care should be taken to ensure that blood flow is maintained and the filter does not become blocked. Haemodiafiltration uses the same blood circuit, but incorporates a dialysate solution, which is passed through the filter on the permeate side of the membrane, thereby facilitating toxin removal by diffusion. Control of fluid removal may be achieved by adjusting the output of dialysate from the filter in relationship to its input flow. The operating conditions and performance of these two systems are reviewed by Kox and Davies (1992). Further discussion on the use of membranes in artificial organs is given by Leonard (1987).

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Chapter 5

Microfiltration

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5.1 INTRODUCTION

Microfiltration (MF) is the oldest membrane technology, having been used several decades before the first industrial use of reverse osmosis (Glimenius, 1985). However, subsequent development of the technology has been slow. Until recently microfilters were operated in the dead-end mode and were exclusively of the depth-filter type in which particles become trapped within the filter structure, but recent developments have led to membrane-type microfilters, with a narrow pore size distribution, which can be operated in the cross-flow mode. This has led to an increase in possible applications, including clarification of fluids in the food and beverage industries, recovery of cells and cell debris in the biotechnology industries, and the treatment of wastes.

5.2 THEORY, MATERIALS AND EQUIPMENT

Like ultrafiltration (UF), MF is a pressure-driven process employing pressures considerably lower than reverse osmosis. In fact the distinction between UF and MF is somewhat arbitrary and there is no distinction on purely theoretical grounds. The distinction lies in the size ranges of the materials which are separated. UF is considered to involve the processing of dissolved macromolecules, while MF involves separation of dispersed particles such as colloids, fat globules or cells. MF can be considered to fall between UF and conventional filtration, although there is overlap at both ends of the spectrum. A guide to the pore sizes used for MF could be 0.01–10 μm.

For many years, MF has been applied as a dead-end operation using highly microporous symmetric membranes of the depth-filter type. Such membranes retain particles and consequently result in the build-up of a filter cake. This reduces flow, and when the pressure drop has reached a certain level the membrane must be replaced or removed and regenerated. In addition, the presence of a filter cake radically alters the filtering characteristics, effectively acting as a prefilter which removes particles which

could otherwise pass through the membrane itself. On a large scale, therefore, it is only practicable to use this technique when small amounts of particles are present.

Cross-flow MF (CMF) is a development which combines the cross-flow technique, as applied to UF and reverse osmosis, with MF. CMF can be used to minimise (although it should be emphasised, not completely eradicate) the problems encountered in dead-end MF, and thus permit the processing of fluids containing quite large amounts of suspended solids on a large scale. The advantages result from the fact that the build-up of filter cake is avoided due to the shearing effect of the feed stream flowing parallel to the membrane (Fig. 5.1). CMF plants can be operated in the same batch or continuous modes as described in Chapter 3.

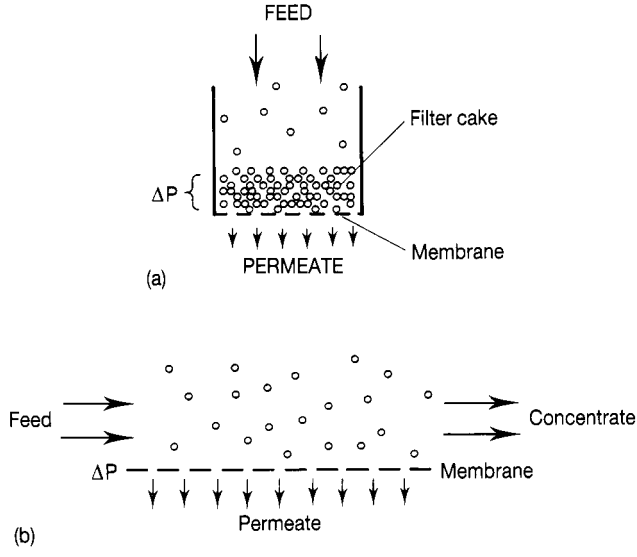


FIG. 5.1. Principles of (a) 'dead-end' and (b) 'cross-flow' filtration.

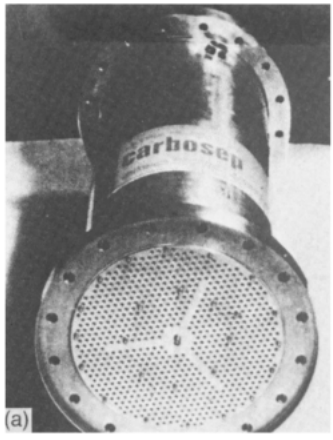
5.2.1 Membrane configurations and characteristics

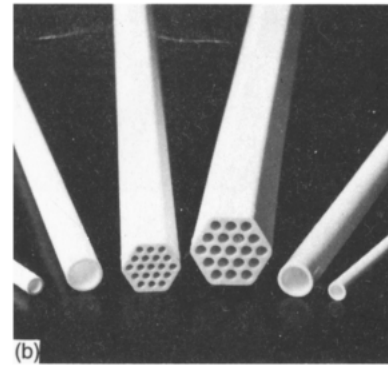
The geometric designs of MF membranes are the same as for UF as described in Chapter 3. Hence the module housings and ancillary equipment are also similar. Also the membrane types are the same as for UF, i.e. cellulose and synthetic polymers (described in Chapter 3) or inorganic. It is notable that the development of inorganic membranes has been towards applications in MF rather than UF and reverse osmosis. In fact some types of inorganic membrane are only available with pore sizes in the MF range.

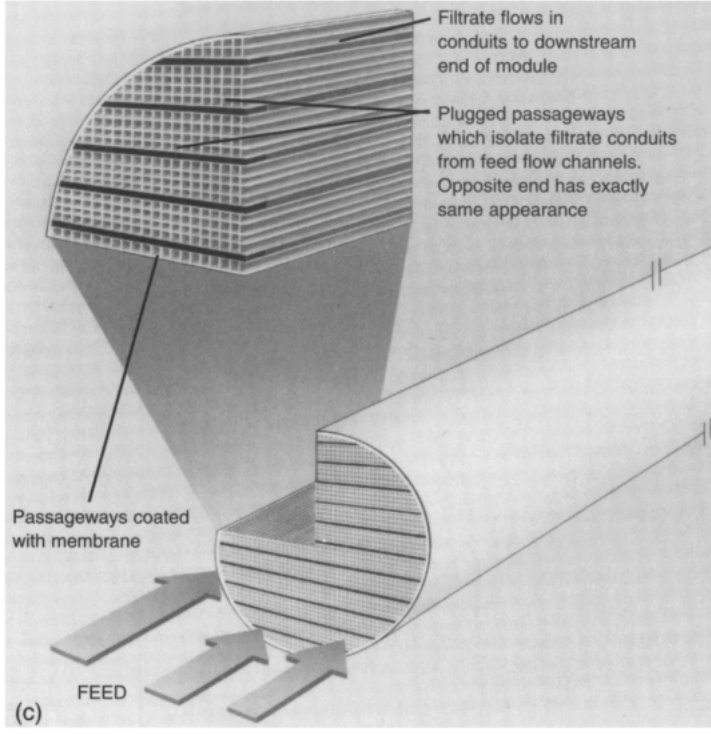
Various inorganic materials have been employed for membrane manufacture including glass, metals and compounds of aluminium, zirconium and titanium, and the geometries can vary radically from conventional membrane design.

The structures and methods of manufacture of inorganic membranes are described in greater detail by Rios *et al.* (1989). Inorganic membranes consist of two parts—a macroporous support and the active membrane coated onto the surface. The supporting

materials must drain away the permeate without any hydrodynamic resistance, and thus have a pore diameter of about 10 μm or more. They are produced from sintered fine powdered materials including alumina, carbon, stainless steel and nickel. The tubular or multichannel geometries of modules (e.g. Fig. 5.2 (a–c)) are produced by extrusion of the







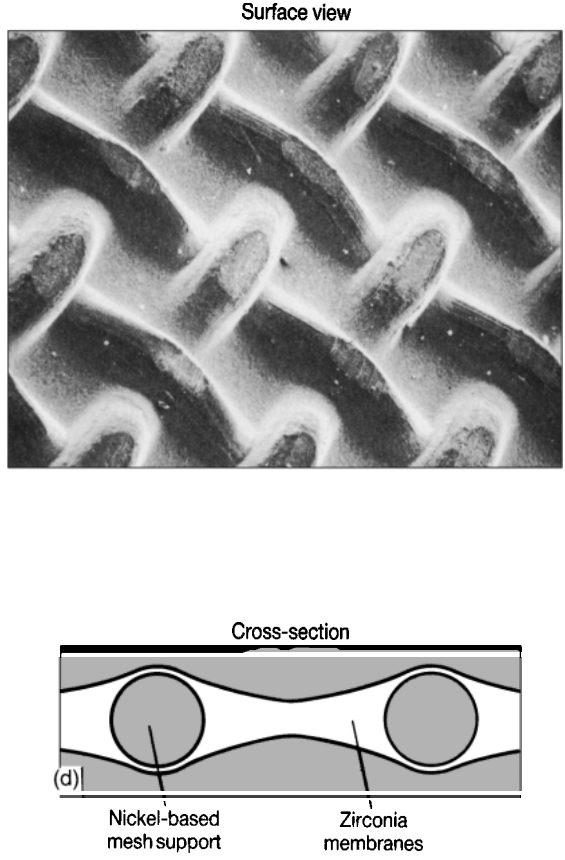


Fig. 5.2. Some designs of inorganic membranes: (a) CARBOSEP membrane composed of zirconia on a carbon support (courtesy of SFEC); (b) different designs of Ceraver alumina membranes (reproduced by permission of Membralox®); (c) alumina membrane module (reproduced by permission of CeraMem); (d) ceramic/metal-mesh composite membrane (reproduced by permission of NWW Acumem Ltd).

powder and binder in aqueous media. The membrane layer may be coated directly onto the macroporous support where pore size is quite large, but for UF and the smaller pore-size MF membranes an intermediate sintered, ceramic layer is necessary due to the surface rugosity of the support. The membrane layer (usually composed of alumina, titania or zirconia) is formed by coating the support with a colloidal suspension and firing at a lower temperature than the firing temperature of the support. To prevent rapid flux decline, the membrane thickness must not exceed a few microns – titanium and zirconium membranes of thickness 3–5 μm have been achieved. Accurate control of the colloidal particle size allows the possibility of producing membranes with an extremely narrow pore size distribution compared to conventional membranes. The final pore size is also related to the sintering temperature. An example of the structure of a sintered ceramic membrane is shown in Fig. 5.3.

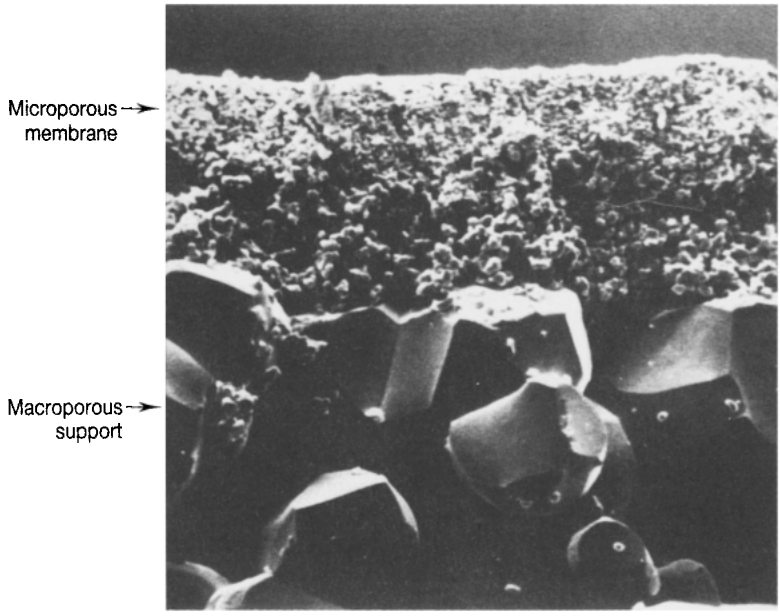


Fig. 5.3. Electron micrograph showing structure of Ceraver sintered alumina membrane with 0.2 μm pore size (reproduced by permission of Membralox®).

Early work on inorganic membranes used glass, but the first reliable cross-flow system was the CARBOSEP membrane made of a microporous layer of zirconia coated onto a macroporous carbon support (Fig. 5.2(a)). New products have subsequently appeared including several designs of alumina membrane (e.g. Ceraver and CeraMem designs – Figs. 5.2(b) and 5.2(c)). A novel design is the composite membrane produced by Acumem composed of a zirconia ceramic membrane with nickel-based superalloy mesh support (Fig. 5.2(d)).

The potential advantages of inorganic membranes result from their greater structural strength and resistance to abrasive degradation, as well as improved chemical and temperature properties. Their rigidity and strength allow the processing of feed materials, such as cheese curd (Mahaut *et al.*, 1982) or particulate materials which would not be possible using conventional designs. Another possibility is that they could be used in conjunction with fluidised turbulence promoters to increase permeate flux, which would be out of the question with the more fragile surfaces of organic membranes. The wide pH ranges of inorganic membranes (e.g. alumina membranes are resistant to pHs ranging from 0.5 to 13.5, although phosphoric and hydrofluoric acids should be avoided) are a great advantage during cleaning and sterilisation. In-place cleaning regimes using high concentrations of caustic soda (3%), nitric acid (2%) and sodium hypochlorite are possible. The modules can withstand temperatures of several hundred °C, which is far beyond the temperatures used for food processing. However, this is an advantage during cleaning cycles and the modules can be sterilised by steam. In practice, operating temperatures are limited by other components such as gaskets, but it could be feasible to

employ high temperatures to defoul individual inorganic membrane modules. The major disadvantage of inorganic membranes is that they are considerably more expensive than polymeric membranes, and while they have some potential advantages, their performance is still limited by concentration polarisation and fouling as with conventional systems.

MF has also been described using membranes produced from woven fabrics or metals.

5.2.2 Performance of microfiltration systems and membrane fouling

The description of flux rate and factors affecting flux rate in UF systems, presented in Chapter 4, applies equally to CMF systems. Engineering aspects and detailed modelling of MF systems are beyond the scope of this chapter, but have been dealt with elsewhere (e.g. Davis and Birdsell, 1987; Ripperger, 1988). In general the larger pore sizes of MF membranes allow very high initial permeate fluxes, although these can decline very rapidly during processing. The flux decline progressively lessens until a near equilibrium filtration rate is observed in most cases (see Fig. 5.4). At this point the shearing action of the cross-flow stream prevents further deposition of material. It is generally assumed that this is actually a state of dynamic equilibrium where particles are leaving and joining the concentration polarisation/fouling layer at the same rate. For any given application it is necessary to carry out experimental trials to establish the most suitable membrane type, and to optimise the pressure and flow characteristics.

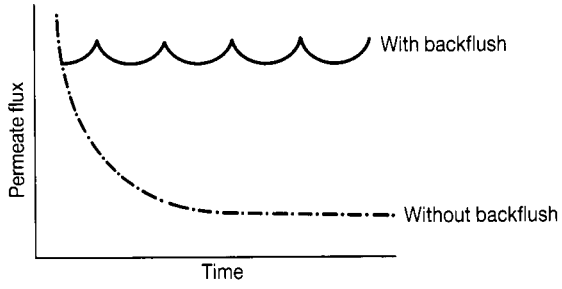


Fig. 5.4. Permeate flux and the effect of backflushing.

As with all membrane processes the permeate flux during MF is limited by concentration polarisation and fouling (see Chapter 4), even with CMF where the formation of filtration cake is prevented or limited. Deep bed clogging of the pores is a particular problem during MF, due to the larger pore sizes. A paradoxical situation can occur in CMF whereby flux may be maintained at a higher level when a tighter membrane (i.e. with lower pore size) is used. This arises when the pores are smaller than the suspended solids in the feed, so that the particles cannot penetrate the pores. An example of this was described by Gatenholm *et al.* (1988), who compared MF with UF membranes for harvesting *Escherichia coli* cells from fermentation broths. The MF membranes showed such a high initial flux decline that the steady state flux was actually higher for the UF than MF. Also Le and Atkinson (1985) noted that permeate flux during CMF of cell suspensions using membranes with a range of pore sizes followed the order 0.45 μm > 0.6 μm > 0.2 μm with respect to pore size. Clearly there is a need to choose

membranes of the appropriate pore size in relation to the characteristics of the particles suspended in the feed. Unfortunately it is very difficult to predict the optimal pore size from first principles, and experimentation is required. However, a recent study of Tarleton and Wakeman (1993) has attempted to understand the relationship between pore size and particle size in flux decline. Clearly the particle size distribution is an important factor, especially when even a small proportion of fines are present in the feed. Many anomalies remain, and further systematic studies are required.

A large amount of effort has been put into devising means of avoiding or counteracting the effects of fouling in MF, and several promising techniques have evolved. The first requirement is to maintain high shear in the retentate and thus reduce the thickness of the boundary layer at the membrane surface and remove deposits from the membrane. It is desirable that turbulent flow be maintained on the retentate side of the membrane. Murkes (1989) has described a new high shear cross-flow method for MF, and claimed an order of magnitude increase in flux rates over conventional CMF.

Forcing a quantity of permeate back through the membrane (i.e. backflushing) can drive deposited particles away from the membrane surface, as well as breaking up concentration polarisation and/or gel layers in the bulk flow. This can lead to regeneration of permeate flux, but it is necessary to ensure that a higher volume of permeate is not used for the backflushing than is gained from the resulting increased permeate flux (Milisic and Bersillon, 1986). However, assuming the correct sequencing and volume of pulsations are used, the technique can be most beneficial (Fig. 5.4). An alternative method of backwashing is to use cleaning sequences consisting of blowing air through the membranes – i.e. gas backflushing (Dietrich *et al.*, 1988; Peters, 1989). This may prove beneficial with problem feed streams. Another approach to maintaining flux is to use the 'uniform transmembrane pressure' mode of operation. This requires simultaneous operation of a retentate pumping loop and a permeate pumping loop, adjusted so that the pressure *drop* across the membrane is small, and is uniform along the length of the membrane. This system has allowed very high fluxes (500–700 l m^{-2} h^{-1}) to be maintained during processing of skimmed milk using ceramic membranes in the 'Bactocatch' system (see Section 5.3.1).

The application of an electric field has been shown to reduce fouling due to colloids and particles during CMF (Visvanathan and Ben Aim, 1989) and thus prevent flux decline (Tarleton and Wakeman, 1988), although the mechanism of action is not fully understood. Bowen *et al.* (1989) reported that electrical pulses were effective in both improving flux rates and restoring flux rates in fouled systems, and could be used as an alternative to backwashing and conventional cleaning techniques.

Other approaches to reducing fouling during MF have included the use of abrasives to break down the fouling layers, and the application of pulsations to the feed stream (Milisic and Bersillon, 1986). Chemical pretreatments of the feed streams can also be used to reduce the problems of fouling (Bedwell *et al.*, 1988), although this is frequently not a suitable option. Pretreatment of the membrane may reduce the flux decline during fouling. Taddei and Howell (1989) reported a 70% improvement of flux during harvesting of yeast cells by conditioning the MF membranes with respect to pH.

The initial challenge to the membrane has been found to affect the subsequent flux during MF. Care must be taken during start-up procedures to prevent too-rapid flux

decline. It is frequently an advantage to start the run in the dynamic mode (i.e. with water circulating prior to introduction of the feed), or with the permeate side filled with solvent.

While the above methods of combating fouling can be very effective, it is essential to combine them with efficient chemical cleaning procedures.

A characteristic of MF is that the separation may alter during processing due to fouling, such that the MF operation, in reality, becomes UF through a 'secondary membrane' formed at the surface of the MF membrane, consisting of macromolecules or colloids. This generally constitutes a problem. However, this property has been used to advantage in the French dairy industry for the concentration of milk. Bennassar *et al.* (1982) described the use of the selective UF properties of the secondary membrane formed while using 0.2 μm membranes. Almost total retention of milk protein was possible with this membrane.

5.3 APPLICATIONS IN THE FOOD AND BIOTECHNOLOGY INDUSTRIES

MF is generally used to separate particles suspended in liquid media, and may frequently be considered as an alternative to conventional filtration or centrifugation. For industrial use the aim is usually to obtain either a clear permeate or the concentrate. Therefore most applications are either clarification, or the recovery of suspended particles such as cells or colloids, or the concentration of slurries. MF is a useful process for the treatment of fermentation broths, frequently as an alternative to centrifugation. During recovery of intracellular components, the cells can be enriched and washed by diafiltration prior to disintegration of the cells, and cell debris can be removed from the products of lysis.

MF membranes can be incorporated in water polishing systems as they are effective in removing suspended solids and bacteria. They may also be applied to a variety of liquid food streams, and may give the accompanying benefit of chemical and/or microbiological stabilisation. Clarification of biochemical or microbiological reaction products or effluents is also possible. MF is a well-established laboratory technique for the production of sterile fluids without the application of heat.

5.3.1 Food industry

Perhaps the majority of applications in the food industry have been in the treatment of juices and beverages. As MF is a purely physical process, it can have advantages over traditional methods involving chemical additives, in terms of the quality of the product as well as the costs of processing.

Finnigan and Skudder (1989) discussed the application of ceramic microfilters to the processing of beer and cider. Very good quality clear permeate was found for both products with high flux rates. Fluxes of 200–400 l m^{-2} h^{-1} were obtained with beer at low temperatures with no rejection of essential components. Recovery of beer from tank bottoms is also possible using ceramic MF, but this is less important in the UK now that duty is not paid at the wort stage of production.

Clarification and biological stabilisation of wine musts and unprocessed wine have been described. This avoids the requirement for fining and, possibly, pasteurisation. Dau *et al.* (1988) used membrane filters with pore size 0.1–5 μm to clarify wine preheated to

30–40°C to increase permeate flux. However, Rios *et al.* (1989) selected ceramic membranes with pore diameter 0.2 μm at temperatures <20°C to avoid sensory changes.

Conventional methods of clarification of fruit juices incorporate the use of diatomaceous earth and enzymes to remove a variety of materials including suspended solids, colloidal particles, proteins and condensed polyphenols. Also they are quite time consuming in that the juice must be stored in settling tanks. Juice clarification using MF is becoming an established commercial practice as it is a continuous process which can result in significant savings in time, materials and labour as well as improved yields. (*Note*: The use of diatomaceous earth results in loss of yield.) Short (1988) described the commercial application of CMF to the processing of a range of fruit juices. A plant producing more than 5×10^5 litres per day of apple juice exists in the USA. This has the major benefit of increasing juice yield from 95 to 97% as well as eliminating the requirement for diatomaceous earth as a pretreatment, and the use of enzymes and fining agents. A saving of $350 000 per annum was estimated. With highly coloured juices, a further benefit of MF is that the colour is retained very well in the permeate. Commercial systems are also known to exist for grape, pear, kiwifruit, pineapple, cranberry and citrus juices. Clarifications of vinegar, vodka, pickling brines and liqueurs by MF have also been described (Le and Billigheimer, 1985; Short, 1988).

It seems likely that CMF could replace carbonatation in the treatment of raw sugar beet juice, although this is not carried out commercially to the author's knowledge. Another section of the industry with several applications is dairy processing. Piot *et al.* (1984) and Merin (1986) have clarified sweet cheese whey using CMF. This has the dual benefit of removing fat and reducing the bacterial population and could eliminate the need for fat separation and heat treatment in the production of whey protein powders prior to UF. The former authors reported that a decimal reduction of microorganisms of 5 could be obtained in the microfiltrate compared to the whey, although some loss of whey protein was observed. Hanemaaijer (1985) described a scheme for whey treatment incorporating MF and UF to produce 'tailor-made' whey products with specific properties for specific applications. The products include whey protein concentrates which are rich in whey lipids, as well as highly purified protein.

Bacterial removal from whole milk by MF is a problem because the size range of the bacteria overlaps with the fat globules, and to a lesser extent with the casein micelles. However, some success has been achieved with skim milk. The 'Bactocatch' system can remove 99.6% of the bacteria from skim milk using ceramic membranes on commercial scale (Malmbert and Holm, 1988). The retentate (approximately 10% of the feed) can then be sterilised by a UHT process, mixed with the permeate and the mixture pasteurised, to give a product with 50% longer shelf-life but no deterioration in organoleptic properties compared to milk that has only been pasteurised. The combination of MF and heat reduces the bacterial numbers by 99.99% (4D). Alternatively the permeate could be used for cheesemaking, or the production of low-heat milk powder (Hansen, 1988). Piot *et al.* (1987) described the use of membranes of pore diameter 1.8 μm to produce skim milk of low bacterial content. Recovery of fat from buttermilk has also been described (Rios *et al.*, 1989).

Membranes have been used to concentrate milk prior to the manufacture of many cheese types. This results in improved yields and other associated benefits such as

reduced requirement for rennet and starter, and the ability to produce much more cheese per vat (Grandison and Glover, 1994). In most cases UF membranes have been used as it is necessary to retain all the protein in the concentrate. An alternative approach is to concentrate the curd after coagulation of the milk in which case a solution of lactose and minerals is removed from the semi-solid protein gel. This can be done using centrifugation in the manufacture of some soft cheese types.

However, the use of MF is an attractive alternative. Rios *et al.* (1989) have carried out extensive trials on this application and concluded that the use of 0.2 μm pore diameter membranes gave a product with better texture and yield than with centrifugation. The choice of ceramic membranes allowed the curd to be contacted directly with the membrane.

Other food applications have been reported with meat and vegetable products including the following.

Devereux and Hoare (1986) described the use of MF to recover precipitated soya protein. This could have advantages over recovery of the dissolved protein using UF.

Gelatin is a proteinaceous material derived by hydrolysis of collagen. This is purified by filtration incorporating diatomaceous earth. The latter process can be replaced by CMF which effectively removes dirt, coagulated proteins, fats and other particulate materials from the feed. Again the CMF method gives higher yields of high quality product on a continuous basis. Short (1988) calculated that incorporating CMF plants for gelatin would have a payback time of 3 years for a capacity of 30 tonnes/h.

5.3.2 Applications for biotechnology

Applications of MF in the biotechnology industry are very promising although the process has not yet made the breakthroughs that may have been expected. The major problem with MF of suspensions of cells or cell debris is the exponential flux decay resulting from adhesion of cells or cell fragments to the membrane – i.e. biofouling. This has limited the application of CMF into biotechnological downstream processing. Solutions to this problem, other than general methods of avoiding fouling, are discussed by Defrise and Gekas (1988) and include choice of biocompatible membrane materials and the use of surfactants and polyelectrolytes. Most biotechnological applications of MF are as a competitor to centrifugation, and MF is becoming recognised as a viable alternative in many cases. CMF may have advantages over centrifugation when containment is required (e.g. when handling pathogenic organisms or in recombinant DNA technology) as aerosols are not produced during the operation (Kroner *et al.*, 1984). The capital and maintenance costs of MF are lower than for centrifugation, although membrane replacement is expensive. Tutunjian (1985) compared the costs of process-scale harvesting and washing of *Escherichia coli* cells using MF with a hollow fibre system to centrifugation. The MF system had about 70% lower capital, and 25% lower running costs than centrifugation. Also virtually 100% recovery of solids is possible with MF, whereas it is often less than 90% with centrifugation.

CMF is a versatile technique for cell harvesting and cell debris removal during recovery of intracellular products. The viability and enzyme content of the cells is unaffected by MF, and it is possible to concentrate cell suspensions to 20–25% dry weight, limited by pumping considerations (Le and Atkinson, 1985). A 0.1 μm

microporous hollow fibre membrane has been used for the purification of bovine growth hormone from *Escherichia coli* lysate (Tutunjian, 1985). Although the growth hormone was partially rejected from the membrane, this could be recovered using diafiltration such that over 99% recovery from the original lysate was possible. Kroner *et al.* (1984) used MF to recover a range of soluble enzymes from cell debris, although they concluded that the separation was not satisfactory due to high retention of the enzymes in the feed. Transmission of enzymes through microporous membranes is rather anomalous. Le and Billigheimer (1985) found that transmission of arylamidase increased rapidly over the first 5–10 minutes of processing and then declined steadily. This was ascribed to adsorption of protein to the membrane surface during the initial phase, to form a monolayer lining. Transmission of the enzyme occurred when the surface was saturated, and subsequently declined due to the formation of a secondary membrane of cell debris and proteinaceous matter. In general, careful manipulation of pressures and flow rates can be used to optimise protein transmission through MF membranes.

Alternatively, valuable extracellular biotechnological products can be separated from cell suspensions by MF. Raehse *et al.* (1986) employed polysulphone tubular membranes with a pore size of 0.3–0.5 μm to separate alkaline protease from a fermentation broth. The ratio of the mean pore diameter of the membrane to the size of the microorganisms was between 0.15 and 0.85, which gave a rapid separation suitable for large-scale applications.

A combination of CMF with affinity chromatography provides an innovative development for purifying biochemicals from fermentation broths or other aqueous media, such as trypsin from pancreatic extract. The principle is to bind the biochemical to a macroligand and remove contaminants by CMF. Following dissociation, the purified material can then be recovered by CMF, and the macroligand recycled (Luong *et al.*, 1987).

It is possible to incorporate MF membranes into membrane reactors. In this way a continuous process can be developed in which the membranes are retentive to the cells (or other biocatalyst), but permeable to the reaction products. Most reported developments in this field have been on laboratory scale, and use UF rather than MF membranes. One promising report with MF is the incorporation of zirconia membranes into a continuous fermentation system with cell recycle for production of alcohol which permitted very high yeast concentrations to be used (Lafforgue *et al.*, 1987). This has led to the possibility of extremely high alcohol production capacity (possibly 150 kg m^{-2} h^{-1}).

5.4 CONCLUSIONS

MF has made significant advances in new applications in the food and biotechnology industries. However, the technique has not yet realised its full potential, largely due to the severe problems of flux decline due to fouling. It is believed that further developments in membrane design and a greater knowledge of fouling mechanisms will result in greater application in the future, especially in the field of downstream processing.

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Chapter 6

Ion-exchange and electrodialysis

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Ion-exchange and electrodialysis are distinct methods of separation, but can conveniently be treated together, as the basic criterion for separation in both cases is the molecular electrostatic charge. While ion-exchange involves retention of ionised solutes on a solid support material, electrodialysis permits the separation of ions using selective ion-exchange membranes.

6.1 ION-EXCHANGE

Ion-exchange methods can potentially be used for separations of many types of molecules such as metal ions, proteins, amino acids or sugars. The technology is utilised in many sensitive analytical chromatography procedures, frequently on a very small scale. On the other hand industrial-scale production operations, such as demineralisation or protein recovery, are possible. This chapter will consider only the larger-scale applications which have current or potential use for production in the food and biotechnology industries.

6.1.1 Theory, materials and equipment

A brief summary of the theory of ion-exchange will be given here. More detailed accounts can be found elsewhere (e.g. Vermeulen *et al.*, 1984; Walton, 1983; Helfferich, 1962).

Solute/ion-exchanger interactions

Ion-exchange could be defined as the selective removal of a single, or group of, charged species from one liquid phase followed by transfer to a second liquid phase by means of a solid ion-exchange material. In practice this involves the process of adsorption – the transfer of specific solute(s) from a heterogeneous feed solution on to the solid ion-exchanger. The mechanism of adsorption is electrostatic, involving opposite charges on the solute(s) and the ion-exchanger. The feed solution is washed off, and this is followed

by desorption, in which the separated species are recovered back into solution in a much purified form.

The ion-exchange solids bear fixed ions which are covalently attached to a solid matrix. There are two basic types of ion-exchanger:

(1) Cation exchangers (sometimes called 'anionic exchangers') which bear fixed negative charges and are therefore able to retain cations, and
(2) Anion exchangers (sometimes called 'cationic exchangers') which bear fixed positive charges.

Ion-exchangers can be used to retain simple ionised species, but may also be used in the separation of polyelectrolytes which possess both positive and negative charges (i.e. amphoteric molecules such as proteins) as long as the overall charge on the polyelectrolyte is opposite to the fixed charges on the ion-exchanger. This overall charge depends on the isoelectric point of the polyelectrolyte and the pH of the solution. At pH values lower than the isoelectric point the net overall charge will be positive and *vice versa*. In some circumstances it is even possible for ion-exchangers to retain macromolecules of like charge, presumably if a portion of the molecule carries a sufficient opposite charge (Peterson, 1970). The main interaction is via electrostatic forces, and in the case of polyelectrolytes the affinity is governed by the number of electrostatic bonds between the solute molecule and the ion-exchanger. However, particularly with large molecules such as proteins, multiple interactions may occur involving steric effects. Size and geometric properties, and the degree of hydration of the ions may affect these interactions, and hence the selectivity of the ion-exchanger for different ions. Charge density may be more important than overall charge in determining the relative selectivity.

Figure 6.1 is a schematic diagram showing a generalised anion exchanger – i.e. bearing fixed positive charges. To maintain electrical neutrality these fixed ions must be balanced by an equal number of mobile ions of the opposite charge (i.e. anions) which are held by electrostatic forces. These mobile ions can move in and out of the porous molecular framework of the solid matrix and may be exchanged stoichiometrically with other dissolved ions of the same charge, and are termed counterions. Ion-exchange systems can be considered to consist of two aqueous liquid phases – one confined within

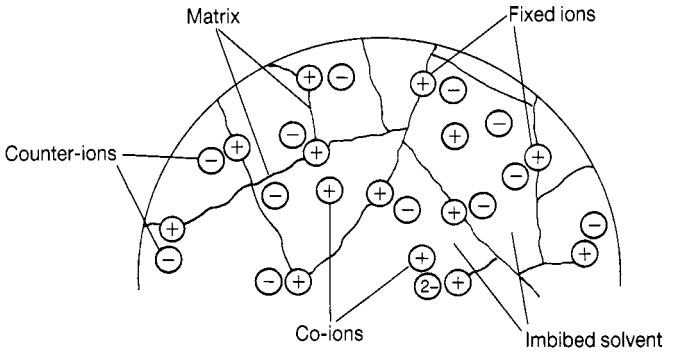


Fig. 6.1. Schematic diagram of a generalised anion exchanger.

the structure of the solid matrix in equilibrium with an outside phase. The interface between the two phases acts as a semipermeable membrane which allows the passage of any mobile ionic species depending on the Donnan equilibrium. This states that the chemical potential of a salt must be the same inside and outside the ion-exchanger – e.g. in the simplest case where the only mobile ions present are Na^+ and Cl^-, then at equilibrium,

$$[Na^+][Cl^-]_{\text{Inside phase}} = [Na^+][Cl^-]_{\text{Outside phase}}$$

Thus a certain proportion of co-ions (mobile ions having the same sign – Na^+ in this example – as the fixed ions) will be present even in the internal phase. Therefore, if an anion exchanger (as in Fig. 6.1) is in equilibrium with a solution of NaCl, the internal phase contains some Na^+ ions, although the concentration is less than in the external phase because the internal concentration of Cl^- ions is much larger.

When an ion-exchanger is contacted with an ionised solution, equilibration between the two phases rapidly occurs. Water moves into or out of the internal phase so that osmotic balance is achieved. Counterions also move in and out between the phases on an equivalent basis. If two or more species of counterion are present in the solution, they will be distributed between the phases according to the proportions of the different ions present and the relative selectivity of the ion-exchanger for the different ions. It is this differential distribution of different counterions which forms the basis of separation by ion-exchange. The relative selectivity for different ionised species results from a range of factors. The overall charge on the ion and the molecular or ionic mass are the primary determining factors, but selectivity is also related to degree of hydration, steric effects and environmental factors such as pH or salt content.

In the adsorption stage, a negatively charged solute molecule (e.g. a protein P^-) is attracted to a charged site on the ion-exchanger (R^+) displacing a counterion (X^-):

$$R^+X^- + P^- \rightarrow R^+P^- + X^-$$

In the desorption stage, the anion is displaced from the ion-exchanger by a competing salt ion (S^-), and hence is eluted:

$$R^+P^- + S^- \rightarrow R^+S^- + P^-$$

Ion-exchangers may be further classified in terms of how their charges vary, with changes in pH, into weak and strong exchangers. The terms strong or weak do not refer to the strength of binding of the ions to the exchanger, or the mechanical strength of the matrix, but to the pH range over which the materials are effective. Strong ion-exchangers are ionised over a wide range, and have a constant capacity within the range, whereas weak exchangers are only ionised over a limited pH range (e.g. weak cation exchangers may lose their charge below pH 6 and weak anion exchangers above pH 9). Thus weak exchangers may be preferable to strong ones in some situations, for example where desorption may be achieved by a relatively small change in pH of the buffer in the region of the pK_a of the exchange group. Regeneration of weak ion-exchange groups is easier than with strong groups, and therefore has a lower requirement of costly chemicals.

Ion-exchange groups

Some common examples of cation exchangers are

$-SO_3^-H^+$ (strong – pK_a 1–2)

$-PO_3^{2-}(H^+)_2$ (medium – pK_a 2–3)

$-COOH$ (weak – pK_a 3.5–8)

Base function is almost invariably present as amines or imines. These are introduced into the matries by chloromethylation, followed by reaction with the appropriate amine to produce weakly to strongly basic ion-exchangers. Some common examples are

$-O-CH_2-CH_2-NH^+-(CH_2-CH_3)_2$ (diethylaminoethyl – DEAE)

$-O-CH_2-CH_2-NH_3^+$ (amino ethyl – AE)

$-O-CH_2-CH_2-N^+(C_2H_5)_2-CH_2-CH(OH)-CH_3$ (quaternary amino ethyl – QAE)

$-CH_2-N^+(CH_3)_3$ (quaternary amine – Q)

Q and QAE are strong anion exchangers while DEAE and AE are weak.

Ion-exchange materials

All ion-exchangers basically consist of a solid insoluble matrix to which are attached the active, charged groups on which ion-exchange occurs. Various terms are used to describe this material including resin, adsorbent, medium, or just ion-exchanger. There is no general agreement on which is correct, and the usage is sometimes confusing – e.g. the term 'resin' is sometimes used as a general term for ion-exchangers, or sometimes specifically for synthetic organic materials, while a resin is strictly a naturally occurring organic compound (Kanekanian and Lewis, 1986).

The solid support must have an open molecular framework which allows the mobile ions to move freely in and out, and must be completely insoluble throughout the process. Most commercial ion-exchangers are based on an organic polymer network, although inorganic materials may be used. The support material does not directly determine the ionic distribution between the two phases, but it is a major factor in determining the physical and chemical stability of the ion-exchanger. Hence this will determine factors such as the capacity, the flow rate through a column, the diffusion rate of counterions into and out of the matrix, the degree of swelling and the durability of the material. The materials tend to be of two main types – xerogels or aerogels. Xerogels are insoluble synthetic polymers containing a cross-linking agent. Their structure and porosity depends on the solvent and degree of solvation and they are compressible to some degree. Xerogels make up the majority of commercially available ion-exchangers including polyacrylamides, polystyrene and dextrans. The pore size of these materials can be controlled by the manufacturing conditions, especially the degree of cross-linking. Aerogels have a much more fixed rigid structure (e.g. porous silica) and are therefore incompressible, which has obvious advantages for production scale.

As the adsorption is a surface effect, the available surface area is a key parameter. For industrial processing the maximum surface area to volume should be used to minimise plant size and product dilution. It is possible for a 1 ml bed of ion-exchanger to have a total surface area >100 m^2. The ion-exchange material is normally deployed in packed beds, and involves a compromise between large particles (to minimise pressure drop) and small particles to maximise mass transfer rates. Porous particles are employed to increase surface area/volume. However, the surface must also be accessible to the solute molecules, and hence materials with an enormous surface area due to the presence of minute pores may be of very limited use, because much of this surface is inaccessible even to small solute molecules. Manufacturers of ion-exchange materials generally quote the exclusion limit of products with respect to molecular size. Particularly in the case of biopolymers, the shape of the pores and the three-dimensional structure of the solute may be a further consideration.

Capacity

The capacity of an ion-exchanger is defined as the number of equivalents of exchangeable ions per kilogram of exchanger but is frequently expressed in meq/g (usually in the dry form), and can be determined by titration of the charged groups with strong acid or base. This property depends on the nature of the fixed ions as well as the available surface area. Most commercially available materials have capacities in the range 1–10 equivalents/kg of dry material.

Blinding and fouling

The operational life of an ion-exchanger, or at least the time between major clean-up campaigns, is limited by blinding or fouling. This is non-specific adsorption onto the matrix surface, or within the pores, which effectively reduces the capacity, and certainly affects the choice of ion-exchanger for a particular separation. The susceptibility of an ion-exchanger to blinding or fouling with a particular feedstock may exclude its use for that function despite having otherwise excellent binding capacity and specificity for the molecules in question. For example, the presence of significant lipid levels in a feedstock may exclude the use of some exchangers for protein separations.

Elution

The choice of method of elution depends on the specific separation required. In some cases the process is used to remove impurities from a feedstock, while the required compound(s) remains unadsorbed. No specific elution method is required in such cases, although it is necessary to regenerate the ion-exchanger with strong acid or alkali. In other cases the material of interest is adsorbed by the ion-exchanger while impurities are washed out of the bed. This is followed by elution and recovery of the desired solute(s). In the latter case the method of elution is much more critical – for example, care must be taken to avoid denaturation of adsorbed proteins.

Elution of the adsorbed solute is effected by changing the pH or the ionic strength of the buffer, followed by washing away the desorbed solute with a flow of buffer.

Increasing the ionic strength of the buffer increases the competition for the charged sites on the ion-exchanger. Small buffer ions with a high charge density will displace

polyelectrolytes which can subsequently be eluted. Altering the buffer pH so that the charge on an adsorbed polyelectrolyte is neutralised or made the same as the charges on the ion-exchanger will result in desorption.

Ion-exchange columns
Fixed bed operations consisting of one, or two columns connected in series (depending on the type of ions which are to be adsorbed), are used in most ion-exchange separations. Liquids should penetrate the bed in plug flow, in either downward or upward direction. The major problems with columns arise from clogging of flow and the formation of channels within the bed. Problems may also arise from swelling of organic matrices when the pH changes.

Mixed bed systems
These may be used to avoid prolonged exposure of the solutions to both high and low pH environments, as is frequently encountered when using anion and cation exchange columns in series (e.g. during demineralisation of sugar cane juice to prevent hydrolysis of sucrose as described below). Cation and anion exchangers are intimately mixed during the adsorption phase so that the feed solution remains at high or low pH only for the time required to pass from one particle to the next. Regeneration is possible on the basis that the two exchange materials have different specific gravities, and thus separate into two layers on backwashing. By the use of a regenerant distributor, strong acids and alkalis may be used to regenerate the resins independently. After rinsing, the ion-exchangers are remixed using compressed air.

Stirred tanks
The flow and swelling problems encountered with fixed beds are obviated by the use of stirred tanks; however, these systems are less efficient and expose the ion-exchangers to mechanical damage as there is a need for mechanical agitation. The system involves mixing the feed solution with the ion-exchanger and stirring until equilibration has been achieved (typically 30–90 min in the case of proteins – Kanekanian and Lewis, 1986). After draining and washing the ion-exchanger, the eluant solution is then contacted with the bed for a similar equilibration time before draining and further processing.

6.1.2 Applications of ion-exchange in the food and biotechnology industries

One method of classifying the applications of ion-exchange could be by industries or commodities. The main areas of the food industry where the process is currently used or is being developed are sugar, dairy and water purification, although sufficient applications occur outside these to render this classification unsatisfactory. Ion-exchange is widely employed in the recovery, separation and purification of biochemicals, monoclonal antibodies and enzymes.

Another way of categorising the applications is by the type of separations attained, for example:

(1) removal of minor components, e.g. deashing or decolorising;
(2) enrichment of fractions, e.g. recovery of proteins from whey or blood;
(3) isolating valuable compounds, e.g. production of purified enzymes.

Alternatively the chemical nature of the adsorbed ions could be used as a basis for classification. Any ionisable component of a foodstuff can potentially be adsorbed on to an ion-exchanger and thus separated.

The following is an attempt to classify applications in food and biotechnology on the basis of the function of the process.

Softening

Softening of water and other liquids involves the exchange of calcium and magnesium ions for sodium ions attached to a cation exchange resin, e.g.

$$R—(Na^+)_2 + Ca(HCO_3)_2 \rightarrow R—Ca^{2+} + 2NaHCO_3$$

The sodium form of the cation exchanger is produced by regenerating with NaCl solution. Apart from the production of softened water for boiler feeds and cleaning of food and processing equipment, softening may be employed to remove calcium from sucrose solutions prior to evaporation (which reduces scaling of heat exchanger surfaces in sugar manufacture), and from wine (which improves stability) (Cristal, 1983).

Demineralisation

Demineralisation using ion exchange is an established process for water treatment, but over the last 20 years it has been applied to other food streams. Typically the process employs a strong acid cation exchanger followed by a weak or strong base anion exchanger. The cations are exchanged with H^+ ions, e.g.

$$2R^-H^+ + CaSO_4 \rightarrow (R^-)_2Ca^{2+} + H_2SO_4$$

$$R^-H^+ + Na^+Cl^- \rightarrow R^-Na^+ + H^+Cl^-$$

and the acids thus produced are fixed with an anion exchanger, e.g.

$$R^+OH^- + H^+Cl^- \rightarrow R^+Cl^- + H_2O$$

Demineralised cheese whey is desirable for use mainly in infant formulations, but also in many other products such as ice cream, bakery products, confectionery, animal feeds etc. The major ions removed from whey are Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, HPO_4^-, citrate and lactate. Ion-exchange demineralisation of cheese whey generally employs a strong cation exchanger followed by a weak anion exchanger (Houldsworth, 1976). This can produce more than 90% reduction in salt content, which is necessary for infant formulae. Lower levels of demineralisation, obtained using a by-pass system, may be adequate for other applications. Due to the high salt content of whey, the system must be regenerated after the treatment of 10–15 bed volumes of whey. This is achieved, following rinsing, by the treatment of cation and anion exchangers separately with strong acids and alkalis respectively. Typically a cycle is about 6 h, of which 4 h are required for regeneration, therefore two or three parallel systems may be necessary. The use of countercurrent regeneration reduces the consumption of regeneration chemicals.

Jonsson (1984) described the SMR (Swedish Dairies Association) process for whey demineralisation, in which the whey first enters a weak anion column in which the whey anions are exchanged for HCO_3^- ions. Following this a weak cation column exchanges the

whey cations for NH_4^+. The whey salts are thus exchanged for ammonium bicarbonate which decomposes to NH_3, CO_2 and water during subsequent evaporation, the NH_3 and CO_2 being recovered. Jonsson and Arph (1987) compared conventional ion-exchange demineralisation of cheese whey to the SMR process and concluded that the requirement for regeneration chemicals and production of waste chemicals are much reduced in the SMR process.

Demineralisation by ion-exchange resins is used at various stages during the manufacture of sugar from either beet or cane, as well as for sugar solutions produced by hydrolysis of starch. In the production of sugar from beet, the beet juice is purified by liming and carbonatation and then may be demineralised by ion-exchange (McGinnis, 1971). The carbonated juice is then evaporated to a thick juice prior to sugar crystallisation. Demineralisation may, alternatively, be carried out on the thick juice which has the advantage that the quantities handled are much smaller, but is limited by the fact that diffusion rates are low at high sugar concentrations. To produce high-quality sugar the juice should have a purity of about 95%. Rousseau (1984) described the 'new demineralisation/demi' process which utilises a mixed bed of weak cationic and weak anionic resins in a batchwise process to treat the thick juice (dry matter 70%). This gives rise to a very pure thick juice with minimum dilution, with the bonus of a decolorisation at no extra cost. A further application in beet sugar production is the Quentin process by which the sugar level of molasses can be decreased. This is achieved by exchanging potassium and sodium ions of the juice prior to the final crystallisation, for magnesium using a strongly acidic cation exchanger. Magnesium is less molassigenic than alkaline ions.

Ash removal or complete demineralisation of cane sugar liquors has been described by Chen (1985). The process is carried out on liquors that have already been clarified and decolorised, so the ash load is at a minimum. The use of a mixed bed of weak cation and strong anion exchangers in the hydrogen and hydroxide forms, respectively, reduces the prolonged exposure of the sugar to strongly acid or alkali conditions which would be necessary if two separate columns were used. Destruction of sucrose is thus minimised.

The cation and anion resins are sometimes used in their own right for dealkalisation or deacidification, respectively. Weak cation exchangers may be used to reduce the alkalinity of water used in the manufacture of soft drinks (Carney, 1988) and beer (Cristal, 1983), while anion exchangers can be used for deacidification of fruit and vegetable juices (Lue and Chiang, 1989; Dechow *et al.*, 1985). In addition to deacidification, anion exchangers may also be used to remove bitter flavour compounds (such as naringin or limonin) from citrus juices (Johnson and Chandler, 1985). Anion or cation exchange resins are used in some countries to control the pH or titratable acidity of wine (Rankine, 1986; Bonorden *et al.*, 1986) although this process is not permitted by other traditional wine producing countries. Acidification of milk to pH 2.2, using ion-exchange during casein manufacture by the Bridel process, has also been described (Pierre and Douin, 1984).

Ion-exchange processes can be used to remove specific metals or anions from drinking water and food fluids, which has potential application for detoxification or radioactive decontamination. For example, procedures have been described for the removal of lead (Brajter and Slonawska, 1986), barium and radium (Snoeyink *et al.*, 1987), aluminium

(Pesavento *et al.*, 1989), uranium (Sorg, 1988) and nitrates (Lauch and Guter, 1986) from drinking water. Removal of a variety of radionuclides from milk has been demonstrated. Radiostrontium and radiocaesium can be removed using a strongly acidic cation exchanger (Tait *et al.*, 1989; Koga *et al.*, 1968), while I^{131} can be adsorbed on to a variety of anion exchangers (Barth *et al.*, 1970). The production of low sodium milk, with potential dietetic application, has been demonstrated (Nakazawa and Hosono, 1989).

Decolorisation

Demineralisation processes may have the added benefit of colour removal. There are, however, other cases where colour removal is required without demineralisation.

Sugar liquors from either cane or beet contain colourants such as caramels, melanoidins, melanins or polyphenols combined with iron. Many of these are formed during the earlier refining stages, and it is necessary to remove them in the production of a marketable white sugar. The use of ion-exchangers just before the crystallisation stage results in a significant improvement in product quality. It is necessary to use materials with an open, porous structure to allow the large colourant molecules access to the adsorption sites. Chen (1985) described the use of strongly basic resins operated in the chloride cycle for decolorisation during cane sugar refining. These are sometimes the only decolorising systems used, but in other cases complement the use of carbon adsorbents. Bohm and Schafer (1969) described the decolorisation of beet sugar juice on an industrial scale using ion-exchange resins.

A new approach to the use of ion-exchange for decolorisation of sugar solutions is the application of powdered resin technology. Finely powdered resins (0.005–0.2 mm diameter) have a very high capacity for sugar colourants due to the ready availability of adsorption sites. The use of such materials on a disposable basis eliminates the need for, and the accompanying disposal problems of, chemical regenerants, as well as removing the problem of sugar dilution which occurs during column operation. However, the advantages must be weighed against the added expense of discarding expensive resins after a single use (Chen, 1985). Colour reduction of fermentation products such as wine has also been described. Brown *et al.* (1988) used a strongly basic anion exchanger to remove colouring matter, followed by a strong cation exchanger to restore the pH. It is claimed that colour reduction can be achieved without substantially deleteriously affecting the other wine qualities.

Protein purification

Ion-exchange can be used successfully in many protein purification processes in the food and pharmaceutical industries. High purity protein isolates can be produced in a single step from dilute solutions containing other contaminating materials. The process compares favourably with competing techniques in terms of cost and efficiency. The amphoteric nature of protein molecules permits the use of either anion or cation exchangers, depending on the pH of the environment. Elution takes place by either altering the pH or increasing the ionic strength. The eluate can be a single bulk, or a series of fractions produced by stepwise or linear gradients, although fractionation may be too complex for large-scale industrial production. Separation of a single protein may take place on the basis that it has a higher affinity to the charged sites on the ion-exchanger

compared to other contaminating species, including other proteins present in the feed. In such cases, if excess quantities of the feed are used, the protein of interest can be adsorbed exclusively, despite initial adsorption of all the proteins in the feed (Kanekanian and Lewis, 1986). Alternatively it may be possible to purify a protein on the basis that it has a much lower affinity for the ion-exchanger than other proteins present in the feed, and thus the other proteins are removed, leaving the desired protein in solution.

One limitation of the process for protein treatment is that extreme conditions of pH, ionic strength and temperature must be avoided to prevent denaturation of the protein.

An area of great potential is the recovery of proteins from whey. It is estimated (van Hoogstraten, 1987) that about 110 million tonnes of whey are produced each year as a by-product of the manufacture of cheese and related products such as casein. Typically whey contains 0.6–0.8% protein, which is highly nutritious and also displays excellent physical properties, yet the vast majority of this is wasted or under-utilised. The Vistec protein recovery process employs carboxymethyl cellulosic anion-exchange materials to produce high purity functional protein from cheese whey (Jones, 1976; Palmer, 1977). The system uses a stirred tank reactor into which the whey is introduced at low pH. Following rinsing of non-adsorbed material, the protein fraction is eluted at high pH, and further purified by ultrafiltration so that the final protein content is approximately 97% (on a dry matter basis). The product is commercially exploited by the Bio-isolates company (Fig. 6.2). Ayers and Petersen (1985) have described a similar process based on sulphopropyl cellulosic materials which is also used for commercial recovery of cheese whey protein. Silica-based ion-exchangers such as Spherosil have the advantage that they are rigid and do not swell or contract when the pH or ionic strength of the environment

Fig. 6.2. Ion-exchange recovery of food proteins (with permission of Bio-isolates plc).

are altered, and are used commercially for whey protein recovery in the Rhône–Poulenc process (Mirabel, 1978). However, the capacity may be lower than the cellulose-based materials. Skudder (1983, 1985) has demonstrated the use of Spherosil QMA to produce fractions of total whey protein as well as further fractionating the proteins into their separate components or groups of components. This approach has the potential of producing protein fractions with a range of functional properties which could be extremely valuable for use in the food industry. The method is not yet carried out commercially to the author's knowledge, possibly due to the complex operating procedures required, and the relatively low capacity of Spherosil. Another application of adsorption of whey protein by ion-exchangers could be to improve the heat stability of milk (Kelly, 1982). The use of ion-exchange to recover or separate the caseins in milk is not carried out commercially, although it has been shown to be feasible – e.g. Ng-Kwai-Hang and Pelissier (1989).

Jones (1976) has described the use of the Vistec system (see above) for continuous recovery of food protein from protein-containing waste streams other than whey. A 10 000 gal d^{-1} pilot plant system, recovering protein from abattoir effluent, had been run continuously for 3 months. The recovered protein could then be incorporated into animal feeds. The same author has also demonstrated the use of this system for recovery of food proteins from waste streams resulting from the processing of soya, fish, vegetables and gelatine production. Such protein fractions could be used as functional proteins in the food industry.

Howell and Lawrie (1983) employed anion-exchange using DEAE-Sephadex to fractionate porcine plasma proteins in an attempt to maximise use of by-products of the meat industry. While the properties of such fractions would clearly be desirable for their use as functional food proteins, the stepwise elution may be too complex for commercial application.

Various other food proteins have been purified or fractionated by ion-exchangers, including pea globulins (Gueguen *et al.*, 1984), gliadin from wheat flour (Charbonnier and Mosse, 1980), egg protein (Parkinson, 1967) and groundnut and soya protein (Satyanarayana *et al.*, 1981).

Purification of proteins from fermentation broths usually involves a series of separation steps and frequently includes ion-exchange. Hammond and Scawen (1989) reviewed the use of ion-exchange in the high-resolution fractionation of proteins in downstream processing. Werner and Berthold (1988) discussed the purification by ion-exchange of a range of recombinant DNA-derived proteins, produced by fermentations, which can be used as active ingredients in pharmaceuticals. The industrial production of recombinant insulin from *Escherichia coli* fermentation involves an ion-exchange purification (Prouty, 1989).

Large-scale purification of a variety of enzymes has been described. For example, α-amylase recovery from *Aspergillus awamori* (Bhella and Altosaar, 1985) or L-leucine dehydrogenase from *Bacillus cereus* (Schuette *et al.*, 1985) may be achieved by ion-exchange. Porter *et al.* (1991) described the large-scale purification of β-galactosidase from soybean meal using a combination of strong anion- and strong cation exchangers.

Ion-exchange is used in the recovery, separation and purification of monoclonal antibodies which are used for high-resolution diagnostic purposes. Duffy *et al.* (1988)

describe the purification of kilogram quantities of monoclonal antibodies by ion-exchange.

Purification of other compounds

Ion-exchange has been used for numerous other separations involving food and biochemicals, which do not fit into the above categories.

Fructose production is of great interest as it is considerably sweeter than sucrose and glucose, and can be used as a natural sweetener at reduced caloric intake. Although present in many natural sources, it is produced commercially from corn starch by hydrolysis to dextrose, which is then partially converted to fructose using the enzyme isomerase. The resulting high fructose corn syrup may be deionised by ion-exchange and then a pure fructose fraction can be recovered with a sulphonic cation exchanger (such as Amberlite IR-140). The separation is based on the fact that such resins, in the divalent salt form, exhibit a slightly higher affinity for fructose than glucose (Kunin, 1979). In practice a small volume of syrup containing both sugars is placed on top of a bed of the ion-exchanger and slowly displaced with water so that the glucose travels more rapidly down the column than fructose. By alternating the feed of syrup with eluting water, a series of 'cuts' of fructose and glucose may be collected. Although the mechanism is complex, the process is carried out successfully in a number of commercial plants.

A further potential application is the production of lactose-free milk. A process using sulphonated cation exchangers has been used to reduce the lactose level of skim milk to <10% of that in the feed, while retaining >90% of protein, minerals and citrate (Harju, 1987).

Many products derived from fermentation processes are purified by ion-exchange. The following are some examples. Kunin (1974) described the recovery of the aminoglycoside antibiotics streptomycin and neomycin directly from a fermentation broth using an expanded bed which permits particulate matter to pass through without clogging the bed. Purification of fermentation-derived ethanol for use as fuel is possible using a combination of strongly basic and strongly acidic ion-exchangers (Rohm and Haas (patent), 1988). The purification of phenylalanine, which may be used in sweetener production, from fermentation broths using cationic zeolite material, has been patented (UOP (patent), 1990).

Ion-exchange may also be used to purify enzymic reaction products, e.g. Heinzler *et al.* (1987) incorporated the process in a system for recovery of flavour constituents from the enzymic degradation of fruit wastes.

6.2 ELECTRODIALYSIS

Electrodialysis (ED) can be used to separate ionic species in the food and biotechnology industries. The process permits the separation of electrolytes from non-electrolytes, concentration or depletion of electrolytes in solutions, and the exchange of ions between solutions.

6.2.1 Theory and equipment

Separation occurs due to electromigration of ions through membranes, which depends on the electrical charge on the molecules, combined with their relative permeability through membranes. Separations are based on the use of ion-selective membranes which are effectively sheets of ion-exchange resins. The membranes are composed of polymer chains which are cross-linked and intertwined into a network, and bear either fixed positive or fixed negative charges. These may be heterogeneous membranes which consist of ion-exchange resins dispersed in a polymer film, or, more commonly, homogeneous membranes in which the ionic groups ($—NH_3^+$ or $—SO_3^-$) are attached directly to the polymer. Counterions are freely exchanged by the fixed charges on the membranes and thus carry the electric current through the membranes, while co-ions are repelled by the fixed charges and cannot pass through the membrane. Therefore cation membranes allow the passage of positively charged ions, while anion membranes allow the passage of negatively charged ions.

In practice the cation and anion membranes are usually arranged alternately with plastic spacers (Fig. 6.3) to form thin solution compartments as shown schematically in Fig. 6.4. In commercial practice 100–200 membranes may be assembled to form a membrane stack (Fig. 6.5), and an ED system may be composed of one or more stacks. Commercial ED membranes may be as large as 1–2 m^2. The spacers must maintain a constant spacing between the membranes, but must not cover a large fraction of the membrane surface or cause stagnation of the fluids. This is usually achieved by either using a netting material which also promotes turbulence, or the use of a tortuous path arrangement which forces the liquid stream into a long pathway before leaving the cell. The basic unit of a membrane stack is called a cell pair and comprises a pair of membranes and spacers as illustrated in Figs. 6.3 and 6.4. A positive electrode at one end

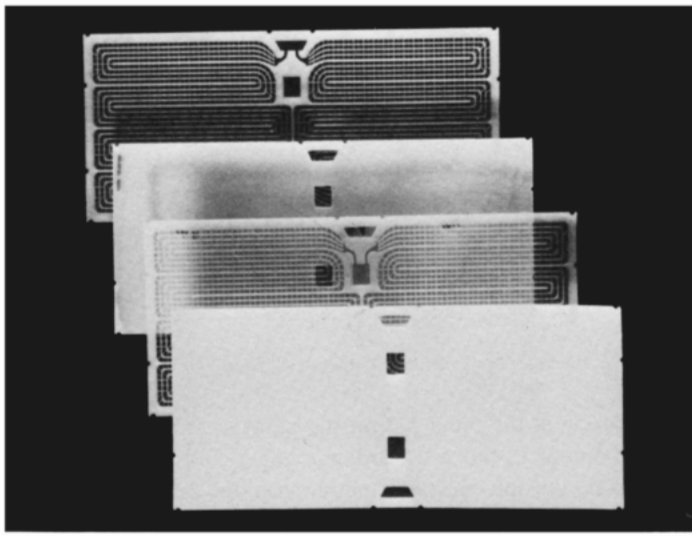
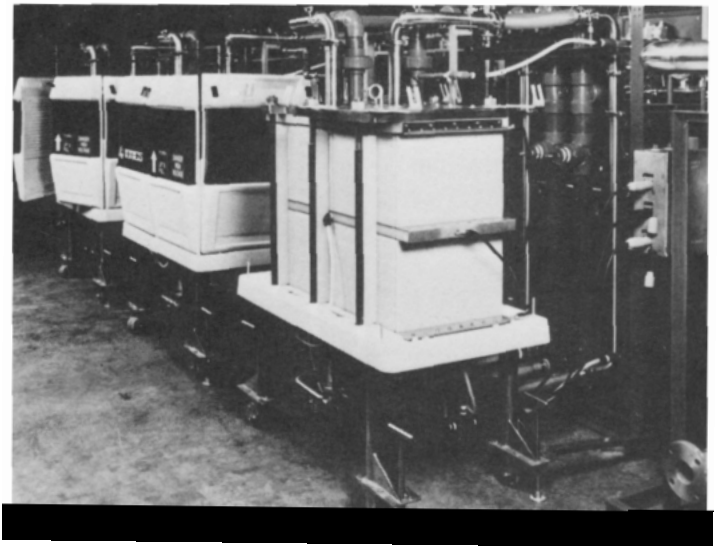


Fig. 6.3 Electrodialysis membranes and spacers (with permission of Ionics Inc.).



- animal – faeces, hair, insects, eggs;
- chemical – pesticides, fertilisers, other contaminants;
- microbial – yeasts, moulds, bacteria and metabolic by-products, e.g. mycotoxins, e.g. patulin in apples.

One of the first preliminary operations must be to remove these. Important considerations are high efficiency of removal, combined with minimising loss and damage and further recontamination of components. A combination of methods is used, including dry cleaning and wet cleaning. Aspiration to remove dust and light contaminants has already been described.

Screening is widely used for removing contaminants considerably different in size to the food being treated. Sieves are available up to an aperture size of 25 mm and screens for larger sizes (see Section 9.3.2). Also disc separators, where the shape of the disc matches the shape of the food, can be used for separating seeds from grain.

It is important to be able to detect the presence of metal fragments and remove them from the raw material, to prevent damage to the food processing equipment and contamination of the final product. Metal detectors may also be incorporated at the end of the packaging line. Magnetic materials can be removed by powerful magnets, which can be permanent magnets or electromagnets. Non-ferrous metals, such as aluminium, are detected by passing the material through a strong electromagnetic field. This field is distorted and initiates a warning signal. X-rays have also been used for products in sealed containers. It has been reported that modern cocoa processing leads to iron contamination levels of 200 mg/kg in cocoa mass and greater than 300 mg/kg in cocoa powder; this arises from hammer mills, impact mills and the agitator blades of rotating ball mills. Between 5 and 15% of the metal was greater than 75 μm. Improvement in design of this equipment was considered to be the best way of reducing this (List and Thiede, 1987). Krishnan and Berlage (1984) looked at the principle of separating walnuts by a magnetic field. Iron dust with gelatin or a magnetic solution was added to the whole nut. This was cracked and the shell separated from the meat using a permanent magnet.

Electrostatic methods for cleaning materials are available, which take advantage of the differences in electrostatic charge of materials under controlled humidity conditions. The solid is fed from a hopper onto a drum, rotating at 70–350 r.p.m., which is either charged to a potential of 5–20 kV or earthed and the oppositely charged particles are separated as they are more strongly attracted to the drum. They are removed from the drum by a scraper. This method can be used to remove dust and stalk from tea fannings and also some unwanted seeds from cereals and oilseeds (Brennan *et al.*, 1990). The Dodder mill uses a roller coated with a velvet-type material, which will attract particles, such as seeds, and remove them from cereals, due to differences in their surface properties.

Wet methods are also widely used for cleaning purposes. Heavily soiled vegetables can be simply presoaked in water; the process helped by agitation. A more efficient process is spray washing, which uses high-pressure sprays and requires smaller volumes of water. The principles of flotation are used for cleaning vegetables; heavy particles such as soil, metal or glass sink, the vegetables are neutrally buoyant and straw and grass float. Screens may also be incorporated to remove oversize material. In wet-processing

applications, the microbial quality of water supply and the additional costs of the water and effluent treatment need to be considered.

Peeling or dehulling, as well as removing the outer layers, will also remove any disorders associated with them.

9.6.4 Sorting and grading

Sorting and grading are important preliminary operations. Sorting is normally reserved for processes which separate foods into categories based on a single physical property, such as size, shape, weight or colour. Grading, on the other hand, is a quality separation and a number of factors may have to be assessed. Some examples are colour, absence of blemishes, flavour and texture. Food grading is usually done manually, by trained experts, because it is not usually possible to link quality with one physical property.

Some examples are meat grading and inspection, fish grading, horticultural products, tea and cheese. However, the food analyst is always seeking for instrumental techniques for assessment of these sensory attributes which contribute towards the character of the food. For example the dielectric properties of fish have been found to change as the fish becomes less fresh. Consequently a wide range of instrumental methods has been evaluated for measurement both on-line and in the laboratory, of properties of foods that correlate with the sensory characteristics of appearance, colour, flavour and texture. Appropriate instrumentation and sensors have been reviewed by Kress-Rogers (1993).

Equipment for size sorting based on rollers and screens, which provide either a fixed or a variable size aperture, are discussed in more detail by Fellows (1990) and Brennan *et al.* (1990). Sorting by weight is important for high value products such as eggs, and some tropical fruits. Image analysis is being increasingly investigated in this respect.

Colour sorting and grading

Foods can be sorted on the basis of their colour, for example removing discoloured baked beans, prior to them being blanched. One of the most common applications is to pick out miscoloured pieces and the simplest method is by manual inspection, as the food passes by the inspectors, on conveyor belts.

Colour sorters have been available for over 40 years and one widely used application range is for particles in the range 2–10 mm. Some examples are: rice, baked beans and other legumes, peanuts and roasted coffee beans. Throughputs range from about 10 kg h^{-1} up to 10 000 kg h^{-1}, with many applications between 100 and 1000 kg h^{-1}.

These are based on a sensor located above the conveyor. The feed is divided into lanes or channels. Mohsenin (1984) summarises the sequence of operations as singularisation into discrete units, acceleration to present a substantial number of units to the system per unit time, presentation of each unit before the sensors, evaluation and comparison to some predetermined standard, and segregation to separate each unit according to its colour or other specified standard. Figure 9.12 illustrates the layout of such a sorter.

Colour sorting relies on the optical properties and reflectance of the samples. The principle is that the light source is directed on the material and the reflected light is measured by a photodetector and compared with preset standards. Materials outside the range are rejected. The incident light and reflected light may or may not be filtered, to

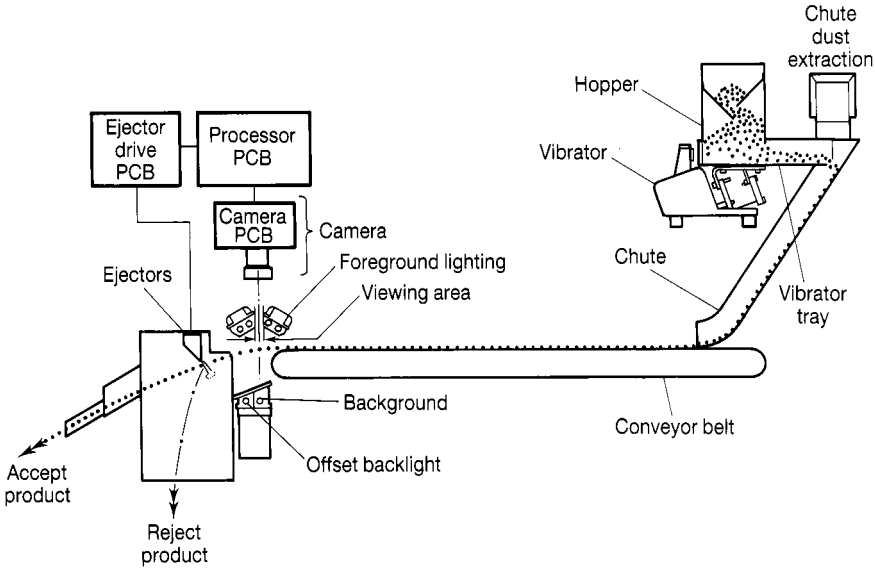


Fig. 9.12. Diagram of colour sorter (with courtesy of Sortex Ltd).

allow only selected wavelengths to reach the detector. Monochromatic sorting uses only one selected waveband range and is used where there is sufficient difference between the reflectivity of acceptable and unacceptable products within the selected waveband. Unfortunately it is not always possible to find a single section of the spectrum where this is the case, so it is necessary to resort to a more complicated procedure which involves measuring at two selected wavebands. This is known as bichromatic sorting. Usually a ratio of the signals from the two wavebands will facilitate sorting of the materials. In some cases dual monochromatic sorting is used, where it might be important to reject more than one type of defect. A discrepancy in either signal will cause the item to be rejected. More detail is provided by Low and Maughan (1993). Other factors to be considered are the natural variations in colour that are found for each product. The feed rate is also important and there may be problems operating in a dusty or humid environment.

As in most applications, such machines are never 100% efficient in terms of either removing all defective items or rejecting acceptable items. Improving the sensitivity of the detection unit increases the efficiency for removing defective items, but also means that a greater proportion of acceptable items will be rejected. There is also a recommended flow rate range for each machine. Increasing the flow rate within this range usually leads to a greater loss of acceptable material. In most cases, the overall performance improves if the material has been cleaned and size graded prior to colour sorting. Some typical examples of removal efficiencies for defective items are

- green coffee: removing defective beans and foreign material, 90% at 900 kg h^{-1};
- white beans: removing discoloured beans and foreign material, 97% at 1500 kg h^{-1};
- frozen peas: removing foreign material, 99% at 10 000 kg h^{-1}; (Low and Maughan, 1993).

Other applications are for sorting of fruit, picking out bruised, damaged or mouldy fruit. Colour measurement is used for control purposes; for example controlling the energy input into baking ovens to ensure a product of uniform colour from the oven. Transmittance methods form the basis of egg inspection and have been used to distinguish between cherries with and without pips.

Future developments will combine colour sorting with vision analysis, whereby sorting will be based on colour differences, size and shape.

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